Knowledge and Skills in China's K-12 Population:
An Inquiry into "Knowledge Capital" in the PRC

Working Paper

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Abstract

In this study, we analyze evidence on the knowledge, skills and capabilities of the cohort of young Chinese in primary and secondary schooling. To go by internationally standardized tests of achievement for pupils from this age group, China would look to be very favorably positioned indeed. According to successive rounds of global testing in reading, mathematics and science, pupils from China are Number One in the world—and across the board. We use both quantitative and qualitative evidence to survey the field and emphatically reject the PISA-based assessment that places China first in the world. Instead, we come away with the working hypothesis that the knowledge capital of China's K-12 population is on par with that of Turkey.

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Introduction

In this study, we analyze evidence on the performance and capabilities of the cohort of young Chinese in primary and secondary schooling. It is a huge population: nearly 300 million boys and girls under the age of 18 live in China today, and something like 180 million of them are 6-15 year olds, for whom the PRC's nine years of schooling is obligatory.[2]

The training, scholastic preparation, and academic achievement of this rising generation will bear directly on China's future economically, and indirectly on Beijing's strategic options and possibilities. To go by internationally standardized tests of achievement for pupils from this age group, China would look to be very favorably positioned indeed.

According to successive rounds of global testing in reading, mathematics and science, pupils from China are Number One in the world—and across the board. These stunning, but recurrent, findings come from PISA, the Programme for International Student Assessment (conducted under the aegis of the OECD, or Organization for Economic Cooperation and Development) and tests 15 year olds from over 100 countries, places and sub-territories.[3] In the most recent round of PISA testing in which it participated, China pupils came in first in all three subjects among the nearly 80 participating countries and economies.[4] China's tested students outperformed counterparts in famously high-performing Singapore in every field, and left Estonia, PISA's top non-Asian performer, even further behind.

The China students' mean math and science scores were over 100 points higher than the corresponding averages for OECD countries. This implies that Chinese students were two or more years ahead of OECD pupils in these fields. And since the PISA tests are scaled such that a 100-point difference corresponds to one Standard Deviation, this means that fully *half* of China's students were testing at the same level in math and science as the OECD's *top sixth*—with *over six times* as many China students earning the scores achieved by the OECD's highest two and a half percent of test takers.

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[2] These estimates are derived from the United Nations World Population Prospects, 2022, https://population.un.org/wpp/; projections for year 2023
[3] OECD, "PISA Participants", https://www.oecd.org/pisa/aboutpisa/pisa-participants.htm
[4] The 2018 round cf. Wikipedia, Programme for International Student Assessment, https://en.wikipedia.org/wiki/Programme_for_International_Student_Assessment China did not participate in the subsequent 2022 PISA round, whose results were recently released, owing to its ongoing COVID lockdowns at the time.

The reading gap between test takers in China and the OECD international mean was not quite as overwhelming—but the difference of 68 points still qualified as extraordinary: comparable to the gap on the downside between that OECD average and the mean for, say, Jordan.

These stunning findings for China in the PISA tests have been validated and repeatedly defended by PISA authorities and OECD administrators. In this paper we assess these results ourselves—so as to validate them, or challenge them, through both evaluation of the PISA results for China *per se*, and also through bringing to bear additional quantitative and qualitative evidence on knowledge and skills for China's pupils—covering as much as possible of the PRC's enormous and vastly diverse pupil population.

In this paper, our assessment of the scores awarded to "China" students by PISA authorities leads us to *reject* those results. We find them to be implausible—clearly unrepresentative of true levels of knowledge and achievement of pupils in China. We demonstrate the reasons for our skepticism in the following pages.

The "China" tested in PISA is by design <u>not</u> the country of 1.4 billion persons—instead it is a cherry-picked selection of the PRC's most affluent and highly developed areas. But even for these specially chosen provinces (Beijing, Shanghai, Jiangsu, and Zhejiang in 2018), there is reason to be dubious that PISA results accurately reflect the knowledge and skills of a genuinely random sample of students.

Neither the PRC Ministry of Education nor OECD staff in charge of the PISA exam and database will permit outsiders unfettered access to the "China" findings. Unlike PISA scores for the USA, for example, where we can look at performance on a state by state basis—we do this for Massachusetts as a comparator with pupils from China in this study—neither PRC nor PISA will allow independent researchers to examine their "validated" scores from China on a province by province basis. Yet even without that transparency, some of the quirks in the PISA data for "China" are so peculiar as to fail the "laugh out loud test" on first viewing.

PISA is the only international dataset on student achievement in which Beijing officially participates. But there are other officially approved and ostensibly standardized data sources on student achievement for China: these include Renmin University's CEPS (China Educational Panel Survey); Peking University's CFPS (China Family Panel Survey), and the national *gaokao* examinations, the mass testing system for young men and women applying to PRC colleges and universities. We examine all of these to look for clues about the true profile of knowledge and skills for pre-college youth in the PRC today.

We were struck—but also intrigued—by the remarkable degree of opacity we encountered in these PRC-generated data on student achievement in China. After working with these datasets intensively, we were tempted to wonder whether the published results had been expressly reviewed internally before release with an eye to avoiding revealing insights into the true status of knowledge and skills for PRC pupils.

CEPS is impenetrable with regard to the “universe” of students it is sampling—it simply refuses to share information on the counties or locations examined with outside researchers. As will be seen, our detective work offers some reasonable surmises about just which counties and districts in China might have been included in this survey. But our reasoned suspicion is that these counties and districts are completely unrepresentative of China as a whole. What CEPS draws its samples from are clearly the most privileged spots within the entire country—places where student achievement is comparable to, indeed almost equal with, with the three districts CEPS admits to be part of Shanghai. It is not even clear that any of the locations included in CEPS represent any *genuinely rural* students—even though rural areas continue to account for a very large fraction of China’s population, and still accounted for a majority of youth 6-17 years of age in 2010, a few years before the CEPS was conducted.

Gaokao college entrance tests in China are deliberately incomparable from one province to the next, conveniently thwarting outsiders attempting to assay the true range and distribution of student achievement in China. These province-level “affirmative action”-style adjustments to test scores from more “backward” provinces, to compensate for the poorer results of their relatively disadvantaged student bodies, are of course highly revealing government interventions in and of themselves.

In the course of its transformation from an elite test in the 1990s for a small select share of PRC youth to a mass test today—by our estimates, about 40% of China’s 17 year olds apply each year for a perch in the explosively expanded PRC higher education archipelago—it became politically impossible to maintain a single national *gaokao* standard. That would have highlighted the yawning gaps separating student knowledge and skills across China in recent years—and would have further underscored how poorly prepared and improbably suited a great many would-be young PRC collegiates actually were. Even with compensatory "plus points", so to speak, the nominal distribution of *gaokao* scores in more "backward" provinces (such as Guizhou) stand out as markedly weaker than for China's richest and most developed areas (such as Beijing and Shanghai). That is, even after "grading on a curve" to raise scores in lower-achievement provinces by appreciable but officially secret adjustments, the distribution of achievement in the *gaokao* scores for the regions included in PISA testing is distinctly higher than those not included.

We cannot know just what the true nationwide distribution of *gaokao* scores would look like if Beijing were instead to re-institute a single nationwide standard. But we can be certain that the dispersion of results would be *much* more uneven than the all-China distribution of results that the PRC Ministry of Education officially reports—with the mean all-China level of achievement substantially lower than what available, officially-doctored, *gaokao* numbers report. And we also know that, as a practical matter, *gaokao* scores are basically unusable as a measure of student performance by overseas colleges and universities considering PRC applicants.

This is not to say *gaokao* tests are never accepted by international educational institutions. To the contrary, and as we show, they are indeed used in the application processes of quite a few Western colleges and universities nowadays in their evaluation of applicants from the PRC. But *gaokao* has proven to be a highly problematic measure abroad: sometimes only acceptable from certain provinces; and other times prone to bring surprises for the Western schools receptive to them. Suffice it to say that the Educational Testing Service (ETS), the research and evaluation organization responsible for the SAT and ACT college admission tests in the US, does no work on, and will not work with, the *gaokao* test; for ETS, the current *gaokao* is not a reliable measure of student achievement.

For PISA's "China" scores to be converted into something approximating all-China scores comparable to tested students from other participating countries, systematic adjustments—downward adjustments—will be required. But just how major these adjustments should be—and just what countries would be comparable with all-China scores after such adjustments—is the critical question.

We come up with a range of possible adjustments to the official PISA "China" to convert them into plausible all-China scores, relying upon testing data for PRC students and other indicators. The boundaries delimiting such plausible adjustments, however, are strikingly wide.

At the upper end are adjustments utilizing the CFPS province-specific student achievement scores as a "bridge." As we will see, there are evident shortcomings to the CFPS data (not least of these the missing, poorer, provinces not included). But assuming the CFPS results were true and accurate reflections of achievement by province, and accepting as well the PISA "China" results to be true reflections of student performance in the provinces tested, we obtain an adjustment that would still leave all-China as one of the world's very top performers for pre-college pupils.

Our hybrid, CFPS-based adjustments would place all-China science and math performance up in the lofty realm of Singapore—the country with the top all-nation performance in PISA—and on par for reading with South Korea, ranked 9th out of 79 in PISA 2018 for that subject.

At this upper bound of adjustments, all-China pupils would still qualify as formidably well trained; knowledge and skills for pre-college youth in China, by such computations, would appear to be much better developed than for the US—or for the international average of OECD countries.

We note here that our calculations here turn out to be close to separate estimates by Stanford's Eric Hanushek, the renowned economics of education specialist and "knowledge capital" pioneer, who independently settled on the same approach in attempting to come to a more realistic assessment of an all-China placement in PISA-world.

But this high upper-bound is very high indeed—so high that it may seem to demand suspension of disbelief for many readers. And we cannot blame them. A little back of the envelope arithmetic suggests that, even after these adjustments, such all-China numbers might be suggesting that the bottom half of China's test takers—more or less, the rural population—could be out-performing the national populations of countries like the UK, Australia, and Germany. That sounds an awful lot like a laugh-out-loud result.

There are a number of alternative quantitative approaches to estimating academic achievement for PRC that arrive at lower proposed levels of national performance—some of them very much lower.

The lowest of them, at least for now, comes from World Bank researchers associated with the Bank's Human Capital Index (HCI) project. They have assembled a global database of student achievement agglomerating and somehow concording results from a number of different tests, a resource they call the Harmonized Learning Outcomes (HLO) dataset. (The HLO's approach to aggregating results from different achievement datasets, we should note in passing, is innovative and unorthodox—and not accepted by all experts in this academic niche.)

By their adjustments to their test taking ingredients from the PRC, they conclude that the all-China mean level of student achievement is radically lower than PISA reports. On the PISA scale HLO assigns all-China a mean score of 456, as against the overall averaged science, math and reading scores of 579 for 2018 that has been "validated" by OECD and PISA. (Later—ironically, in light of the PISA 2018 information for "China"—the HLO team revised their all-China estimate—downward even further, to 441, for a difference of 138 points.)

Differences of those magnitudes would amount to the equivalent of about three years of classroom learning by conventional measures for such testing. They would also knock all-China performance down from the notional ranks of Singapore and South Korea in our upper-boundary estimates to the much less exceptional level of comparators like Turkey[5] (or slightly below Turkey)—an arguably creditable showing for a higher middle-income developing country, as the PRC in fact is.

At this lower bound estimate, all-China mean pupil achievement would fall below the overall PISA 2018 international average for OECD countries (441-456 vs 488) or for the USA (441-456 vs 495). But it would still be higher than for such Southeast Asian economies as Malaysia (431) and Thailand (413).

Furthermore, such a mean level of all-China achievement would be consonant with very high international levels of "knowledge capital" for a very large number pupils from China.

Given that a 50 PISA point differential is meant to be equivalent to one half a Standard Deviation in statistical terms, a normal distribution would have up to 30 percent of all-China pupils performing 50 or more points above the HLO's notional estimates of mean all-China pupil achievement scores. That is to say: an

[5] Officially in Turkish the country is *Türkiye*. Anakara today prefers this variant and now uses it exclusively; increasingly UN and other international publications are using it as well. We will stick with describing it as "Turkey" since this is still the most familiar version for English language readers.

all-China HLO score of 441-456 would imply that almost 30 percent of all-China pupils would be performing at the achievement level of 506 or above. To put that number in context: it would be consistent with the proposition that nearly a third of all-China pupils were performing above the mean achievement level of the country of, for instance, Sweden—and that something like a third were above the mean level for the United States.

The upper and lower boundary of existing quantitative estimates of all-China student achievement *relying on test data from China*—that qualifier is important, we are only talking here about adjustments based off actual in-China pupil achievement data—thus spans a range of about 85 PISA points. That range of uncertainty is, for analytical or policy purposes, maddeningly wide—equivalent to a difference of almost two full years of schooling, or to roughly the mean tested performance gap in PISA 2018 between the international OECD average and Brazil, a relatively poor international performer.

But we can parse this dramatic estimated achievement gap with other quantitative and qualitative evidence that can help us understand what China's true distribution of knowledge and skills for its pre-college population may look like.

Utilizing the HLO dataset, we modeled international correspondences between socio-developmental patterns around the world and mean national scores for student achievement. Our simple models "predicted" an all-China HLO for 2015 of 468, based on what would be expected for a country with China's 2015 levels of income, educational attainment, urbanization, etc.

These are "if-then" calculations *simulating*, rather than *adjusting*, China achievement score data. Our simple models ask: "if" we go by international patterns, "then" what would a country with China's developmental profile look like? And our simple models suggest a "simplified China"—a country with some but obviously not all of its characteristics—would have a much lower level of student achievement than PISA 2018 avers, although somewhat higher than HLO suggests.

This simple model would suggest China's true level of mean overall pupil achievement would be much closer to HLO than to CFPS-adjusted PISA, much less PISA's reported "China" scores. A China score of 468 would be about nine-tenths of the way to HLO in the HLO-PISA testing differential, and about six-sevenths of the way to HLO for the differential between HLO and our CFPS-adjusted PISA scores. An overall mean score of 468 would be close to what Belarus actually earned in PISA 2018.

Our simple model, however, took no account of “cognitive” factors. Controversial as those may be—psychometrics is regarded as a “forbidden science” in many respectable intellectual circles nowadays—we nevertheless know that cognitive capabilities vary widely, both within countries and around the world—and that they have a powerful influence on student achievement outcomes. When we added a cognitive variable to our simple aforementioned models—using international datasets purporting to represent average national IQ around the world—China’s “predicted” score shot up by 30 points. Taking both socio-developmental conditions and reported IQ figures into account, the revised “if-then” estimate for pupils in a country like China would be closer to 500—a bit higher, in other words, than the overall mean international level reported for OECD countries in PISA 2018.

Including the “cognitive factor” adjustment, our revised model now “predicts” all-China scores to be almost midway between the “China” PISA claims and the HLO calculations—and closer to our CFPS-adjusted PISA scores (about two thirds of the distance between those and HLO). By such calculations, our revised model would have China scores looking like France or Germany in PISA 2018--OECD countries with slightly above average academic achievement. Remember, though, that neither of our international models contain any actual data on testing in China—and that the impressive results in this model rely critically on the assumption that IQ for the all-China population ranks among the highest in the world, as a number of international psychometric databases report.

This is as far as the *quantitative* data was able to take us concerning “knowledge capital” for pupils in China. But there are additional, *qualitative* data that can also help inform us about realities in China, even if they do so without decimal points.

Three realms of qualitative evidence, we believe, should have a direct bearing on our evaluation of “where the needle should end up” in the big range of quantitative uncertainties outlined above for academic achievement for China’s pupil population.

1) The first of these is *test-cheating* and *test-beating*. China’s history of national test cheating traces back to the earliest national attempts at standardized testing, the storied imperial examination system. Some of the earliest uses of mass communication in China were for publication and

dissemination of answer booklets for these tests—notwithstanding official prohibitions and strictures against their circulation. In Chinese civilization test-cheating is a time-honored practice, and it has been honed exquisitely over time into a national art. It is alive and well in the PRC today, as we shall describe in the following pages. Furthermore, test-beating is not only a national pastime, but a preoccupation of many within officialdom—including, we have reason to believe, officials in China engaged in the very PISA testing that the OECD has "validated".

Given the tradition of test-cheating and test-beating in China, and the importance of the PISA test results to local and national officialdom in the PRC, it would seem optimistic in the extreme to assume the results reported for "China" reflect the true level of achievement for a randomly sampled population of un-coached pupils. Rather, the numbers claimed for "China" test scores should more properly be regarded as something approximating the maximum possible level of achievement that could be attained by administrators willing to bend every rule, without explicitly violating them. Taking the "cheating factor" into account, PISA 2018 results for "China" are almost surely biased upward to a serious degree. Faithfully administered test results for the provinces of China included in the PISA waves would almost certainly have shown much lower scores than those officially recorded—although we cannot know just how much lower they would have been.

2) The second is *cognitive differences within China*. IQ testing in China is limited, there is scant history to this practice in China before the Communist Revolution, and the Marxist-Maoist ideological prejudice was to disdain psychometrics as pseudoscience at best—or as subversive and sinister intellectual machinations. Only just recently, apparently, have Chinese experts finally been developing their own China-specific IQ test.

   The paucity of cognitive testing in China in practice means that IQ testing to date has taken place in settings convenient for professionals administering these tests: large urban areas. These represent the "universe" from which the various sampled populations for China's IQ and cognitive batteries have been drawn. These urban Chinese have consistently tested with above-average results. With IQ tests by definition scaled against a norm of 100, China populations are recorded with mean levels in the 104-107 range. Since

IQ is normed so that every 15 points is equal to a Standard Deviation, this would imply that China's tested population has a cognitive capacity, as reflected in such testing, almost up to half a Standard Deviation higher than the international average: meaning in rough terms that the top half of that sampled population would be performing like the top sixth elsewhere.

But IQ testing in China has largely neglected examination of cognitive capabilities in China's rural regions—and there is reason to believe that conditions may be very different in the hinterlands. Pioneering studies by Stanford economist Scott Rozelle and colleagues uncovered the fact that populations of rural children in China consistently test well below the international mean. Rozelle and colleagues surmise, in fact, that up to a third of China's children—urban plus rural together—could have tested IQs of 90 or below: that is, two thirds of a Standard Deviation below normed international averages.

Recall that our simple "if-then" models for "predicting" all-China pupil test scores were extremely sensitive to the suggestion from international IQ datasets that China's average IQ was definitely above international tested averages. If we take account of the evidence for rural China, the presumption of an all-China cognitive advantage may not turn out to be correct. Adjusting for the urban bias in China cognitive soundings, consequently, should also make for the expectation that PISA results for "China" should be scaled down that much more to reflect the true terrain of cognitive/development advantage and disadvantage within that enormous country.

3) The third qualitative factor to be born in mind concerns China's *demography and family structure*. Given the enduring realities of *hukou* policy, a disproportionate share of China's children—including those of age for primary and secondary schooling—live in the countryside. Irrespective of their cognitive capabilities, their educational opportunities are likely to be far more limited in the countryside than would be the case for their urban counterparts. Availability of schooling is more limited in rural areas. Quality of education is almost universally poorer in the country than the city. And costs of education are much harder for rural people to shoulder. Although the classrooms themselves are open at no direct cost to rural children, much

of what is needed to attend (from books to uniforms to equipment) must be paid for out of pocket by students' families themselves—and income levels in the country are on average only about a third of urban levels. (Additional demographic factors may also have an impact, as we shall see.) Given the tremendous remaining differences between rural and urban education in China, consideration of the demographics of China almost certainly weighs toward a serious downward adjustment of PISA-style reported test results.

Not all evidence we analyze weighs in favor of a significant downward adjustment of PISA-style test scores of all-China pupils. The strongest and most interesting evidence in favor of the possibility of genuinely high nationwide achievement rankings comes from the country of Vietnam. Vietnam is poorer than China; it is also looks to be significantly less urbanized. Yet according to PISA 2018, Vietnam's overall mean test score was over 20 points higher than the OECD's overall international mean; higher than the overall means for such places as Switzerland and Scandinavia (Denmark, Finland, Norway, Sweden). There are so far no serious questions from international specialists about the accuracy or authenticity of these results for Vietnam.[6]

Vietnam would appear to be an example proof that Asian populations can indeed "punch far above their weight"—as suggested by income and development indicators—when it comes to pupil achievement. And this is an example that should not be taken lightly: there is no doubt much to learn here.

But there is also a deep and important difference between the China and the Vietnam cases. Vietnam is regarded by PISA authorities, and by other education specialists, as a case where improved quality of education—on a nationwide basis—has made a decisive impact on national achievement. In other words, Vietnam is thought to be a case where educational policy, and attention to school quality in rural areas, has directly helped to account for extraordinary national performance. There can be no corresponding presumption for China. Although the neglect of rural education is, by various accounts, being addressed by PRC leadership nowadays, it is still a matter worthy of concern. And the very fact that PRC authorities did not permit PISA to test China's vast rural hinterlands at all

[6] The 2022 round of PISA testing shows an across-the-board drop for Vietnam scores in Science, math and reading—but Vietnam's 2022 scores nonetheless remain above the PISA 2022 overall international OECD mean.

may tell us all we need to know about what they thought such tests would reveal about the true China.

When we take all this evidence into account, we are left with an inescapably normative task. Weighing these many quantitative and qualitative factors together requires a measure of subjective judgment; informed observers could arrive at different conclusions assaying these same facts and figures. But an exercise of this sort invites exactly such a judgement. If worth the effort that went into it, an exercise of this sort virtually demands the investigator to offer such a judgement. It is not a mark of distinction, or badge of courage, to shrink from that invitation.

Given what we have learned about academic achievement for China's pre-college population, and what we have learned about conditions of education and living for those many scores of millions of Chinese boys and girls across that vast country, we are fairly confident that the true overall level of achievement is substantially overstated by our "high bound" CFPS-adjusted estimate.

Or tentative surmise—or working hypothesis—would be that a true reported level of knowledge and skills, as represented by HLO/PISA metrics, would place all-China pupil scores rather closer to the "lower boundary" than the "upper boundary" estimates discussed here. That is to say: we are inclined to suspect that the overall mean level of achievement for China is not as high as the corresponding overall international level for OECD countries reflected in their PISA scores not nearly as high.

Just how much below these mean aggregate OECD levels might we expect to find faithfully administered all-China pupil scores? IF all-China cognitive readings proved to be substantially lower than the above-average figures so commonly taken as true, we could imagine nationwide levels very close to the HLO's assigned figure for China—or to our own simple "if-then" model's "prediction".

We can't "prove" a hunch—that is the nature of a hunch. But our hunch is that the all-China terrain of tested student achievement, if we could obtain the data to describe this, might be akin to the readings we actually can obtain from a country like Turkey.

On its face, after all, Turkey is not such a bad developmental comparator for China in other ways, too. Like China, it is a high-middle-income country in the World Bank's classification. Like China, it has big urban-rural gaps (though more of Turkey's people live in its modernized urban areas). Demographic reconstructions from the UN Population Division (UNPD) suggest that life expectancy at birth in China and Turkey are virtually identical—and have been for a generation.

None of this proves that "knowledge capital" for children in the two countries would be similar, of course—but Turkey might be a "baseline" of sorts for comparisons with China in this realm as well. If reliable all-China testing revealed that the nation's cognitive profile for children were closer to that of Turkey than received professional wisdom currently holds, such a results would perhaps be less surprising. But, lest it go unsaid, that is a very big—and as yet untestable—"if".

And at the risk of repetition: even theoretically, with an HLO "low boundary" level of all-China pupil achievement, a presumptive normal distribution of achievement, and a primary/secondary school age population of China's scale—second in size only to India's; larger than Europe's and America's and Japan's and Canada's and Australia's, combined—the PRC would contain an absolutely immense pool of high performing, high-achieving, high "knowledge capital" boys and girls.

Recent work by Eric Hanushek and colleagues on deficits of "global universal basic skills" (defined as inability even to achieve the minimum "level one" standard set in the PISA tests) underscores and formalizes this point.

Slide 3

Illustrative Estimates of Youth on Track for Basic Skills

(applying Hanushek et al GUBS basic skills estimates to 2022 youth population)

Country/Region	0-17 Population (million)	Percent with Basic Skills	Total in Millions
USA	74	75	56
Russian Federation	30	77	23
Europe minus Russia	110	70	77
Japan	18	87	16
India	434	11	49
Latin America&Caribbean	183	35	64
Sub-Saharan Africa	557	6	33
China: Low Skills Estimate	295	31	92
China: High Skills Estimate	295	81	239
China: Midpoint Estimate	295	56	165

Sources: https://population.un.org/wpp/ ; https://www.sciencedirect.com/science/article/abs/pii/S030438782300161X?via%3Dihub

Even with this "lower bound" estimate, we can be confident China would have by far the largest pool of high-achievement pupils of any country on earth.[7] If just a fifth of Chinese pupils performed at or above the mean South Korean level of achievement (an inference one might draw from the HLO figures), there would still be 8 times as many pupils achieving above the ROK mean in China than in South Korea. The sheer mass of talent implied for China by such rough calculations may afford the society, the economy, and the government's leadership opportunities of scale that could not be reaped elsewhere.

But at the end of the day, we are in a position where we must speculate—and guess—about the true terrain of knowledge and skills in China: and we find ourselves in this position because the PRC and the CCP resolutely resist transparency on these matters. One may wonder just how much Beijing itself actually knows about the "knowledge capital" of its pupils.

The regime in China apparently has no great incentive to illuminate the contours of knowledge and skills for its rising generation—but some considerable motivation to cloak or obscure those realities.

The interest of knowledge and skills for China's boys and girls would surely be served by a fresh new dose of "truth from facts". Whether the same can be said for the interest of the regime that rules them is another question altogether.

We proceed in five sections. The first—and largest—will analyze and assess available quantitative data bearing on scholastic achievement by pre-college pupils in China. The second will examine the phenomenon—should we perhaps say hallowed tradition—of test cheating in China. The third will address the issue of cognitive performance for China's children, and what available evidence suggests about national differentials and their bearing on China's youth "knowledge capital". The fourth will look at the ways in which demographic patterns and trends in family structure might influence opportunities for and constraints against scholastic achievement for China's pre-college population. The final section will

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[7] Even though India has more primary- and secondary-age boys and girls than China—and has since roughly the beginning of the 21st Century—levels of general academic achievement in India are so low that their pool of high-performers must be smaller than China's. Surprisingly enough, to go by estimates of Hanushek and colleagues, the USA may actually have a *larger* number of pupils with "basic skills" than India—even though India has roughly six times as many children under the age of 18. Cf. Sarah Gust, Eric Hanushek, Ludger Woessmann, "Global universal basic skills: Current deficits and implications for world development", *Journal of Development Economics*, Volume 166, January 2024, 103205, https://www.sciencedirect.com/science/article/abs/pii/S030438782300161X?via%3Dihub

offer concluding observations, questions, speculations, and suggestions for next steps.

1. **Test Scores of Pupil Achievement in China: A Statistical Analysis**

In this central portion of this paper, we present results of our exploration of quantitative indicators bearing on knowledge and skills for boys and girls in China today. This section includes our own exhaustive statistical analysis of four separate of datasets that compile and score achievement for children from China. Three of the four are represented as randomly sampled surveys with standardized achievement measures; one of these three reports results for an internationally standardized achievement test. The fourth offers complete all-China data on the college admission test taken by all aspiring collegiates in China—a “sample” of millions.

We also look to see the possibilities for “linking” these datasets, examining statistical “bridges” to connect the domestic China data to international data, to give a better perspective on the showing of pupils in China against a global canvas.

We also gather analyses that add to the understanding of China’s performance in an international perspective. These include the PRC government’s own NAEQ (Ministry of Education’s National Assessment of Educational Quality); the World Bank’s Human Capital Index (HCI) project and its Harmonized Learning Outcomes (HLO) database; and the new work by Stanford Professor Eric Hanushek, progenitor of the “knowledge capital” concept, and his colleagues on what may be learned from PISA and other corresponding international datasets concerning patterns of “universal basic skills” revealed from such testing.

We know the academic testing of just reading, math and science skills overlooks other skills essential to personal and national success: entrepreneurship; leadership; ability to cooperate; financial competence; “grit” and motivation, and more. We are also mindful of the limitations of standardized achievement tests per se—and of the complications of extending standardized achievement tests internationally, across countries and cultures.

Yes—but we have to work with the data that exist, not the data we wish existed. And while existing standardized achievement tests perforce ignore numerous factors that arguably relate to long term personal accomplishment, they do in fact measure critical knowledge and skills—which is why they are so widely accepted and used in education, and why social scientists have been able to demonstrate their predictive powers for both individual and national economic performance.

1-A) PISA[8]

Internationally standardized testing of student academic achievement have relied on three main exams for a generation. The first is the OECD's aforementioned PISA. The two others, conducted under the aegis of the International Education Association (IEA), are PIRLS (the Progress in International Reading Literacy Study) for reading comprehension and TIMSS (Trends in International Math and Science Study) for math and science.

In numbers of countries participating and percentage of world population represented, PISA coverage is greater than coverage for either PIRLS or TIMSS. In part this is because of the convenience of testing all three major subject areas with a single exam rather than two separate exams. In part it is because PISA claims to be the superior exam for evaluating students' ability to apply their knowledge, rather than to report what they have learned—and educators around the world seem to be agreeing. Although PIRLS and TIMSS are still widely used, PISA seems to be on the path to becoming the "gold standard" for international evaluation of pupil achievement. (PISA also collects socioeconomic information on its examinees and data on their teachers, making it an attractive and comprehensive data source for educational policymakers in countries all around the world to refer to in evaluating their own programs.)

The triennial PISA exam evaluates students 15 years of age from dozens of countries on a range of subjects, including reading, math, science, problem-solving, and financial literacy. We will be discussing only results for the three main PISA subjects: reading, mathematics, and science.

As for the nature of the examination process itself: all of these exams are sit-down tests that include a mix of multiple-choice- and constructed-response-style questions. Over time, PISA, PIRLS, and TIMSS have each gradually transitioned from paper- to computer-based assessments, PISA doing so in 2015. Each examination takes around two hours to complete, with the actual evaluation time for the examinees set about 2 hours for PISA. None of these tests contain fewer than 20 questions per subject area; the total number questions to be answered by the testees in PISA is roughly 100.

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[8]This section of the paper refers to Slides 16-40 in the Online Appendix.

The first PISA exam was administered in 2000, and the most recent exam was administered in 2022. In total, there have been 8 such waves conducted over the last quarter century. We will not discuss the PISA 2022 scores here since “China” did not report results.

Slides 16 and 17

International Standardized Student Achievement Exams: An Overview of Datasets And Pre-Covid Era Availability

Exam	Number of Participating Samples (Countries) Covered	Years Covered	Subjects Tested	Age Group(s)	Test Score Ranges	Number of Questions	Sample Size
PISA Programme for International Student Assessment	2000: 43 (41) 2003: 41 (39) 2006: 57 (53) 2009: 75 (58) 2012: 68 (62) 2015: 76 (72) 2018: 79 (78)	2000 to 2018, every 3 years	Reading, Mathematics, Science	15 years old	~ 0 – 1000	~35 questions per subject	2000: Over 250,000 2003: Over 250,000 2006: 400,000 2009: 470,000 2012: 510,000 2015: 540,000 2018: 710,000
PIRLS Progress in International Reading Literacy Study	2001: 37 (32) 2006: 47 (37) 2011: 58 (43) 2016: 50 (47)	2001 to 2016, every 5 years	Reading	Grade 4	~ 0 – 1000	2 passages, about 30 questions	2001: 150,000 2006: 210,000 2011: 334,446 2016: 346,852
TIMSS Trends in International Mathematics and Science Study	1995: 54 (38) 1999: 55 (36) 2003: 52 (43) 2007: 67 (57) 2011: 78 (57) 2015: 64 (51)	1995 to 2015, every 4 years	Mathematics, Science	Grades 4 and 8	~ 0 - 1000	25 questions per subject	1995: N/A 1999: Nearly 500,000 2003: Over 360,000 2007: 433,785 2011: 608,641 2015: 604,950

3

Total Population Represented by IEA and OECD Exams (incl. China) pre-Covid			
Exam	Total Countries	Total Pop. (in millions)	Pop. As Share of World Pop.
PISA (OECD)	82	2,923.9	37.6%
PIAAC (OECD)	28	1,055.3	13.5%
PIRLS (IEA)	60	1,896.2	24.3%
TIMSS (IEA)	77	2,474.5	31.7%
At Least One Test	**97**	**3,099.6**	**39.8%**
Total Population Represented by IEA and OECD Exams (excl. China)			
Exam	Total Countries	Total Pop. (in millions)	Pop. As Share of World Pop.
PISA (OECD)	81	2,743.9	35.2%
PIAAC (OECD)	28	1,055.3	13.5%
PIRLS (IEA)	60	1,896.2	24.3%
TIMSS (IEA)	77	2,474.5	31.7%
At Least One Test	**96**	**2,916.0**	**37.4%**

Note: Out of the exams listed above, China only participated in PISA in 2012, 2015 and 2018. Specifically, the administrative regions of Beijing, Shanghai, Jiangsu, and Zhejiang (B-S-J-Z) were represented in the 2018 PISA exam. Therefore, we employ a total population size for B-S-J-Z (China) equal to the total population of the four regions, or roughly 183,549.9 million in 2018.
Source: National Center for Education Statistics, U.S. Department of Education, International Data Explorer, https://nces.ed.gov/surveys/international/ide/, (accessed January 31, 2020); and United Nations, Department of Economic and Social Affairs, Population Division (2019). World Population Prospects 2019, Online Edition. Rev. 1, https://population.un.org/wpp/, (accessed March 20, 2020).

2

In the 2018 PISA wave, 78 national populations participated.[9] The scoring scales adopted by each of the PISA subject exams follows a scheme similar to TIMSS and PIRLS, allowing for standardized comparisons of performance across time and space. The OECD describes the details of its scoring system below:

In each test subject, there is theoretically no minimum or maximum score in PISA; rather, the results are scaled to fit approximately normal distributions, with means for OECD countries around 500 score points and standard deviations around 100 score points. About two-thirds of students across OECD countries score between 400 and 600 points.[10]

In order to maintain intertemporal comparability, PISA maintains a constant performance scale based on student performance in the very first wave of each subject test (2000 for reading, 2003 for mathematics, and 2006 for science).

1-A-i) What the Official Results Show About Scores for Pupils from China

In 2019 the world's leading authority on the testing and evaluation of educational performance all around the world, the Programme for International Student Achievement (PISA)[11], announced stunning results from its latest round of global examinations of 15 year olds: China was number one, all across the board. So striking to these educators was China's recorded academic aptitude that PISA began its entire 5 volume report with the following paragraph:

> *…our PISA 2018 assessment shows that 15-year-old students in the four provinces/municipalities of China that participated in the study – Beijing, Shanghai, Jiangsu and Zhejiang – outperformed by a large margin their peers from all of the other 78 participating education systems, in mathematics and science. Moreover, the 10% most disadvantaged students*

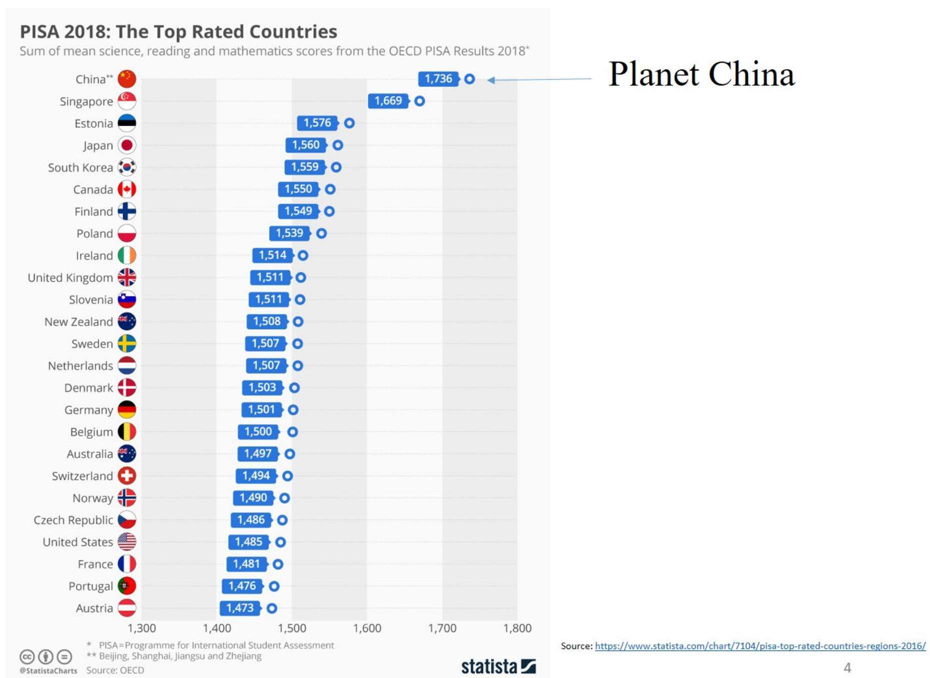
[9] Note, however, that participating countries do not always administer *all* the tests that the examining authority prepares for a given year. Thus for PISA 2018 a total of 78 countries participated in the math and science exams, while 77 countries took the reading exams. Analogous discrepancies are evident in some years for TIMSS and PIAAC, too.

[10] PISA FAQ – Results: "What do the test scores mean?" http://www.oecd.org/pisa/pisafaq/#:~:text=PISA%20ranks%20participating%20countries%20and,be%20used%20to%20rank%20countries.

[11] "What is PISA," The Organisation for Economic Co-operation and Development, https://www.oecd.org/pisa/.

in these four jurisdictions also showed better reading skills than those of the <u>average </u>student in OECD countries, as well as skills similar to the 10% most advantaged students in some of these countries. True, these four provinces/municipalities in eastern China are far from representing China as a whole, but the size of each of them compares to that of a typical OECD country, and their combined populations amount to over 180 million. What makes their achievement even more remarkable is that the level of income of these four Chinese regions is well below the OECD average. <u>The quality of their schools today will feed into the strength of their economies tomorrow</u>.[12](Emphasis added)

Slide 18



In math and science, China's test-takers outscored the average student from Western countries (i.e., members of the OECD—the Organization for Economic Cooperation and Development[13]) by over 100 points—the equivalent, in effect, of well over two additional years of schooling. (In reading, the Chinese student's

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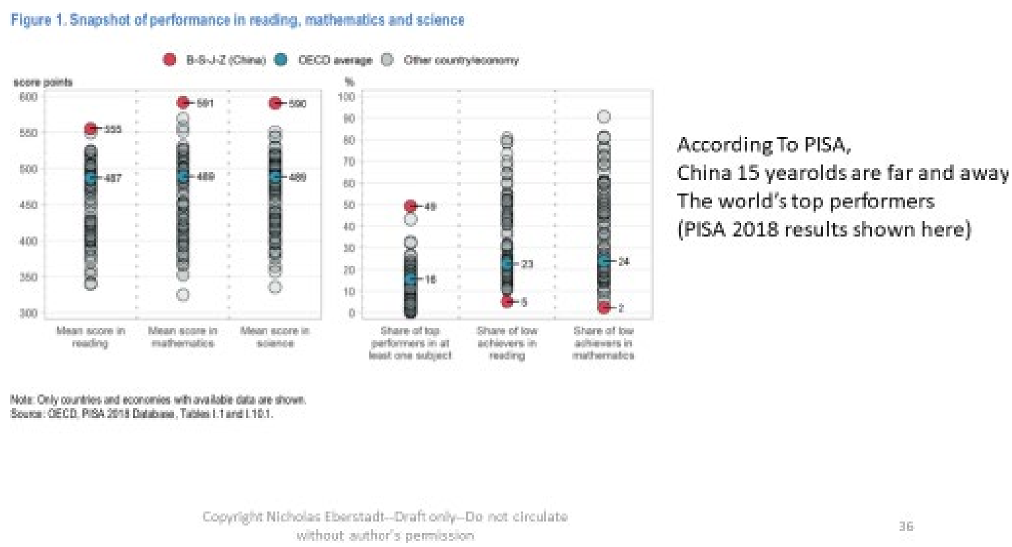
[12] OECD, "Preface", in *PISA 2018 Results (Volume II): Where All Students Can Succeed*, 2020, p. 3, https://www.oecd-ilibrary.org/docserver/b5fd1b8f-en.pdf?expires=1642118019&id=id&accname=guest&checksum=09B08AFB55A336C068668F43AE0AF6EE.

[13] OECD, "The OECD," 2008, https://web-archive.oecd.org/2012-06-15/154289-34011915.pdf

edge over their Western counterparts was "only" about 70 points, or a little less than two years of class-time equivalent.)

Slide 19



1-A-ii) Impressive? Implausible? Impossible? What to Make of PISA 2018 Results for "China"

China's PISA 2018 results were indeed extraordinary—and they have been growing more extraordinary round by round of PISA. What we cannot know, unfortunately, is *why* they have grown ever more extraordinary. But the extensive and exhaustive statistical analysis that we have conducted on the PISA results for tested pupils from China provides some illumination.

The charts from our examination of China's PISA results for 2015 and 2018 can be seen in the Online Appendix, Slides 20-204. They are voluminous and may be perused by the reader at leisure. We refer to only a small number of them below, in the interest of space.

A summary of published PISA results to date for tests conducted in China may be seen below. For simplicity's sake, we have averaged the mean results for the three tests—reading, math and science—into a single overall mean score for the year;

despite the simplification, this overall mean represents performance in the achievement test quite serviceably.

Slide 20

Mean Combined PISA Scores Over Time In China

PISA Test Year*	Province/s Tested	Mean Overall Pisa Score	Percent PRC Population (As of 2022)
2009	Shanghai	577	1.7%
2012	Shanghai	588	1.7%
2015	Beijing-Shanghai-Jiangsu-Guangdong (B-S-J-G)	514	18.1%
2018	Beijing-Shanghai-Jiangsu-Zhejiang (B-S-J-Z)	579	13.8%

* B-S-J-Z participated in PISA 2022, but tests were not conducted due to Covid-19 pandemic school lockdowns

As we see, PISA and Chinese authorities have experimented over time with a number of configurations for representing "China". We will return to that point in a moment. For now, we may note that the "China" students tested from four select provinces in 2018 earned an overall mean score higher than the one recorded just for Shanghai, the country's most developed province, nine years earlier—while covering an overall population roughly eight times larger than Shanghai's.

We may further note that between the 2015 and 2018 rounds of PISA testing, "China" students reported an overall leap in mean scores of 65 points—that is to say, roughly two thirds of a standard deviation. To put that number in perspective—it is approximately the same as the mean PISA 2018 gap in reading scores separating the Netherlands from Mexico.

As we delve into the PISA 2018 scores for China, we should keep four criteria in mind in our evaluations: 1) transparency; 2) replicability; 3) credibility; 4) generalizability.

To begin: outside researchers cannot gain access to the province-level data for "China" from either PISA 2015 or PISA 2018. Neither the PRC nor PISA permit outsiders to work with the raw data or even to see the overall mean test scores except on an aggregated basis with all four provinces combined (although the roster of provinces differs from 2015 to 2018). It is thus impossible to compare performance for Shanghai 2015 with Shanghai 2018 or Beijing 2015 with Beijing 2018—or even to guess what their mean performance levels in those rounds might have been. And of course it is impossible to chart any longer term performance for Shanghai 2009-18, even though that municipality was tested in each wave.

Neither Beijing nor PISA have explained the reasoning in altering the roster of test taking provinces between 2015 and 2018. It is clear, nonetheless, that the intention was *not* to increase national coverage of PRC students. By swapping out Guangdong, the country's most populous province, for Zhejiang, a province with roughly half its population, Beijing and PISA actually *reduced* implied coverage—substantially. Between 2015 and 2018, the share of China's population in PISA-tested provinces dropped by about a quarter. And by substituting Zhejiang for Guangdong, Beijing and PISA managed to include four out of the five most developed provinces of China as the testing "baseline" for "China" for PISA 2018.[14] Thus, PISA 2018 opted for a smaller and less representative set of Chinese provinces for 2018 than the already limited and unrepresentative one it had been working with in 2015.

Although OECD notes that PISA 2018 covered a swath of China accounting for more than 180 million persons, in fact the PISA sample for China that year involved just 361 schools and twelve thousand students—about 33 students per school selected. We do not know the process by which those schools were selected, much less the identities of the schools themselves, or the protocols observed in conducting (and preparing for) the tests. Suffice it to say there would appear to be plenty of scope in this mysterious process for non-random results. We deal with some of these possible ramifications in our section on "test cheating" in this paper.

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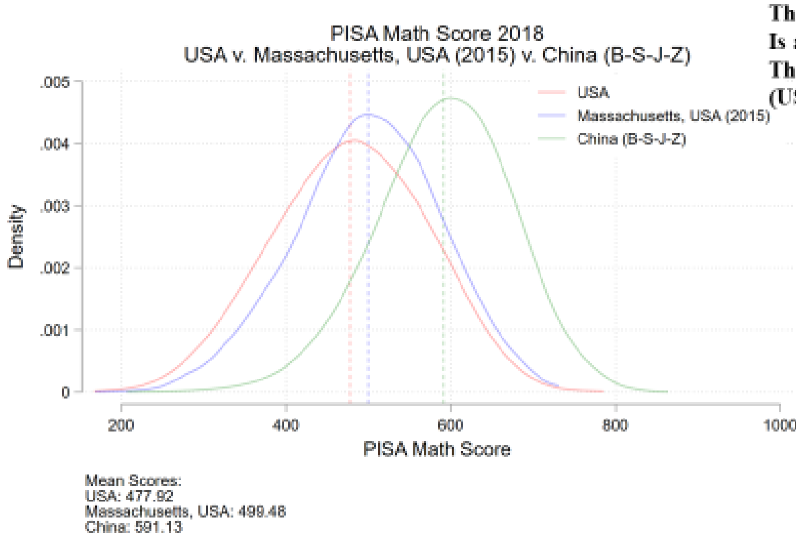
[14] Global Data Lab, "Subnational HDI (v7.0)", China 2018, https://globaldatalab.org/shdi/table/shdi/CHN/?levels=1+4&years=2018&interpolation=0&extrapolation=0

Although we highlight the anomalies and curiosities in the “China” PISA 2018 data in much greater detail in the accompanying tables in the Online Appendix, there are several aspects worthy of attention in this text.

First: the comparison between PISA’s “China” and the US state of Massachusetts.

Slide 21

According to OECD’s PISA Tests,
The typical 15 year old in 4 China Provinces
Is almost a standard deviation better at math
Than the typical Massachusetts student
(USA’s top test taking state)

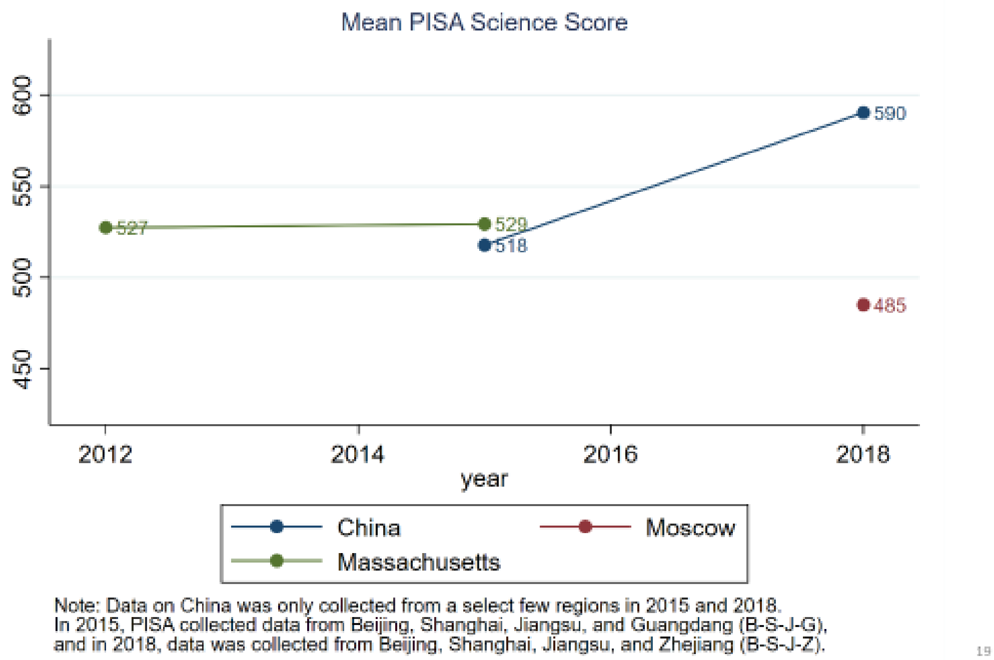


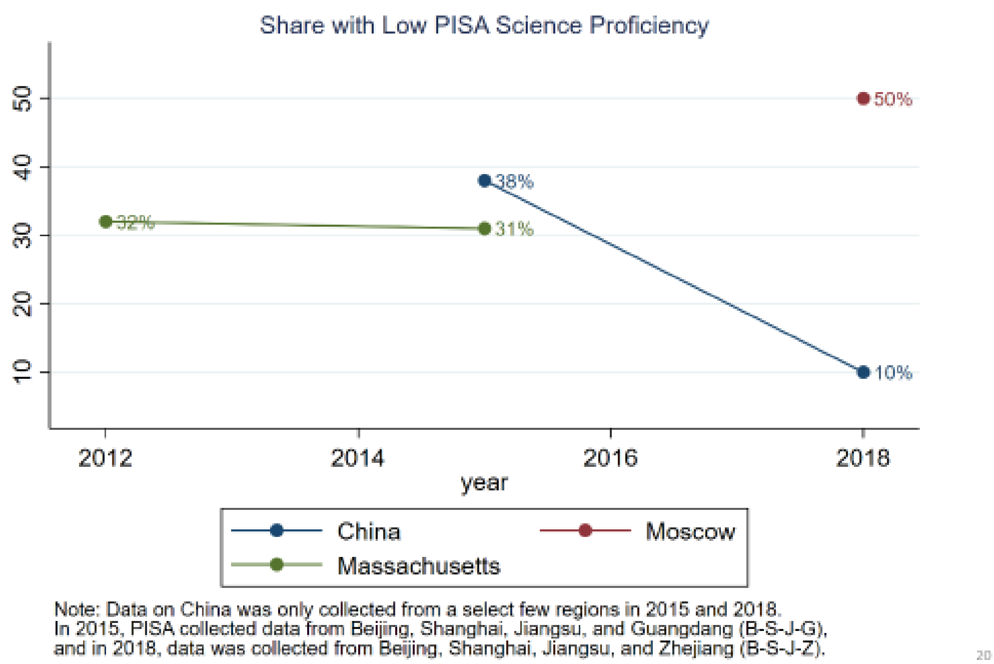
Jurisdiction	% Above 600
USA	9.6%
Massachusetts, USA	11.6%
China (B-S-J-Z)	47.8%

12

It is intriguing. In 2015, Massachusetts scores consistently exceed “China” in reading and science, with higher shares of high performers and lower shares of low performers too. By 2018, however, a redefined “China” vaults ahead of Massachusetts across the board. Whereas Massachusetts’ scores do not change dramatically over the short time span under consideration, “China’s” surge. For example: between 2015 and 2018 “China’s” share of low performing students plummeted from 43% to 19% in reading; from 32% to 9% in math; and from 38% to 10% in science.

Slides 22 and 23



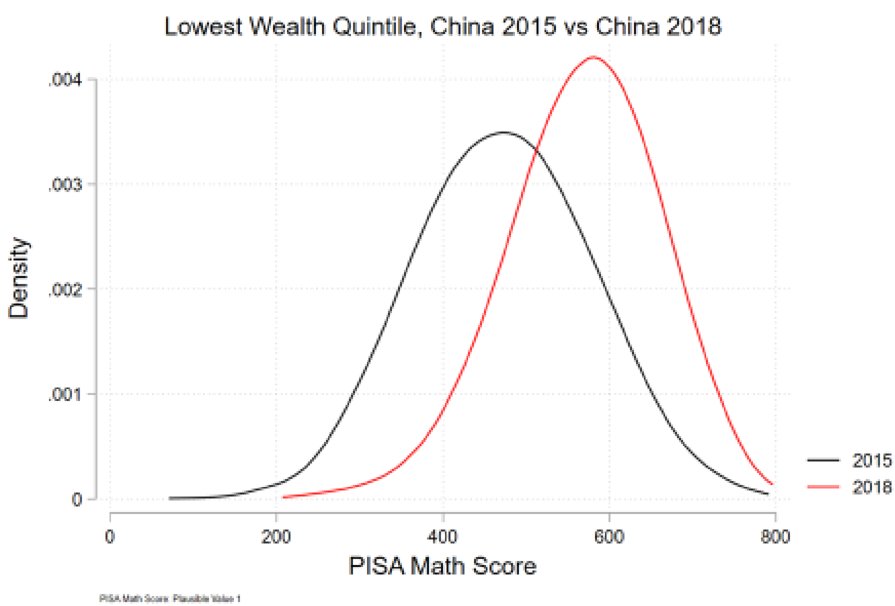


Note that, unlike student results in "China", results for Massachusetts—and the Unites States, for that matter—are generally fairly stable from one PISA round to the next. This is also true for most of the other countries tested by PISA; while some fluctuations do occur (some of these even qualifying as trends of improvement or retrogression), nothing like these gyrations are recorded elsewhere.

Then there is the matter of performance for socioeconomic subcomponents of population. PISA allows examination (at the overall “national” level for “China”) of testing differences by socioeconomic strata and household wealth level. What occurred between 2015 and 2018 was most remarkable.

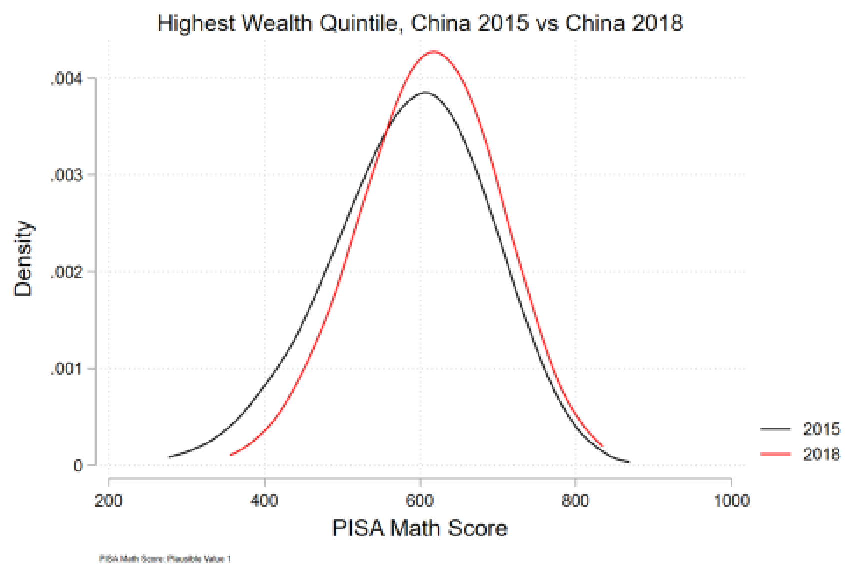
Between 2015 and 2018—just three years—the performance of students from “China’s” lowest wealth quintile surged:

Slide 24



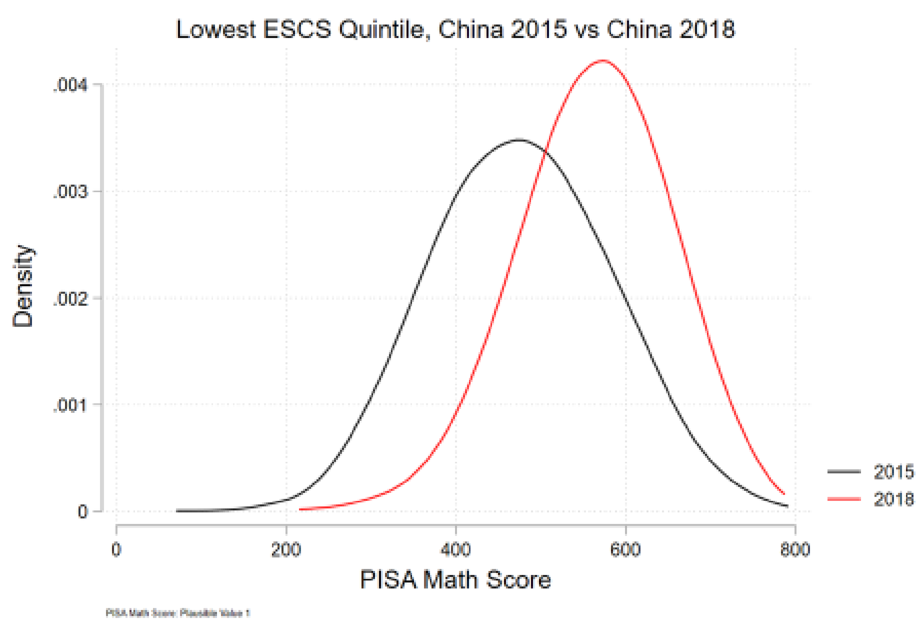
By 2018, indeed, there was virtually no difference between test results for the poorest and the richest of these test-takers from "China":

Slide 25



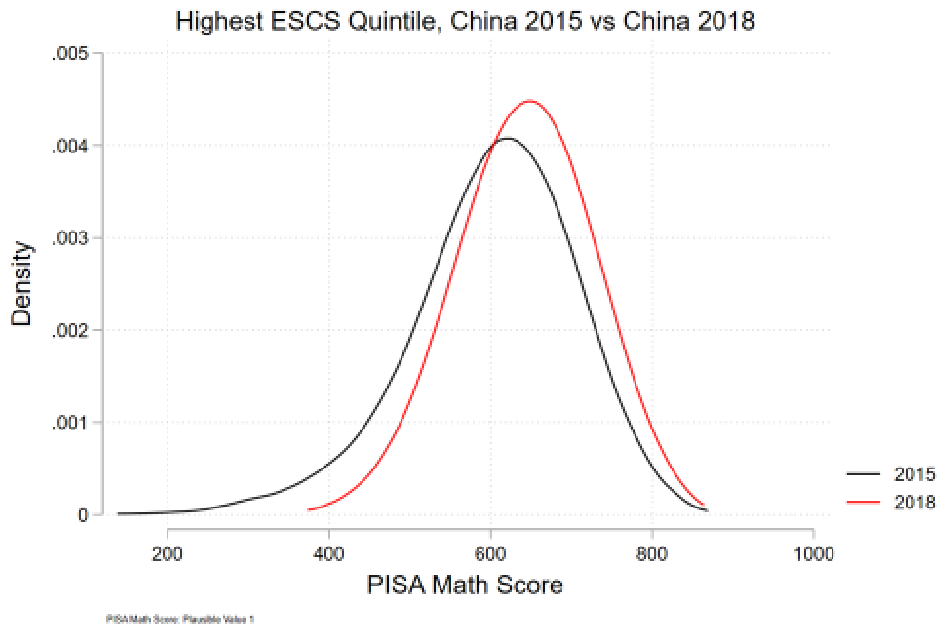
The same was true for socioeconomic status—witness the leap for the least advantaged in reported "China" data:

Slide 26



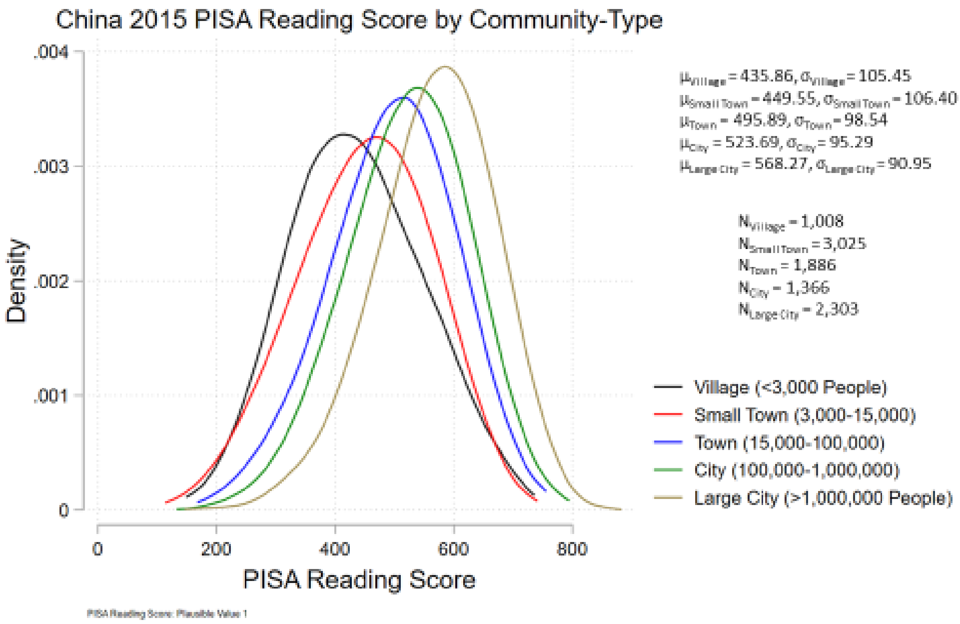
Here too, the least advantaged has come to perform like the most advantaged—in just three years!

Slide 27



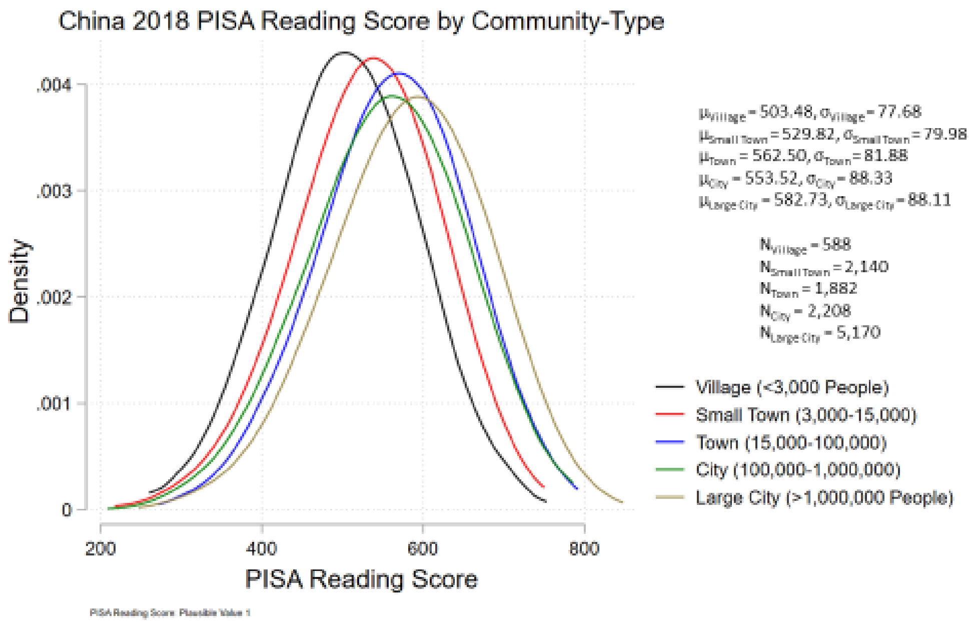
In 2015, villagers—rural students—were manifestly disadvantage in test performance:

Slide 28



Not so much just three years later! If you believe the data: mean scores for reading jumped by almost 70 points in villages of less than 3000 people, and by almost 80 points in places with populations of 3 thousand to 15 thousand—as against a much more modest 14 point rise in cities of over one million.

Slide 29



We compared these cheering patterns of sudden social equity in pupil achievement for "China" with evidence from a variety of other countries: not just Massachusetts and the US, but Brazil—another country, like China and America, with vast geographic expanses and some well discussed socioeconomic differentials; and Vietnam—a poor Asian country that continues to perform far above socioeconomic expectations in PISA (despite some questions concerning the authenticity of its reported results).[15] Nothing quite like "China's" implied social revolution between 2015 and 2018 showed up in those data, as may be seen from the accompanying charts in the Online Appendix.

What to make of all this? We are inclined to believe that the Confucian tradition places a premium on study and academic excellence. We are prepared to believe that overall academic achievement could, for this reason, appear more favorable for China than our own simple statistical analysis, based only on a country's socioeconomic indicators, might "predict". The example of Vietnam may be a case in support of such a predisposition.

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[15] Hai-Anh Dang, Paul Glewwe, Jongwook Lee, Khoa Vu, "What Explains Vietnam's Exceptional Performance in Education Relative to Other Countries? Analysis of the 2012, 2015, and 2018 PISA Data", *Economics of Education Review*, Volume 96, October 2023, 102434, https://www.sciencedirect.com/science/article/pii/S027277572300081X

But as we have shown here, the results for "China" strain credulity. It does not help that they are irreproducible and unverifiable—at least for those of us outside the innermost circles of Beijing NAEQ and OECD-PISA.

1-A-iii) PRC NAEQ

The PRC's National Assessment of Educational Quality[16] is the current State and Party answer to the question of "who tests the testers", a perennial issue in the long history of the imperial examination system. NAEQ "modernizes" those control and evaluation functions that long ago might have fallen under the purview of Imperial Inspectors or Censors.

For 1500 years before the Communist Revolution, China's testing system was designed to identify and rank the top 10 (or 100 or 1000) test-takers from across the empire: the state's interest was in seeking out and minting genuine Mandarins and Confucian literati to serve the Emperor.

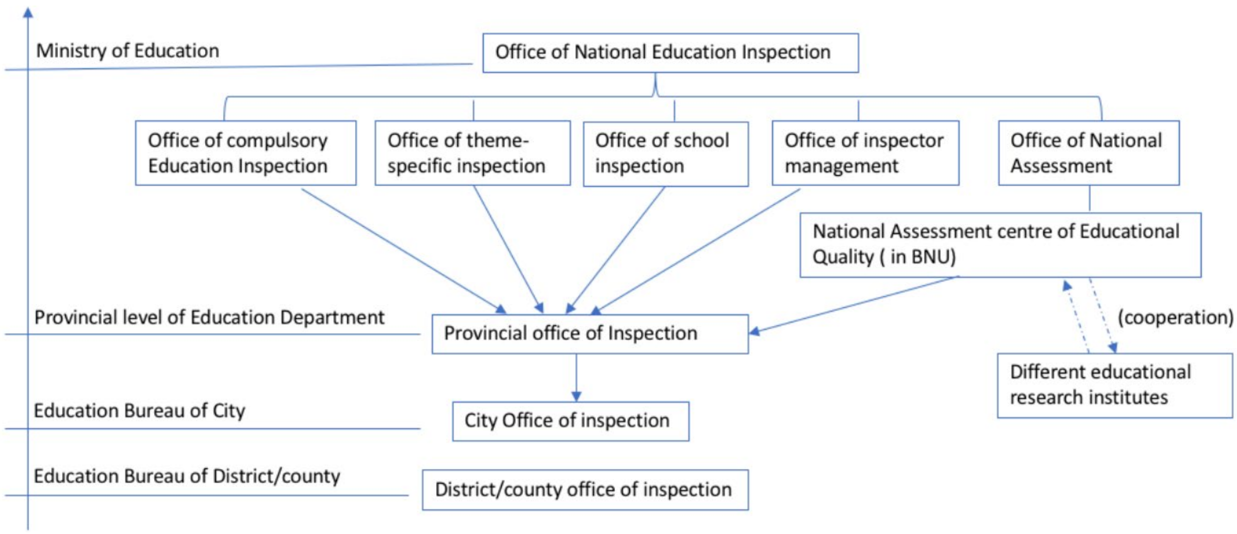
For a modern society with mass education, overall quality of the student body necessarily assumes higher priority for the Chinese government than specifically naming the person who earns the very highest scores. (To be sure: given its Confucian tradition, the prestige and status of being a top test-taker is hardly diminished for those who win this accolade in such mass tests as *gaokao*, the competitive college entrance exam for all PRC aspirants—just as top-test takers win life-long honor, regardless of their later accomplishments, throughout the Sino-sphere, Korea, and Japan.)

The NAEQ is part of the PRC's Ministry of Education, under the MOE's Office of National Education Inspection, per the organizational chart below.[17]

[16] For an overview, see Yu Jiang, Jiahui Zhang, and Tao Xin, "Toward Education Quality Improvement in China: A Brief Overview of the National Assessment of Education Quality", *Journal of Education and Behavioral Statistics*, Volume 44, Issue 6, https://doi.org/10.3102/1076998618809677

[17] Xingguo Zhou, "Changed and Unchanged: The Transformation of Educational Policies on Assessment and Evaluation in China," Doctoral Dissertation at University of Turku, 46, https://core.ac.uk/download/pdf/275894117.pdf.

Slide 30



NAEQ also is connected to the Beijing Normal University (BNU) Faculty of Education[18] as an "interdisciplinary center"—reportedly one of 17 such[19]. This special center, however, operates in *xitong* fashion at the university: as a special research organization or think tank drawing on non-government expertise for exclusively government purposes. Unless otherwise indicated, every study and research effort from NAEQ is confidential, for top levels of state and party alone to see, or as one recent study puts it, "only available to the highest levels of government".[20]

This *xitong* arrangement created a sort of one-way portal for the regime: in which it established an academic interface that would allow it to obtain information and cooperation from experts—most especially international academic experts—without compromising government secrecy through conventional academic mores of reciprocity and transparency.

[18] Faculty of Education, Beijing Normal University, "National Assessment of Education Quality (NAEQ)", http://fe.english.bnu.edu.cn/t003-c-1-73.htm

[19] Faculty of Education, Beijing Normal University, "Overview of Inter-disciplinary Centers" http://fe.english.bnu.edu.cn/t003-c-1-23-0.htm

[20] Yan, Y. (2022). Policy Instruments for Educational Governance: A Comparative Investigation of India and China. In: Lee, W.O., Brown, P., Goodwin, A.L., Green, A. (eds) International Handbook on Education Development in Asia-Pacific. Springer, Singapore. https://doi.org/10.1007/978-981-16-2327-1_98-1

NAEQ was established in 2007—a little more than a decade after global standardized achievement testing took off with PIRLS and TIMSS, and seven years after the launch of OECD's first wave of PISA.[21]

It was "established almost entirely from scratch" in the words of one paper on it[22]:

> Having closely followed international trends since the late 1970s, Chinese scholars started to raise concerns about the shortage of coherent and trustworthy data on the education system's performance. Against this backdrop,…the assessments conducted by the National Assessment of Education Quality (NAEQ), was born. An interviewee explained:
>
> *[The NAEQ assessments were established] because there is no data. Because the data in Chinese education is largely a mess and there are many discrepancies. So, we need data from a place that we can trust. We need to get first-hand data.*

NAEQ was charged with a very different sort of brief from any before in the long history of educational testing and evaluation in China; naturally "testing and evaluation with Chinese characteristics":

> Part of this includes moving beyond the practice of only using enrolment rates in the evaluation of schools to using NAEQ results as part of school inspection (MoE, 2015b). In other countries this would be seen as an element of a high-stakes practice, but in China it is seen as a way to provide feedback for improvement. However, whether that goal can be met is unclear, given that NAEQ outcomes are not reported at the school level….[23]
>
> …As China has a long history of ranking the quality of education with examination scores or other administrative methods, this new model of testing, along with the popularisation of student-centred pedagogy, is

[21] William Smith and Xiao Xu, "Building public trust in national assessment: The example of China's NAEQ" *European Journal of Education* 2023; 58:23-25 DOI: 10.1111/ejed.12539 https://onlinelibrary.wiley.com/doi/pdf/10.1111/ejed.12539

[22] Nelli Piattoeva, Vera Gorodski Centeno, Olli Suominen, and Risto Rinne, "Governance by data circulation? The production, availability, and use of national large-scale assessment data" in *Politics of Quality in Education: A Comparative Study of Brazil, China, and Russia*, Routledge, 2018, https://library.oapen.org/bitstream/handle/20.500.12657/30180/1/648937.pdf

[23] Yan, Y. (2022). Policy Instruments for Educational Governance: A Comparative Investigation of India and China. In: Lee, W.O., Brown, P., Goodwin, A.L., Green, A. (eds) International Handbook on Education Development in Asia-Pacific. Springer, Singapore. https://doi.org/10.1007/978-981-16-2327-1_98-1

> expected to use quality evaluation to reduce ranking and ruthless competition in schools.
>
> The report is delivered by the NAEQ's national centre to avoid different provinces or localities competing with each other, and it aims to provide comprehensive information to policymakers so they can learn the real level of educational development and problems in schools. The performance report is sample-based, and since 2015, all Chinese provinces and municipalities have received an annual assessment….[24]

"Building trust"[25] was a large part of NAEQ's brief—trust with the existing MOE educational apparatus; trust within China in the new approach to evaluating educational quality; trust with international testing authorities, PISA principal among them—and trust with all of these parties despite maintaining almost complete secrecy and control over internal information and its uses at the NAEQ for Beijing's central state and party.

NAEQ reportedly settled into the convention of testing quality of education (via student achievement scores) at the 4th and 8th grade level—just as PIRLS and TIMMS had been doing in the years immediately before its existence.

Testing of 4th and 8th grade students in select counties and schools in China now reportedly occurs regularly[26]--possibly at other grades too, although just how much is not public information.

Note as well that at the end of 4th grade, NAEQ also administers a National Large Scale Assessment (NLSA) to many hundreds of thousands of pupils across China, in a test that may possibly approximate national representativeness.[27] These data are not readily available but have on occasion been shared with outside

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[24] Galina Gurova, Helena Candido, and Xingguo Zhou, "Effects of quality assurance and evaluation on schools' room for action" in *Politics of Quality in Education: A Comparative Study of Brazil, China, and Russia*, Routledge, 2018, https://library.oapen.org/bitstream/handle/20.500.12657/30180/1/648937.pdf

[25] William Smith and Xiao Xu, "Building public trust in national assessment: The example of China's NAEQ" *European Journal of Education* 2023; 58:23-25 DOI: 10.1111/ejed.12539 https://onlinelibrary.wiley.com/doi/pdf/10.1111/ejed.12539

[26] Yajing Xue, "A Comparative Study of Compulsory Education Quality Assessment System between China and the United States", *Frontiers in Education Research*, ISSN 2522-6398 Vol. 2, Issue 9: 74-80, DOI: 10.25236/FER.2019.020912 https://francis-press.com/uploads/papers/2ojI4hal13vTX0AeJRsRRHWr7zqS97UdD258qHPP.pdf

[27] Nelli Piattoeva, Vera Gorodski Centeno, Olli Suominen, and Risto Rinne, "Governance by data circulation? The production, availability, and use of national large-scale assessment data" in *Politics of Quality in Education: A Comparative Study of Brazil, China, and Russia*, Routledge, 2018, https://library.oapen.org/bitstream/handle/20.500.12657/30180/1/648937.pdf

authorities—they are, for example, the basis for the World Bank's "learning poverty" comparisons between China and other countries, about which more in the following pages.

And NAEQ quickly began cooperating with PISA to coordinate an exploratory and initial limited testing regimen within China. In 2009, PISA was allowed to test pupils under NAEQ supervision: in Shanghai, where results were shared internationally and officially published by PISA; reportedly for 11 other provinces too[28], though this information cannot be confirmed outside the confines of PISA and NAEQ. In the 2012 wave of PISA, NAEQ and PISA reportedly collaborated on internationally standardized PISA testing in a dozen provinces again. But only the results for Shanghai were divulged—to this day even the names of the other provinces remain an official secret, one that PISA has agreed to keep as well.

As already mentioned above, by 2015 NAEQ consented to allow PISA testing in four provinces to be conducted, with results published: Beijing, Shanghai, Jiangsu and Guangdong ("B-S-J-G"); in 2018, for reasons never publicly explained, the roster dropped Guangdong and added Zhejiang in its place ("B-S-J-Z").

We go on at some length about NAEQ here precisely because so little is actually known by outsiders about the organization, its activities, the information it collects, or how it evaluates and uses such data.

Hearsay has it that counterparts on the PRC side "managed" their relationship with PISA to do their best to ensure that top provinces, schools within provinces, and classes within schools would be the "random sample" tested.[29] The NAEQ's leverage here was, as often happens with overseas transactions, the allure of the "enormous China market". PISA was establishing itself as the leading international instrument for standardized student achievement; including China, if possible, was an important objective to this end. (PISA likewise explored collaboration with India on an exploratory two-province basis.) Leveraged cooperation may explain why PISA has been so complacent, and quiet, about the undue level of opacity concerning PISA "China" test results. (Well-regarded independent experts on international education, such as then-Brookings Institute scholar Tom Loveless, were openly outraged about what they saw as PISA's unseemly acquiescence in

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[28] Valerie Strauss, "China is No. 1 on PISA – but here's why its test scores are hard to believe", The Washington Post, December 4, 2019 https://www.washingtonpost.com/education/2019/12/04/china-is-no-pisa-heres-why-its-test-scores-are-hard-believe/
[29] Private conversation, Zoom, January 2024.

improper demands by Beijing to slant the test in China, and wrote scathing critiques of PISA's China tests for 2009 and 2012.[30]) Or there may be other explanations that have little or nothing to do with "China franchise" considerations. There is no way for outsiders to know at this point.

One of the few tidbits that NAEQ has dangled, however, is the chart below—shared at a UNESCO meeting in a presentation by the Deputy Director of NAEQ in his capacity as BNU Professor.[31]

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Source: Tao Xin, "A Glance of National Assessment of Education Quality in China", 2017, 监测数据采集 (unesco.org)

To go by this presentation: PISA testing in 2012 took place under NAEQ auspices in 10 provinces in China; NAEQ's apparently never-yet-disclosed-domestic testing instrument correlates very closely in its results with PISA; and this 10-province-

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[30] Tom Loveless, "Lessons from the PISA-Shanghai Controversy", 2014 Brown Center Report on American Education, March 18, 2014, https://www.brookings.edu/articles/lessons-from-the-pisa-shanghai-controversy/ ; Tom Loveless, "PISA's China Problem", Brookings Institution, October 9, 2013, https://www.brookings.edu/articles/pisas-china-problem/ ; Tom Loveless, "Attention OECD-PISA: Your Silence on China is Wrong", Brookings Institution, December 11, 2013, https://www.brookings.edu/articles/attention-oecd-pisa-your-silence-on-china-is-wrong/; Tom Loveless, "PISA's China Problem Continues: A Response to Schleicher, Zhang, and Tucker", The Brookings Institution, January 8, 2014, https://www.brookings.edu/articles/pisas-china-problem-continues-a-response-to-schleicher-zhang-and-tucker/

[31] Tao Xin, "A Glance of National Assessment of Education Quality in China," May 2017, 31, https://tcg.uis.unesco.org/wp-content/uploads/sites/4/2018/08/Session-8a_China.pdf.

China for NAEQ-PISA testing would have ranked 10^{th} in PISA 2012 international ranking.

To go by those revelations: "10 province China" would have been testing just below the level of Belgium (country number nine in the ranking that year) or parallel to Germany (country number 10).

If we assume NAEQ was testing the provinces it knew would look most favorable, this would seem to suggest, at a minimum, that a fair share of China pupils were testing at a high European standard a little over a decade ago.

How large a share? One can make guesses by looking at provincial population totals. Suffice it say there are 31 provinces in China—three of which are giant municipalities (Beijing, Shanghai, Tianjin), all likely candidates. It would not be unreasonable to imagine that a third of China's population (per the 10 out of 31 provinces) was being tested in that "trial run". Possibly the share could have been somewhat higher. We will not know for sure unless and until either NAEQ or PISA enlighten us.

In December 2019, upon the official OECD launch of its full PISA 2018 study, the PRC Ministry of Education sent out a "victory lap" press release, reviewing "China's" astonishing performance.[32] Not only did B-S-J-Z students take top marks in reading, math, and science, the MOE noted, but educational equity was very high; teachers were doing a top performance job, and "China" pupils were more interested in reading than counterparts from any country.

The release did caution that the provinces sampled for the PISA results were "more developed than other regions across the country in terms of economic and educational development" but implied the reason B-S-J-Z were selected was not to skew China's test scores, but rather logistic, since these were areas "with high IT application in education where students can complete computer-based tests".

To go by the release, educational policymakers at Beijing's highest levels sounded highly pleased by the PISA results and satisfied by their authenticity. This begs the question of exactly what NAEQ chieftains think they know about the academic performance of pupils in China; what they tell their superiors at MOE about it; and just what their superiors inform the leaders of the State Council and the CCP about

[32] Ministry of Education, the People's Republic of China, Press Release: "Students in B-S-J-Z (China) rank first in PISA 2018 survey", http://en.moe.gov.cn/news/press_releases/201912/t20191210_411536.html

the state of knowledge and skills for China's rising youth cohort. This is a point to which we will return in our concluding observations.

1-A-iv) PISA and the World Bank Research on Student Achievement in China in International Perspective: The Human Capital Index (HCI), Harmonized Learning Outcomes (HLO), and "Learning Poverty"

The World Bank (formally—the International Bank for Reconstruction and Development or IBRD) has the largest professional staff working on economic and social development policy research issues of any organization in the world. "The Bank" and its affiliated employees and consultants publish more peer-reviewed studies on development than any university, NGO, international institution, or government, and it also produces an immense amount of technical papers and studies bearing on its mandate "*to reduce poverty by lending money to the governments of its poorer members to improve their economies and to improve the standard of living of their people*."[33]

The Bank has sponsored research relating to "knowledge capital" in China, knowledge and skills for China's pupils, and the reliability of PISA's "China" soundings as a measure of both these quantities. Unlike OECD's PISA efforts, the Bank's research on these issues does not seem to be conditioned by the need to cooperate with, or seek approval from, Chinese authorities; furthermore, the Bank's studies on these questions are "open source", with data transparency the rule, and accessibility of information routine for outside scholars. Bank research on these matters, in other words, appears to be genuinely independent of Chinese authorities—free from both direct PRC pressure and from indirect pressure, i.e. the need to consider ramifications of research in light of possible consequences for relations with representatives of Beijing.

Three overlapping Bank research efforts merit discussion here: the Human Capital Index (HCI); the Harmonized Learning Outcome (HLO) database; and the "Learning Poverty" Database.

[33] The World Bank, "Getting to Know the World Bank", July 26, 2012, https://www.worldbank.org/en/news/feature/2012/07/26/getting_to_know_theworldbank#:~:text=The%20World%20Bank%20is%20an,of%20living%20of%20their%20people.

All three efforts originated in 2017[34], with the launch the Human Capital Project[35], a new Bank initiative, and fall within the purview of that program.

Human Capital Index:

The HCI was first published in 2018 and subsequently updated in 2020[36]. It is a synthetic measure, intended to offer a single "overview" number for a country's human capital endowment for its entire population. The possible HCI score for a country ranges between 0.00 and 1.00.

The HCI's three conceptual components are survival to school age; education, and health. HCI uses the following statistical indicators to cover those components: 1) probability of surviving to age 5; 2) expected years of school; 3) learning-adjusted years of school; 4) Average Harmonized Test Scores—i.e., the Harmonized Learning Outcomes (HLO) Database, about which more below; 5) adult survival rates for ages 15-60; and 6) fraction of children not stunted.

In the 2020 HCI report, the Bank researchers calculated HCI for 174 countries (or as the report actually states, "economies": the term used to finesse the political status of Taiwan; the West Bank and Gaza; etc.). The HCI includes PRC among these.

The HCI's score for China was 0.65. The country with the highest HCI was Singapore, at 0.88; Central African Republic was lowest, at 0.29.

To place China's HCI in international perspective: the HCI divides the rankings for the 174 countries it covers into six subdivisions; if we call the top performers the first of these six groupings, then China falls in the third. That grouping includes 29 countries, ranging in scores from countries such as Peru and Mexico (0.61) to countries such as Vietnam and Greece (0.69). Russia, Thailand, Malaysia, and (improbably) Luxembourg are also among the countries in this grouping.

There are three other countries assigned an HCI score of 0.65, in addition to China: these are Bahrain, Chile—and, interestingly enough, Turkey.

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[34] World Bank Live, "Jim Yong Kim at Columbia University: Building New Foundations of Human Solidarity", October 5, 2017, https://live.worldbank.org/en/event/2017/human-capital-project

[35] The World Bank Human Capital Project, "A Project for the World" https://www.worldbank.org/en/publication/human-capital

[36] World Bank. 2021. The Human Capital Index 2020 Update: Human Capital in the Time of COVID-19. Washington, DC: World Bank. doi:10.1596/978-1-4648-1552-2. https://documents1.worldbank.org/curated/en/456901600111156873/pdf/The-Human-Capital-Index-2020-Update-Human-Capital-in-the-Time-of-COVID-19.pdf

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Table 1.2: The Human Capital Index (HCI), 2020

Economy	Lower Bound	Value	Upper Bound	Economy	Lower Bound	Value	Upper Bound	Economy	Lower Bound	Value	Upper Bound
Central African Republic	0.26	0.29	0.32	India	0.49	0.49	0.50	Mauritius	0.60	0.62	0.64
Chad	0.28	0.30	0.32	Egypt, Arab Rep.	0.48	0.49	0.51	Uzbekistan	0.60	0.62	0.64
South Sudan	0.27	0.31	0.33	Guyana	0.48	0.50	0.51	Brunei Darussalam	0.62	0.63	0.63
Niger	0.29	0.32	0.33	Panama	0.49	0.50	0.51	Kazakhstan	0.62	0.63	0.63
Mali	0.31	0.32	0.33	Dominican Republic	0.49	0.50	0.52	Costa Rica	0.62	0.63	0.64
Liberia	0.30	0.32	0.33	Morocco	0.49	0.50	0.51	Ukraine	0.62	0.63	0.64
Nigeria	0.33	0.36	0.38	Tajikistan	0.48	0.50	0.53	Seychelles	0.61	0.63	0.66
Mozambique	0.34	0.36	0.38	Nepal	0.49	0.50	0.52	Montenegro	0.62	0.63	0.64
Angola	0.33	0.36	0.39	Micronesia, Fed. Sts.	0.47	0.51	0.53	Albania	0.62	0.63	0.64
Sierra Leone	0.35	0.36	0.38	Nicaragua	0.50	0.51	0.52	Qatar	0.63	0.64	0.64
Congo, Dem. Rep.	0.34	0.37	0.38	Nauru	0.49	0.51	0.53	Turkey	0.64	0.65	0.66
Guinea	0.35	0.37	0.39	Fiji	0.50	0.51	0.52	Chile	0.64	0.65	0.66
Eswatini	0.35	0.37	0.39	Lebanon	0.50	0.52	0.53	Bahrain	0.64	0.65	0.66
Yemen, Rep.	0.35	0.37	0.39	Philippines	0.50	0.52	0.53	China	0.64	0.65	0.67
Sudan	0.36	0.38	0.39	Tunisia	0.51	0.52	0.52	Slovak Republic	0.66	0.66	0.67
Rwanda	0.36	0.38	0.39	Paraguay	0.51	0.53	0.54	United Arab Emirates	0.66	0.67	0.68
Côte d'Ivoire	0.36	0.38	0.40	Tonga	0.51	0.53	0.55	Serbia	0.67	0.68	0.69
Mauritania	0.35	0.38	0.41	St. Vincent and the Grenadines	0.52	0.53	0.54	Russian Federation	0.67	0.68	0.69
Ethiopia	0.37	0.38	0.39	Algeria	0.53	0.53	0.54	Hungary	0.67	0.68	0.69
Burkina Faso	0.36	0.38	0.40	Jamaica	0.52	0.53	0.55	Luxembourg	0.68	0.69	0.69
Uganda	0.37	0.38	0.40	Indonesia	0.53	0.54	0.55	Vietnam	0.67	0.69	0.71
Burundi	0.36	0.39	0.41	Dominica	0.53	0.54	0.56	Greece	0.68	0.69	0.70
Tanzania	0.38	0.39	0.40	El Salvador	0.53	0.55	0.56	Belarus	0.69	0.70	0.71
Madagascar	0.37	0.39	0.41	Kenya	0.53	0.55	0.56	United States	0.69	0.70	0.71
Zambia	0.38	0.40	0.41	Samoa	0.54	0.55	0.56	Lithuania	0.70	0.71	0.72
Cameroon	0.38	0.40	0.42	Brazil	0.55	0.55	0.56	Latvia	0.69	0.71	0.72
Afghanistan	0.39	0.40	0.41	Jordan	0.54	0.55	0.56	Malta	0.70	0.71	0.72
Benin	0.38	0.40	0.42	North Macedonia	0.55	0.56	0.56	Croatia	0.70	0.71	0.72
Lesotho	0.38	0.40	0.42	Kuwait	0.55	0.56	0.57	Italy	0.72	0.73	0.74
Comoros	0.36	0.40	0.43	Grenada	0.55	0.57	0.58	Spain	0.72	0.73	0.73
Pakistan	0.39	0.41	0.42	Kosovo	0.56	0.57	0.57	Israel	0.72	0.73	0.74
Iraq	0.40	0.41	0.41	Georgia	0.56	0.57	0.58	Iceland	0.74	0.75	0.75
Malawi	0.40	0.41	0.43	Saudi Arabia	0.56	0.58	0.59	Austria	0.74	0.75	0.76
Botswana	0.39	0.41	0.43	Azerbaijan	0.56	0.58	0.59	Germany	0.74	0.75	0.76
Congo, Rep.	0.39	0.42	0.44	Armenia	0.57	0.58	0.59	Czech Republic	0.74	0.75	0.76
Solomon Islands	0.41	0.42	0.43	Bosnia and Herzegovina	0.57	0.58	0.59	Poland	0.74	0.75	0.76
Senegal	0.40	0.42	0.43	West Bank and Gaza	0.57	0.58	0.59	Denmark	0.75	0.76	0.76
Gambia, The	0.39	0.42	0.44	Moldova	0.57	0.58	0.59	Cyprus	0.75	0.76	0.76
Marshall Islands	0.40	0.42	0.44	Romania	0.57	0.58	0.60	Switzerland	0.75	0.76	0.77
South Africa	0.41	0.43	0.44	St. Kitts and Nevis	0.57	0.59	0.60	Belgium	0.75	0.76	0.77
Papua New Guinea	0.41	0.43	0.44	Palau	0.57	0.59	0.61	France	0.75	0.76	0.77
Togo	0.41	0.43	0.45	Iran, Islamic Rep.	0.58	0.59	0.60	Portugal	0.76	0.77	0.78
Namibia	0.42	0.45	0.47	Ecuador	0.59	0.59	0.60	Australia	0.76	0.77	0.78
Haiti	0.43	0.45	0.46	Antigua and Barbuda	0.58	0.60	0.61	Norway	0.76	0.77	0.78
Tuvalu	0.43	0.45	0.46	Kyrgyz Republic	0.59	0.60	0.61	Slovenia	0.77	0.77	0.78
Ghana	0.44	0.45	0.46	Sri Lanka	0.59	0.60	0.60	New Zealand	0.77	0.78	0.78
Timor-Leste	0.43	0.45	0.47	Uruguay	0.59	0.60	0.61	Estonia	0.77	0.78	0.79
Vanuatu	0.44	0.45	0.47	Argentina	0.59	0.60	0.61	United Kingdom	0.77	0.78	0.79
Lao PDR	0.44	0.46	0.47	St. Lucia	0.59	0.60	0.62	Netherlands	0.78	0.79	0.80
Gabon	0.43	0.46	0.48	Trinidad and Tobago	0.57	0.60	0.62	Ireland	0.78	0.79	0.80
Guatemala	0.45	0.46	0.47	Colombia	0.59	0.60	0.62	Sweden	0.79	0.80	0.81
Bangladesh	0.46	0.46	0.47	Peru	0.59	0.61	0.62	Macao SAR, China	0.79	0.80	0.80
Zimbabwe	0.44	0.47	0.49	Oman	0.60	0.61	0.62	Finland	0.79	0.80	0.80
Bhutan	0.45	0.48	0.50	Thailand	0.60	0.61	0.62	Canada	0.79	0.80	0.81
Myanmar	0.46	0.48	0.49	Malaysia	0.60	0.61	0.62	Korea, Rep.	0.79	0.80	0.81
Honduras	0.47	0.48	0.49	Mexico	0.60	0.61	0.62	Japan	0.80	0.80	0.81
Cambodia	0.47	0.49	0.51	Bulgaria	0.60	0.61	0.62	Hong Kong SAR, China	0.80	0.81	0.82
Kiribati	0.46	0.49	0.52	Mongolia	0.60	0.61	0.63	Singapore	0.87	0.88	0.89

HCI < 0.40   0.40 ≤ HCI < 0.50   0.50 ≤ HCI < 0.60   0.60 ≤ HCI < 0.70   0.70 ≤ HCI < 0.80   0.80 ≤ HCI

Source: World Bank calculations based on the 2020 update of the Human Capital Index (HCI).

Note: The Human Capital Index (HCI) ranges between 0 and 1. The index is measured in terms of the productivity of the next generation of workers relative to the benchmark of complete education and full health. An economy in which a child born today can expect to achieve complete education and full health will score a value of 1 on the index. Lower and upper bounds indicate the range of uncertainty around the value of the HCI for each economy.

Source: The Human Capital Index 2020 Update: Human Capital in the Time of COVID-19 (worldbank.org)

Harmonized Learning Outcomes (HLO):

World Bank researchers developed an aggregate database that combines and concords international standardized pupil achievement test results from the three major international tests (PIRLS, PISA, TIMSS); from regional applications of some of those same tests not conducted under the aegis of the respective testing authorities; and from sub-territorial (local level) tests in India and China. These several thousand national and sub-national observations, covering (at least in principle) well over 95% of the world's population, were then brought into correspondence across datasets with "conversion factors" intended to "harmonize" these data into a single, apples-to-apples, mega-set for global student achievement. The product is known as the "Harmonized Learning Outcomes" database. It is available online through the World Bank's websites[37]. The project's methodology and database have also appeared in *Nature*[38]. Thus the work has passed peer review in one of the world's top science journals. Some experts nonetheless criticize the HLO's method for agglomerating these worldwide achievement scores. (Eric Hanushek is one of these dissenters; his solution to what he sees as the HLO's methodological difficulties is the alternative "Global Universal Basic Skills" approach described below).

The HLO team examined the PISA scores for China but found them unrepresentative and potentially misleading. They expanded the HLO dataset for China to include an important study by Stanford's Scott Rozelle and Chinese colleagues under the aegis of Stanford's REAP program.[39] In it they conducted the PIRLS reading exam for rural grade school students in three provinces: Guizhou, Jiangxi and Shaanxi. Jiangxi and Guizhou are among China's poorer provinces, with Guizhou ranked as lowest by HDI, excepting only Tibet.[40]

The REAP study concluded that these rural China students were testing at the very bottom of the PIRLS international roster. If their performance could be said to represent rural China, then rural China was performing behind such laggards as Indonesia and Morocco. On a stand-alone basis, more developed Shaanxi fared slightly better, ranking above Indonesia with comparators such as Colombia and Qatar—but Jiangxi and Guizhou ranked behind Morocco, the country with the lowest PIRLS score, and the Guizhou rural students came in dead last.

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[37] Cf. Harmonized Learning Outcomes (HLO) Database, The World Bank, https://datacatalog.worldbank.org/search/dataset/0038001

[38] Noam Angrist, Simeon Djankov, Pinelopi Goldberg and Harry Patrinos, "Measuring Human Capital Using Global Learning Data," *Nature* 592**,** 403–408, 2021, https://www.nature.com/articles/s41586-021-03323-7

[39] Qiufeng Gao, Yaojiang Shi, Hongmei Yi, Cody Abbey, and Scott Rozelle, "Reading Achievement in China's Rural Primary Schools: A Study of Three Provinces", Rural Education Action Program Working Paper 325, November 2017, https://fsi9-prod.s3.us-west-1.amazonaws.com/s3fs-public/325_-_reading_achievement_across_different_regions_in_rural_china_final.pdf

[40] Global Data Lab "Subnational HDI (v7.0)", China 2021 https://globaldatalab.org/shdi/table/shdi/CHN/?levels=1+4&years=2021&interpolation=0&extrapolation=0

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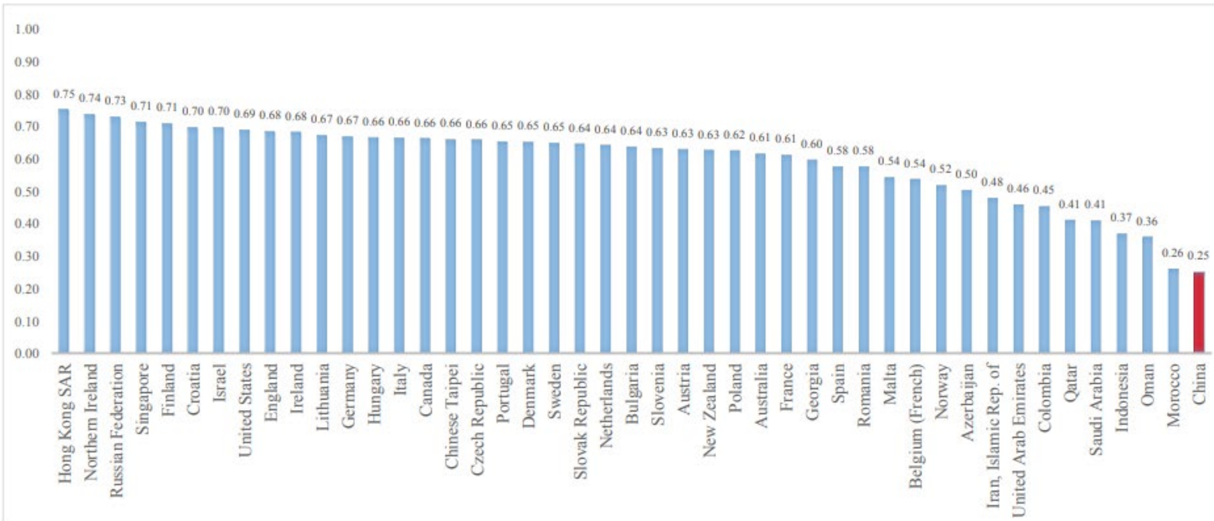

Figure 1: Reading test scores of students in our full rural Chinese sample compared to those of students from other countries/regions

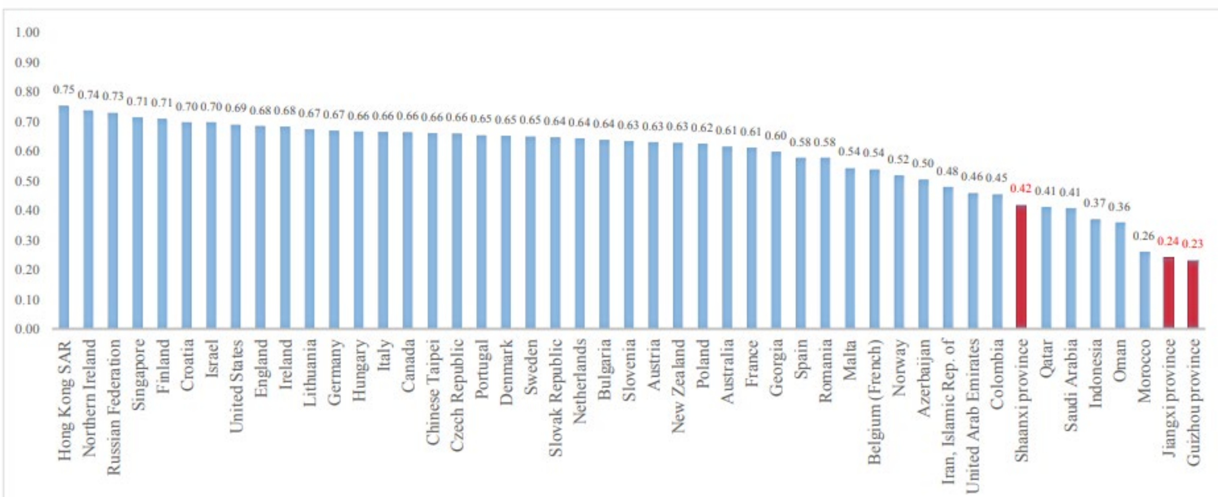

Figure 2: Reading test scores of students in our subsamples from Shaanxi, Guizhou and Jiangxi provinces compared to those of students from other countries/regions

Utilizing their HLO "conversion factors" and socioeconomic data on China's provinces, the World Bank HLO team re-estimated an all-China mean level of academic achievement for pre-college boys and girls. The results of their calculations are shown below:

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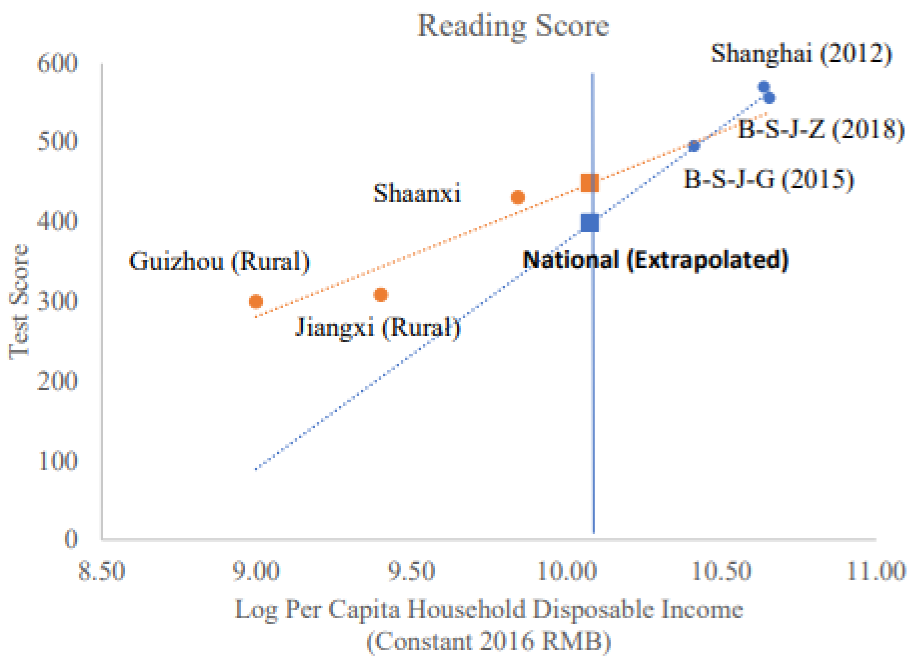
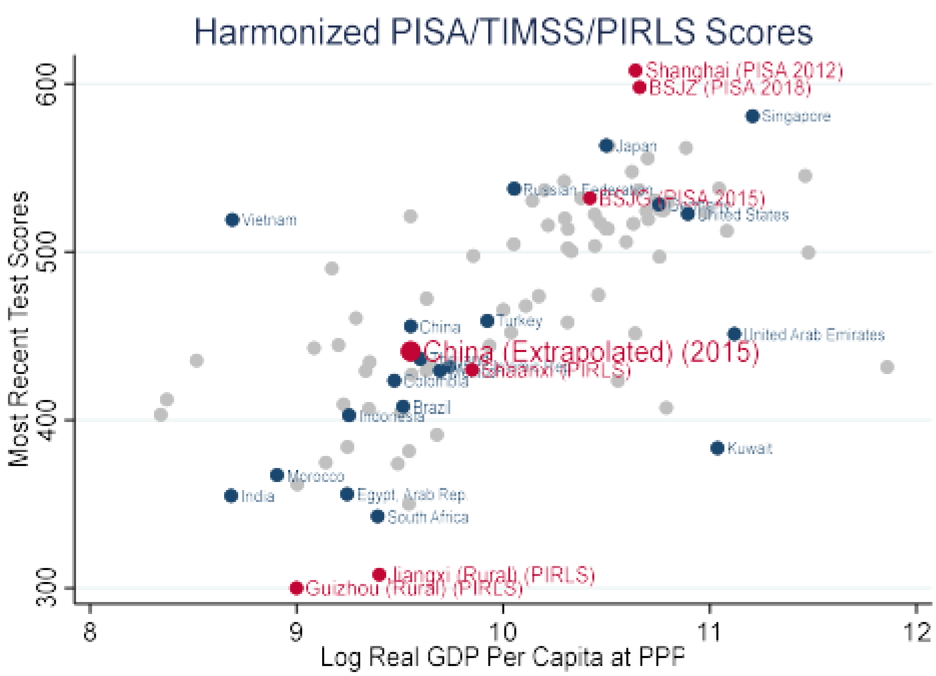
Figure 1: Comparing PISA and PIRLS Extrapolated Reading Scores



Note: Orange dots represent PIRLS, blue dots represent PISA.

Figure 2: China's Extrapolated National HLO In International Perspective



Ironically, the 2018 PISA results for China, which were not available to the World Bank team for their initial 2018 HCI estimates for China, led them to *revise lower* their all-China HLO figure, due to the higher implied slope connecting the performance of more developed regions to those of less developed regions within China:

> *In summary, extrapolating PISA scores based on income differences across provinces suggests a national level HLO of 432, while doing the same using PIRLS scores suggests a national-level HLO of 449. For the 2020 HCI, the HTS is calculated as an average of these two figures, 441, as the best estimate of the national-level HLO for China, with the values of 432 and 449 as lower and upper bounds….*
>
> *This updated HTS estimate of 441 is slightly lower than the HTS estimate of 456 for China reported in Patrinos and Angrist (2018) that was used in the 2018 HCI, and was based on similar extrapolations. This value should not be interpreted as a decline over time in learning as measured by the HTS. Rather, this is an updated estimate of the overall level of the HTS based on the limited available internationally-comparable learning data for China based on the additional information available in the 2018 PISA assessment.*[41]

What would a score of 441 in the HLO database signify? Very roughly: it would situate China in between Mexico and Turkey (although other countries could also be used for the comparison).

Learning Poverty:

"Learning poverty" is a new statistical construct, intended to bring attention to the lack of basic knowledge and skills early in life for boys and girls. It is an estimate of the proportion of 10-year-olds who cannot read a text (in practice, often a single sentence).[42] The index is called "learning poverty" because being unable to read critically constrains other school-based learning.

Estimates of the prevalence of "learning poverty" combine the fraction of 4th grade students (or 10 year olds) who cannot manage basic reading with the proportion

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[41] Noam Angrist, Syedah Aroob Iqbal, and Aart Kraay, "Updating China's Harmonized Test Scores for the Human Capital Index", September 10, 2020, https://thedocs.worldbank.org/en/doc/333161600277389753-0140022020/original/HarmonizedTestScoresforHCI2020China.pdf

[42] The World Bank Brief, "What is Learning Poverty?", April 28, 2021, https://www.worldbank.org/en/topic/education/brief/what-is-learning-poverty

who are not in school at the grade 4 level (all these are presumed to be unable to read a text) for a total estimate of the percentage of 10-year-olds who lack basic reading skills. Data on reading proficiency for 4th graders come from either standardized achievement tests (such as PIRLS) or from national large scale assessments (NLSAs).

The Bank has released two statistical assessments on the state of "learning poverty" around the world: the first in 2019[43]; the second in 2022[44].

Intriguingly, China participated in the "learning poverty" report—providing domestic NLSA data from PRC NAEQ not available through other auspices. It is impossible for outsiders to know just how comprehensive or representative the numbers from China's NLSA actually were. But they indicated a very different situation for knowledge and skills from the profile offered by the PISA 2018 report.

According to World Bank reports on "learning poverty", 18.2% of PRC children could not read a basic text in 2016[45]. That compares with 2.8% for Singapore in 2016—the country supposedly outperformed across the board in reading, math and science by B-S-J-Z, according to PISA 2018. For the US in 2016, the corresponding estimate was 4.9%.

The figure below places China's estimated learning poverty in international perspective. If World Bank estimates are accurate, China's "learning poverty" rate, at 18%, would be half as high as expected for a country of its income level, and about a third as high as India's (55%). But China's rate of "learning poverty" would also be far higher than rates for developed countries, and completely incompatible with PISA estimates of achievement in "China".

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[43] The World Bank, "Ending Learning Poverty: What will it take?" https://openknowledge.worldbank.org/server/api/core/bitstreams/1ef8a794-710b-584e-a540-a3923ad7ea90/content

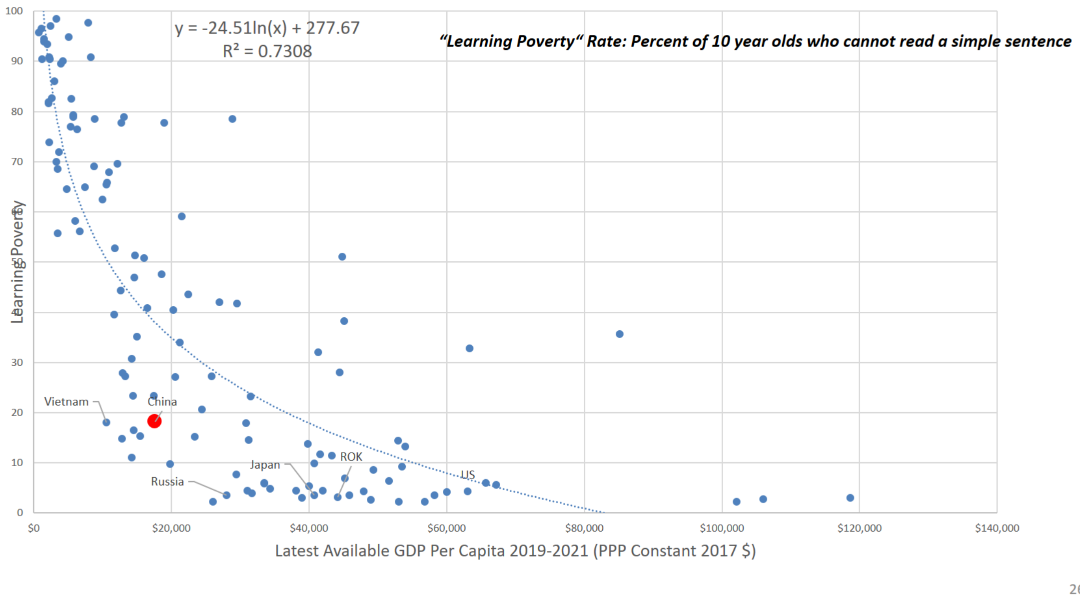
[44] The World Bank, "The State of Global Learning Poverty: 2022 Update", June 23, 2022, https://thedocs.worldbank.org/en/doc/e52f55322528903b27f1b7e61238e416-0200022022/original/Learning-poverty-report-2022-06-21-final-V7-0-conferenceEdition.pdf

[45] The World Bank, "The State of Global Learning Poverty: 2022 Update", June 23, 2022, Annex 5: Detailed 2019 country learning poverty data, p. 66 https://thedocs.worldbank.org/en/doc/e52f55322528903b27f1b7e61238e416-0200022022/original/Learning-poverty-report-2022-06-21-final-V7-0-conferenceEdition.pdf

Slide 36

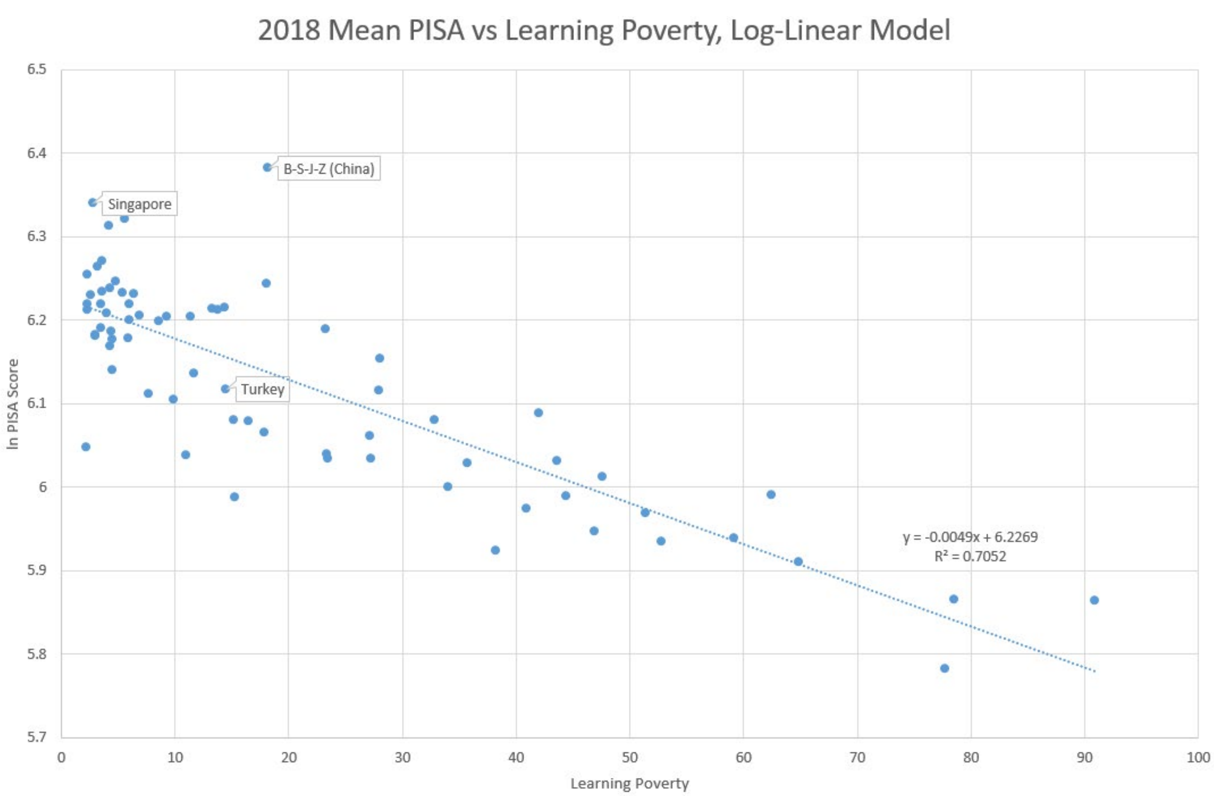
"Knowledge Capital and Student "Learning Poverty":

"Learning Poverty" vs. GDP per Capita, China and Other Countries (World Bank Estimates)



It is also informative to compare World Bank estimates for "learning poverty" with PISA 2018 mean academic achievement scores. We do this in the figure below.

Slide 37



We see several things in Slide 37, above. One of the most telling comparisons is for China and Turkey. According to the Bank, "learning poverty" is distinctly lower in Turkey[46]: under 15% (14.7%), as against China's 18%.[47] Once again, Turkey and China look to be in the same league—this time, with Turkey seemingly performing above China with this indicator of pupil "knowledge capital".

By the simple bivariate regression model in the chart here, a country with China's level of "learning poverty" would be predicted to have a mean national HLO score of 459—very close to what we estimated on the basis of international development patterns, and to what the HLO team estimated using in-country test scores for China that included scores from rural areas. And once again: this "predicted" score for China comes very close to the actually reported score for the country of Turkey.

Among the many questions that arise from this World Bank research is a bureaucratic one: why didn't the Bank's Executive Director from the PRC object to the Human Capital Project's highly unflattering portrait of human capital, student learning, and "learning poverty" in China—especially in light of the PRC MOE's self-congratulatory pronouncements after the release of PISA 2018? It is unlikely that the discrepancy escaped notice of PRC authorities as this research was being circulated internally at the Bank for pre-publication vetting and review. Yet according to researchers involved in these overlapping initiatives, they were aware of no communications from the Executive Director for China on their work, and never were placed in the position of having to defend this work under pressure from higher ups.

We do not have an answer for why this particular bureaucratic dog did not bark. We can think of many potential reasons. Of particular interest, however, is the speculation of the World Bank researcher in charge of managing this project. He hypothesized that the Executive Director for China recognized the downside of any potential interference of that sort—since, he mused, this would only bring more attention to the glaring contradiction between China's PISA claims and its (in his

[46] The World Bank, "Turkey Learning Poverty Brief", June 2022, https://documents1.worldbank.org/curated/en/099805407182217390/pdf/IDU00274a60c0e0ba04e890906f0b3c2ae973a0f.pdf
[47] The World Bank, "China Learning Poverty Brief", June 2022, https://documents1.worldbank.org/curated/en/099627407152234705/pdf/IDU0d1aca3ef03efb04a820a452005f0c51ff4ca.pdf

view) true on-the-ground educational realities—and wisely avoided the embarrassment any such interventions would have courted.[48]

1-A-v) Eric Hanushek and Colleagues on PISA and Student Achievement in China in International Perspective

In a series of recent papers, Stanford's Eric Hanushek and his German colleagues Sara Gust and Ludger Woessmann come up with an ingenious and important approach to making an "end run" around the thorny question of harmonizing standardized achievement testing from separate global databases, to arrive at an arguably better common metric for judging "knowledge capital" for pupils all around the world.[49]

Hanushek and others objected to the World Bank's HLO methodology for aggregating test scores from PISA, PIRLS and TIMSS on statistical grounds, arguing the approach could introduce biases and distortions into the overall HLO database.

Their "work around" for circumventing such potential problems is to *skip the scores* in these tests and *focus instead on the skill levels indicated*. These tests, they write, work with a common agreed conception from international educators for judging "skill levels"—a proficiency scale ranging from "basic" (Level 1) through advanced (Levels 5 and 6). They examine performance in math and science only, leaving out reading competence (and thus PIRLS results). They also use results on math and science tests conducted by regional authorities using TIMSS handbooks and tests, but not actually ratified by the IEA (which oversees TIMSS).

Gust, Hanushek and Woessmann therefore go straight to the question of "basic skills". They use data not only from PISA but from the other international standardized tests for pupil achievement to calculate the proportion of test takers who fail to meet the benchmarks for "basic skills" in these respective tests. Then they look at the proportion of students who do not complete basic pre-college

[48] Personal conversation, World Bank official, 2022.

[49] Sarah Gust, Eric Hanushek, Ludger Woessmann, "Global universal basic skills: Current deficits and implications for world development", *Journal of Development Economics*, Volume 166, January 2024, 103205, https://www.sciencedirect.com/science/article/abs/pii/S030438782300161X?via%3Dihub

schooling (presuming this additional group by definition lacks basic skills)[50]. The sum of these two quantities gives the number that Hanushek et al. arrive at for the proportion of a society's youth who lack basic skills. And since their aggregated collection of databases cover 98 percent of the world's population—with two very large asterisks in the cases of India and China, where only PISA tests of certain limited regions connect populations in question to the rest of the dataset—they present their findings as a profile of the current state of "Global Universal Basic Skills" (or GUBS) and draw global inferences about what this profile means for the future performance of the world economy.

Given the realities of the data with which Hanushek et al. must work for their GUBS, it is inevitable that some methodological and practical problems should remain. The aggregated datasets, for example, are testing children of different ages: age 15 for PISA versus grades 4 and 8 for TIMSS. This is mixing apples and oranges, so to speak, and in light of the possibility of different trajectories of skills development in different places for boys and girls over the 10-15 age period, it makes for some indeterminate distortions. Remember—"basic skills" will be rather different for a 4th grader, and 8th grader, and a 15 year-old. There is also the question of whether test-makers at OECD (who manage PISA) and the IEA (who run TIMSS) really do hold to the same standard in fashioning their respective metrics for the "basic skills" threshold—and the issue of whether the results from the regional tests conducted in the TIMSS format would actually have been ratified by the TIMSS authorities. Notwithstanding these irresolvable caveats, it is nevertheless clear that the GUBS approach is the best one yet devised for a full global assessment of knowledge and skills for pre-college populations all around the world.

In attempting to assess the true level of "basic skills" for pupils for the whole of China, Hanushek et al. confront the same dilemma that everyone else wrestling with the limits of the PISA testing for "China" has faced.

They take the PISA results for 2018 for B-S-J-Z on its face, accepting the scores as faithful and authentic representations of the capabilities of randomly selected students in those provinces.

[50] Setting this at 9th grade, the grade level in which PISA exams are held. This is notionally age 15. In many contemporary societies compulsory schooling ends at age 15 and young men and women may work (or are expected to work) at age 16.

We note that Hanushek and his coauthors arrive at their estimates of the proportion of students in China failing to achieve basic skills by converting two all-China panel surveys (CEPS and CFPS) onto a PISA-scale (a methodology we thoroughly exploit for slightly different purposes elsewhere in this paper) by using Shanghai as a "linking county" between the datasets. In particular, as detailed in our section on linking CEPS to PISA, they assume that CEPS Shanghai scores and PISA Shanghai scores are two different random samples drawn from an underlying "skill distribution". Making this assumption, along with the assumption that the underlying skills are normally distributed, allows them to place academic achievement scores from the all-China panel surveys onto a PISA scale.

As designated by the OECD, the cutoff between Levels 1 and 2 for basic skills is 420 points on the PISA math scale and 410 points on the PISA science scale[51]. By converting all-China panel surveys on student achievement onto a PISA scale, therefore, they are able to estimate the proportion of students achieving basic skills in China. The validity of this transformation requires the assumption that PISA scores are truly representative of Shanghai, and that the panel surveys are true representations of all-China academic achievement. We have will have more to say about these assumptions later, but suffice it to say that if they don't hold we can interpret Hanushek, et al.'s results as a lower bound on the estimated proportion of Chinese students failing to achieve basic skills.

Given the stratospheric scores reported for B-S-J-Z for PISA 2018, and the high levels of compulsory enrollment in China, they calculate that 18 percent of these tested youths lack basic skills.[52]

But then Hanushek et al. make an adjustment. They posit that the figure they arrived at only pertains to *urban* China—and that *rural* China, which they say contains 65% of all China's pre-college pupil population, remains untested.

How to come to an estimate for all-China? They take a "sensitivity analysis" approach— "what if" calculations based on bounding the possible estimate for rural China's performance with the top performance for rural pupils in a largely

[51] OECD (2019), PISA 2018 Results (Volume I): What Students Know and Can Do, PISA, OECD Publishing, Paris, https://doi.org/10.1787/5f07c754-en.

[52] These are revisions from an earlier version of their study, in which they calculated that just 6.5% of these B-S-J-Z pupils lacked basic skills—a stratospheric reading far more favorable than for any full national population examined. In that earlier paper, by way of comparison, the corresponding share without basic skills for Singapore was put at 8%, and 22% for Switzerland. The estimates for Singapore and Switzerland were not revised—nor were any others for national populations.

rural low-income East Asian country, and also at the other end by the poorest performance by such a country.

For the high bound they pick Vietnam (whose remarkable punch-above-weight performance we have already examined in detail). For the low bound, their country is Cambodia. For rural Vietnam, an estimated 19% of pupils lacked basic skills; for Cambodia, the corresponding figure was about 95%. Based on those parameters, Hanushek et al. calculate that all-China's true share of pupils lacking basic skills would fall somewhere between 19% and 69%, and may be seen in the table immediately below.

Slide 38

Table A5: Sensitivity of skill estimates: Alternative bounds on India and China

	Baseline	India		China	
		Based on Tamil Nadu	Based on Himachal Pradesh	Based on rural Vietnam	Based on rural Cambodia
	(1)	(2)	(3)	(4)	(5)
India	0.888	0.886	0.901		
China	0.180			0.190	0.689
World	0.672	0.672	0.675	0.676	0.741
By income group					
Lower-middle-income countries	0.858	0.858	0.864		
Upper-middle-income countries	0.423			0.472	0.638
By region					
South Asia	0.892	0.890	0.901		
East Asia & Pacific	0.354			0.360	0.633

Notes: Estimated share of children (incl. those currently out of school) who do not reach at least basic skill levels in math and science (equivalent to PISA Level 1). Col. 1: baseline results (see col. 3 of Table 2). Col. 2 and 3: assume India achieves at level of Tamil Nadu and Himachal Pradesh, respectively. Col. 4 and 5: assume that China baseline value applies only to urban children (35%) and that rural children (65%) achieve at the level of rural Vietnam and rural Cambodia, respectively.

Source: Gust, Hanushek, Woessmann (2023). *Global Universal Basic Skills: Current Deficits and Implications for World Development*, Journal of Development Economics (forthcoming) GUBS 231016.pdf (stanford.edu) Table A5

While methodologically elegant, the calculated range is in practical terms unsatisfyingly vast. Comparator countries with roughly a 19% share of pupils without basic skills by the Hanushek et al. reckoning would include Denmark and the Netherlands; at 69%, we would be comparing the All-China pupil population with Ecuador, Iraq, and Indonesia. Note that Hanushek et al. do not think their upper boundary for China is implausible.

Yet even with this achingly wide and seemingly imprecise range we can make a couple of meaningful comparisons.

First: by the GUBS studies, pupil skill levels, even at the most unfavorable lower bound estimate, would be vastly more favorable for China than for India.

Like China, India has only been tested by PISA in a few preliminary sub-territories: in the India case, the states of Tamil Nadu and Himachal Pradesh. Since Tamil Nadu is famous for its unusually low levels of illiteracy and high levels of health (despite mediocre income levels in the India context), it was selected as a test case for PISA presumably for the same reasons that Shanghai was chosen in China. Yet the tested results for Tamil Nadu were dreadful—89% of pupils failing to reach even the basic skills level.

The stunningly low performance of pupils in India raises a number of important questions—including how India can muster such an impressive policy, intellectual, and scientific-technical elite from such a seemingly disadvantaged pupil population.

For our purposes here, however, the point is that even at Hanushek et al.'s most "pessimistic" reading, the share of pupils possessing basic skills would be roughly three times higher in China than India—and over six times higher at the upper "optimistic" bounding.

By the "pessimistic" reading, China would have more pupils with basic skills than any other country on earth—twice as many as India, the runner up. It would also have a pool of boys and girls with basic skills of roughly the same size as all of Europe, including Russia—and a larger pool than EU-27. The pool would be 75 percent larger than the corresponding pool for the US. And all this is using the most "pessimistic" lower boundary—on any other set of assumptions, the pool of pupils with basic skills would be even larger for China: quite possibly much larger.

Further, by the most "pessimistic" reading, the prevalence of basic skills for pupils in China would look more or less similar to what Hanushek et al. calculate for two regions, Latin America and the Caribbean; and MENA (Middle East and North Africa). More optimistic readings correspondingly raise the implied levels of development for the regions to which China would be compared.

Slide 39

Table 2: Basic skill deficits on a global scale

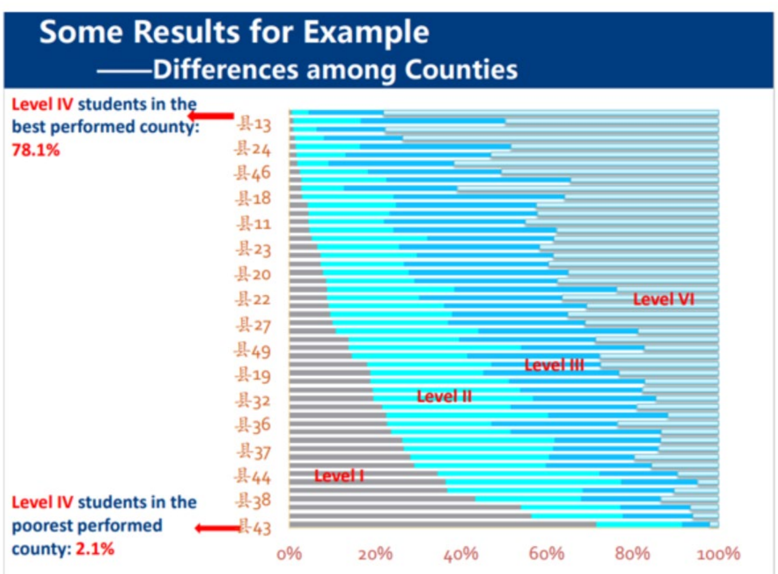
	Share of students below basic skills (1)	Share of children not enrolled in secondary school (2)	Share of all children below basic skills (3)
World	0.631	0.355	0.672
By income group			
Low-income countries	0.905	0.693	0.956
Lower-middle-income countries	0.813	0.440	0.858
Upper-middle-income countries	0.383	0.189	0.423
High-income countries	0.239	0.069	0.255
By region			
Sub-Saharan Africa	0.893	0.665	0.941
South Asia	0.850	0.402	0.892
Middle East & North Africa	0.639	0.195	0.679
Latin America & Caribbean	0.612	0.210	0.652
Central Asia	0.400	0.094	0.421
East Asia & Pacific	0.311	0.219	0.354
Europe	0.259	0.102	0.284
North America	0.222	0.069	0.239

Notes: Col. 1: Estimated share of current students who do not reach at least basic skill levels in math and science (equivalent to PISA Level 1). Col. 2: One minus net secondary enrollment rate (from WDI and own imputations). Col. 3: Estimated share of children (incl. those currently out of school) who do not reach at least basic skill levels in math and science. See section 3 for methodological details. Country groups follow World Bank classification.

Source: Gust, Hanushek, Woessmann (2023). *Global Universal Basic Skills: Current Deficits and Implications for World Development*, Journal of Development Economics (forthcoming) GUBS 231016.pdf (stanford.edu) Table 2

At this juncture it may be helpful to return to the PRC's NAEQ and its presumably very carefully released information concerning basic skills for the students it tested in different (of course un-named) counties in China.

Slide 40



Source: Tao Xin, "A Glance of National Assessment of Education Quality in China", 2017, 监测数据采集 (unesco.org)

NAEQ uses its own standards for categorizing scholastic achievement of PRC pupils. Theirs is a scale running from Level I to Level IV—with Level II qualifying as "basic skills" in the NAEQ taxonomy.[53]

It is perhaps informative in and of itself to know that PRC authorities set "basic skills" at level 2 on a scale running from 1 to 4, while PISA and others have a 1-6 scale in which basic skills are set as level 1.

Nevertheless, while we lack any basis for making direct comparisons between NAEQ and PISA, we may note that even in the data NAEQ was comfortable disclosing publicly, an absolutely tremendous range of student achievement profiles were apparent. In the disclosed PRC county data, the range of students failing to achieve NAEQ "basic skills" range from near-zero to over 70 percent!

We have no precise way to apply that information to the Hanushek et al. dataset on pupils lacking "basic skills", of course. But let us do a thought experiment.

What if we take a "split the difference" approach, and hypothesize that the all-China figure for pupils lacking basic skills might be the midpoint of the upper and lower boundary in the Hanushek et al. study? The NAEQ county-level skills figures could lend some qualitative support to exploring such a hypothesis.

If we take the mean of those upper and lower boundaries, we would come to a figure of just over 35% of all-China pupils without basic skills.

How would that look next to other countries and regions for GUBS?

That 35% figure would be worse than North America (24%) and Europe (28%), and high-income countries as a whole (26%). But it would be distinctly better than the 42% for upper-middle-income countries—the World Bank income grouping in which PRC falls.

By comparison with specific countries: 35% would be more favorable than for Ukraine (41%), or Serbia (41%), to say nothing of Mexico (56%) or fellow-BRIC member Brazil (66%).

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[53] Tao Xin, "A Glance of National Assessment of Education Quality in China", 2017, 监测数据采集 (unesco.org)

As coincidence would have it: 35% also happens to be the GUBS figure for Turkey's estimated share of youth without basic skills. We are drawn to this comparison—not least because our own previous statistical modeling, and the World Bank's HLO dataset, arrive at a Turkey-PRC rough equivalence in "knowledge capital" for the pre-college population through two additional and independent approaches.

1-B) The China Family Panel Studies (CFPS)[54]

The China Family Panel Studies (CFPS) is a national survey of Chinese individuals and families which began in 2010 under the auspices of Peking University's Institute of Social Science Survey. [55] Nearly 15,000 families and almost 30,000 persons over age 9 within these families were interviewed for that 2010 baseline.[56] The survey includes questions concerning respondents' economic situation, family, education, and communities. Importantly for our current concerns, the survey also includes achievement scores for math and reading. While the survey was purportedly repeated every two years from 2010 through 2020, only data for 2010 are available, so those are the only CFPS data which we used in this analysis.

Hainan, Qinghai, Inner Mongolia, Ningxia, Tibet and Xinjiang are not represented in the 2010 CFPS data. It bears mentioning that all but one of these provinces were in the bottom half of the Human Development Index rankings by province in 2010.[57] Even without these likely low performers, the math and reading scores in CFPS do exhibit a significant range of achievement. On a scale of 0 to 24 for the math test, the difference between the highest provincial mean (13.84 in Shanghai) and the lowest (9.01 in Guizhou) is over a fifth of the test's range. Stated slightly differently, Shanghai's mean math score is nearly a quarter higher than the national mean while Guizhou's is 20% below that national average (see Slide 206). Again, these are province-level *means*, not worst and best cases within the provinces.

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[54] This section of the paper refers to Slides 206-217 in the Online Appendix.
[55] "Introduction," China Family Panel Studies, https://www.isss.pku.edu.cn/cfps/en/about/introduction/index.htm.
[56] *Ibid.*
[57] Global Data Lab, Subnational HDI (v7.0), https://globaldatalab.org/shdi/table/shdi/CHN/?levels=1+4&interpolation=0&extrapolation=0.

Shanghai and Guizhou are at their respective extremes, but the divergence is by no means small in other provinces: Anhui's average math score is almost a fifth better than the national mean, while Guangxi and Fujian averaged more than 10% below it. The spread of reading scores (on a scale of 0 to 34) is similar: Shanghai's and Tianjin's mean scores sit roughly a fifth ahead of the national average (with Anhui and Beijing close behind) while Guizhou's mean is less than 80% of the national average.

Slide 206

Province	Avg. Math Score	Math Avg/ National Avg.	Avg. Reading Score	Reading Avg./ National Avg.	Adjustment Factor
Anhui	13.25	1.178	25.54	1.171	1.175
Beijing	12.54	1.115	25.20	1.156	1.136
Chongqing	12.20	1.084	24.67	1.132	1.108
Fujian	10.09	0.898	20.36	0.934	0.916
Gansu	11.18	0.994	22.01	1.009	1.002
Guangdong	11.05	0.982	19.67	0.902	0.942
Guangxi	9.91	0.881	20.58	0.944	0.913
Guizhou	9.01	0.801	17.26	0.792	0.796
Hebei	10.88	0.967	22.07	1.012	0.990
Heilongjiang	12.53	1.114	23.68	1.086	1.100
Henan	11.06	0.983	22.40	1.027	1.005
Hubei	12.39	1.102	23.97	1.099	1.101
Hunan	11.96	1.063	23.69	1.086	1.075
Jiangsu	11.72	1.042	23.99	1.100	1.071
Jiangxi	11.20	0.996	21.70	0.996	0.996
Jilin	12.01	1.068	25.07	1.150	1.109
Liaoning	12.33	1.096	24.11	1.106	1.101
Shaanxi	13.14	1.168	24.36	1.117	1.143
Shandong	11.99	1.066	24.04	1.102	1.084
Shanghai	13.84	1.231	26.35	1.208	1.220
Shanxi	12.07	1.073	22.94	1.052	1.063
Sichuan	10.97	0.976	20.18	0.926	0.951
Tianjin	12.73	1.132	25.83	1.185	1.158
Yunnan	10.45	0.929	20.33	0.933	0.931
Zhejiang	13.02	1.157	24.51	1.124	1.141

As important as the absolute magnitude of these national spreads is the variance in CFPS achievement scores relative to the variance of *gaokao* and PISA scores. The structures of these other two exams – *gaokao's* national homogenization of scores (albeit an incomplete homogenization of scores with different subjects) and PISA's selective slice of the Chinese population – both seem configured to present a tighter range of achievement among China's pupils than commonsense suggests is true. A simple statistic, the coefficient of variation, offers a means of comparing the scale of dispersion across the *gaokao*, PISA and CFPS exams in common terms. The coefficient of variation is simply the ratio of the standard deviation to the mean, and the intuition is to show the typical difference from the average as a share of the average.

Slide 207

Exam	**Coefficient of Variation**
CFPS Math, 2010	39.3%
CFPS Reading, 2010	33.1%
Gaokao (overall), 2021	24.3%
PISA Math, 2018	13.7%
PISA Reading, 2018	15.7%

The numbers in Slide 207 clearly grate against the notion that the PISA exams capture the full sweep of Chinese youth's academic ability. With a math coefficient of variation nearly *three times* larger than PISA's math coefficient, the CFPS range suggests a much more disparate talent pool than we are given to believe from PISA. The contrast between *gaokao* and CFPS is not as extreme (roughly 36 to 60 percent larger depending on CFPS subject), but nonetheless reinforces a breadth of academic achievement a great deal beyond the one propounded by official statistics. Of course, the years and age groups are different across exams, but even with qualifications, it seems unlikely that these differences are unimportant.

Knowing that the distribution of academic ability is likely larger is, of itself, useful information. But we can also draw conclusions as to the direction in which the distribution is stretched in CFPS relative to PISA: the 2018 PISA provinces all had 2010 CFPS reading scores more than 10% above the national average, and math scores 10% higher in each except Jiangsu, where they were still 4% higher than the national average. Thus, the expanded range of performance in CFPS is in all likelihood an expansion *downward* from the vaunted PISA results. We attempt to quantify this phenomenon in a section below.

1-B-i) CFPS-Adjusted *Gaokao*

As discussed above, because *gaokao* data are "smoothed" across provinces, the national distribution of scores doesn't offer clear information on the true distribution of academic achievement. Since the CFPS contains academic achievement results without these benignant adjustments for bureaucrat morale between provinces, the distribution of the CFPS results offers a more accurate view of the state of academic achievement among China's pupils.

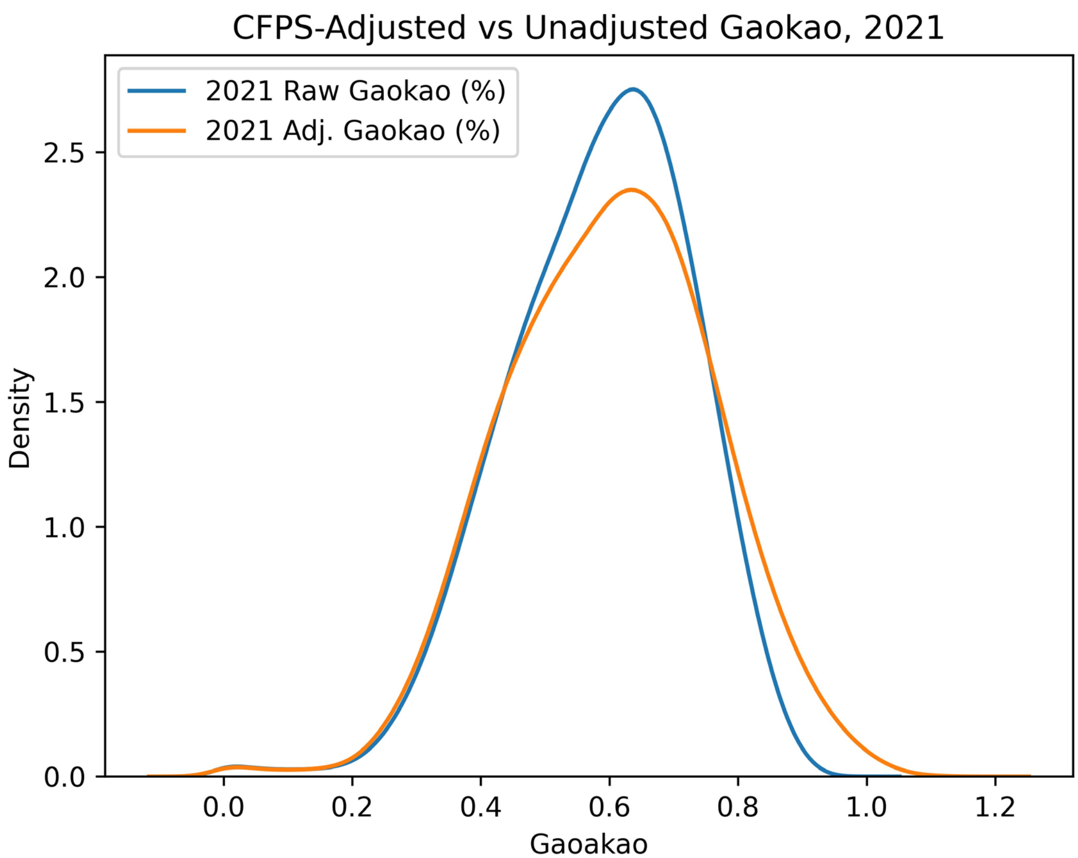
To make for comparisons on a *gaokao* scale, we used the CFPS academic achievement data to derive a province-level adjustment factor, which was simply the ratio of each province's average achievement score to the national average for each subject (math and reading), and then the average of the two subject ratios.[58] For example, in Jiangsu, the average math score was 11.72, versus a national average of 11.25, yielding a ratio of 1.042. The average reading score in Jiangsu was 23.993, versus 21.8 nationally, resulting in a ratio of 1.1. Jiangsu's final adjustment factor is then the average of those two subject ratios, or 1.071. We used these province-level adjustment factors as a crude means of undoing the inter-province flattening of the *gaokao* distribution, simply multiplying all *gaokao* scores by their corresponding provincial adjustment factors (see Slide 206). While the *gaokao* data are from 2021 and the CFPS data from 2010, the pool of CFPS test-takers aged 10-15 in 2010 (and therefore 21-26 in 2021) is not so dissimilar to the pool of primarily 18-year-olds taking the *gaokao* in 2021 as to preclude useful inference from the pairing.

Because the maximum score for the *gaokao* varies across province, we have performed calculations in terms of percent of full score rather than in absolute terms. The national *gaokao* distribution adjusted with CFPS factors is not

[58] These averages, and all figures presented from CFPS unless otherwise noted, were weighted by the 'RSWT_NAT' national sample weights for individuals.

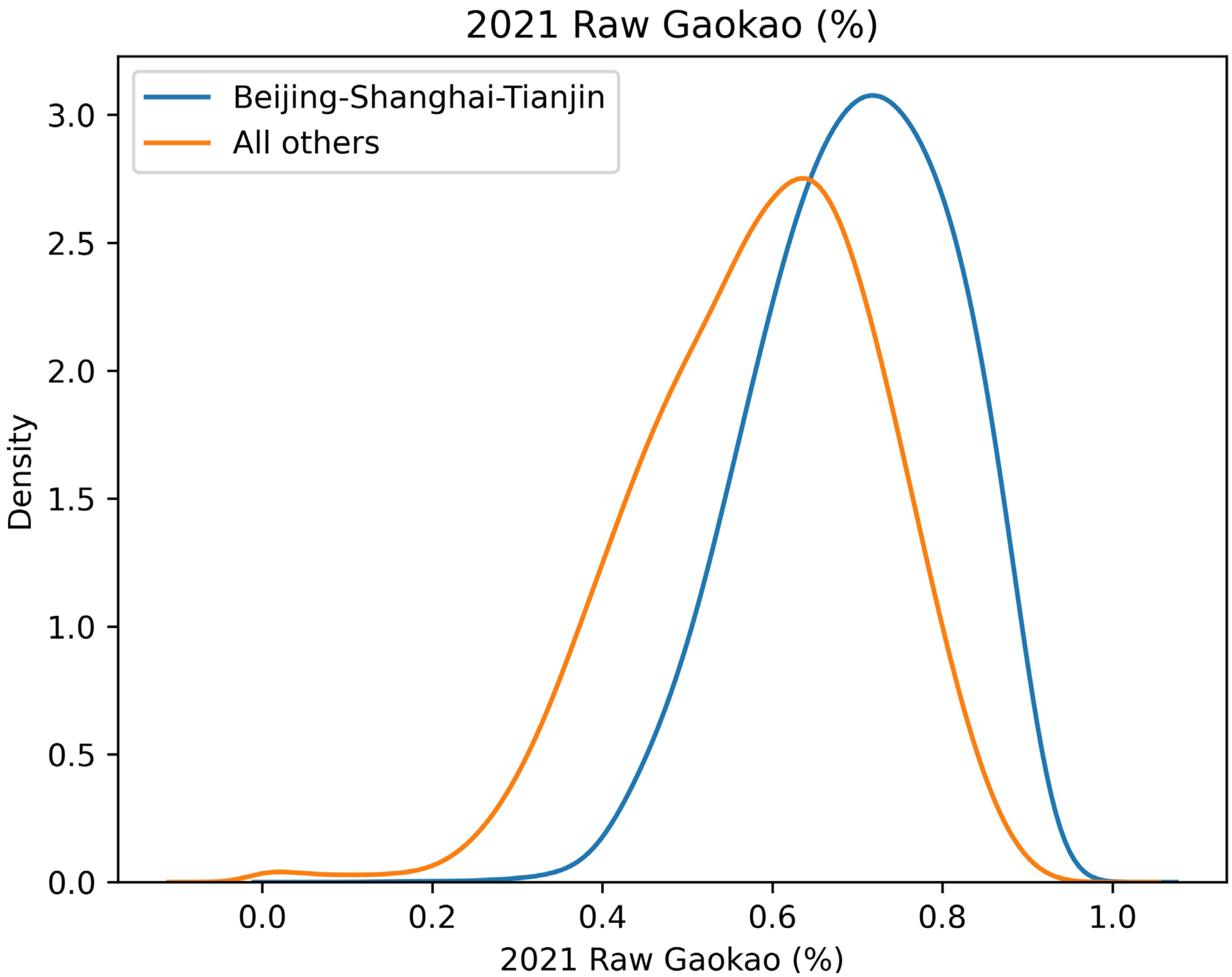
pointedly different from the unadjusted national distribution (see Slide 208) – the adjusted mean and median are slightly higher.

Slide 208



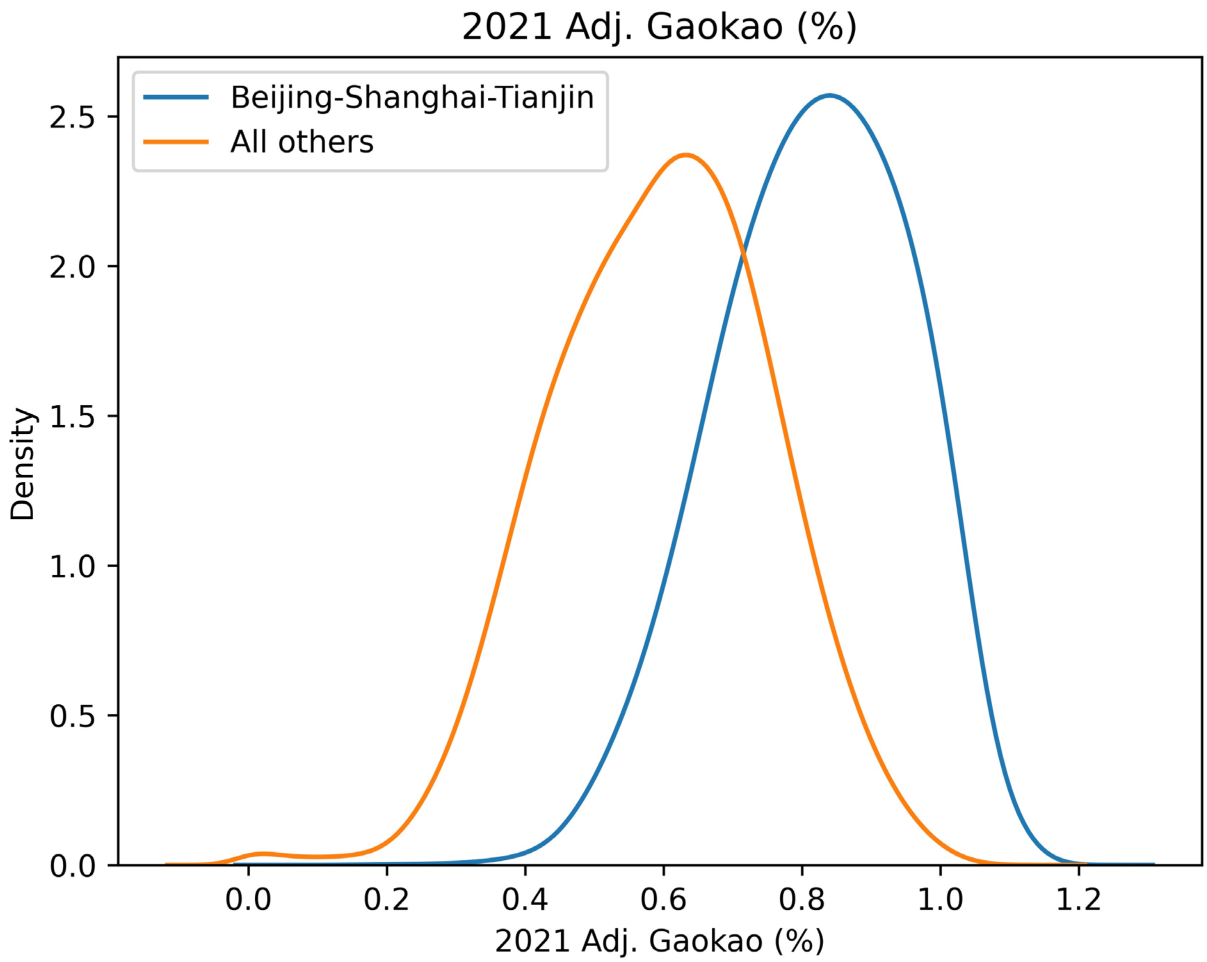
But comparisons of selected high achieving provinces (Beijing, Shanghai and Tianjin) with all others is a bit more interesting. Even in the raw-but-leveled *gaokao* data, the mean and median percentage correct in Beijing, Shanghai and Tianjin (B-S-T) are more than 10 points higher than in the rest of China (see Slide 209). So the administered normalization is at least not comprehensive. But in the CFPS-adjusted comparisons of B-S-T with the rest of China, the gap in means is nearly twice as large, and the spread in median scores between B-S-T and the rest of China more than doubles (see Slides 210 and 211). Precisely because of the inter-province standardization, the spread of scores is much more informative than their levels.[59] These adjusted scores indicate that the mass of China lies much further below its select elite provinces than the first impression from *gaokao* data reveals.

Slide 209



[59] If everyone in the class always gets an A+, we do not ascertain the class's general ability. If the teacher suddenly assigns a few students each an A++, the difference between A+ and A++ is obviously the only metric that matters since A+ meant nothing in the first place. A similar phenomenon is occurring with normalized *gaokao* scores.

Slide 210



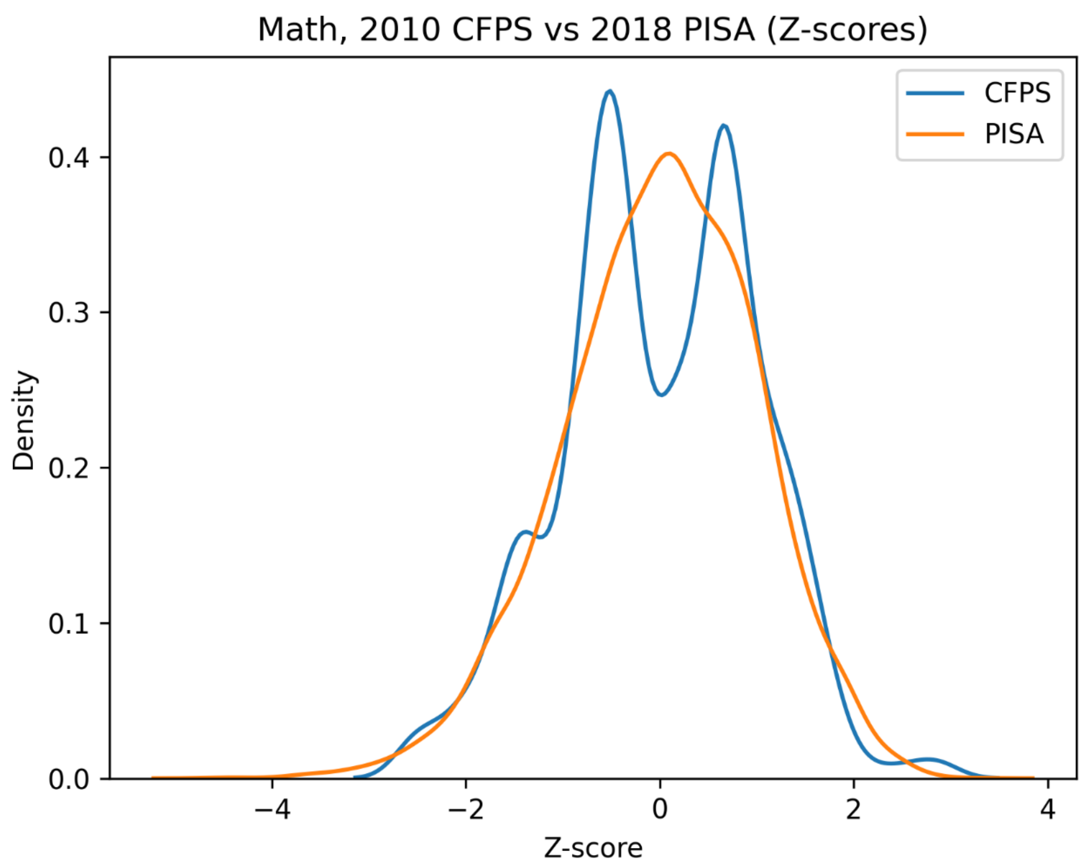
Slide 211

	Unadjusted Gaokao			CFPS-Adjusted Gaokao		
	B-S-T	**All others**	**Pts. Diff**	**B-S-T**	**All others**	**Pts. Diff.**
Wtd. Mean	69.6%	58.3%	**11.3%**	81.5%	59.6%	**22.0%**
Wtd. Median	70.3%	59.5%	**10.8%**	82.3%	60.2%	**22.1%**
Wtd. Std. Dev.	11.5%	14.1%	**-2.7%**	13.8%	15.9%	**-2.1%**
Min	13.3%	0.0%	**13.3%**	15.1%	0.0%	**15.1%**
Max	93.2%	94.4%	**-1.2%**	113.6%	109.2%	**4.5%**
10th	54.1%	39.5%	**14.6%**	63.0%	38.9%	**24.1%**
25th	61.6%	48.8%	**12.8%**	72.0%	48.4%	**23.6%**
75th	78.5%	68.7%	**9.8%**	92.1%	70.8%	**21.2%**
90th	84.3%	75.7%	**8.5%**	99.2%	79.7%	**19.5%**

1-B-ii) CFPS-Adjusted PISA

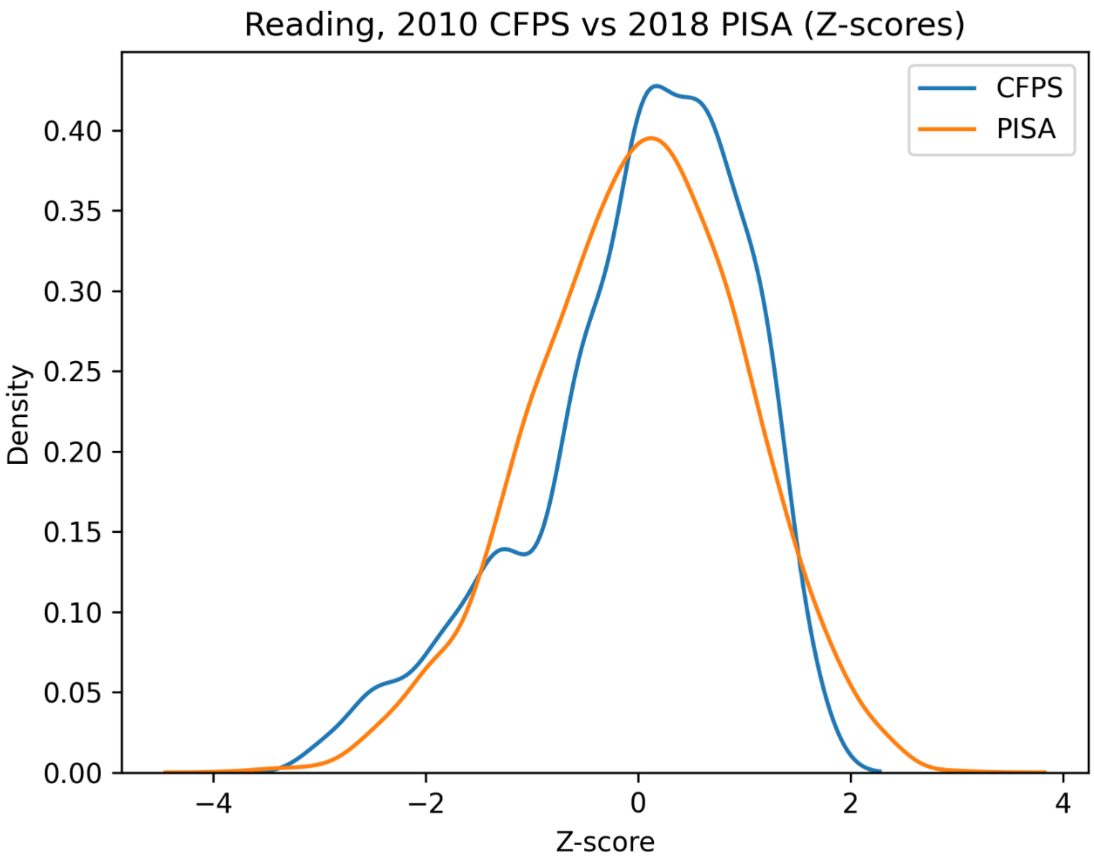
2010 CFPS achievement scores were also used to fill in the national distribution of China's 2018 PISA scores, which, as previously discussed, are less than representative of China as a whole, with observations from only four provinces (Beijing, Jiangsu, Shanghai and Zhejiang, or B-J-S-Z). Fortunately, the two CFPS subject areas – math and reading – overlap with the PISA exam's subjects, and therefore, subject-wise comparisons are possible (see Slides 212[60] and 213 for comparisons on a standardized scale, the Z-score or number of standard deviations from the mean).

Slide 212



[60] The bimodal distribution in CFPS math scores is peculiar, but it is indeed the shape of the data.

Slide 213



The method employed here is to impute a national distribution of PISA scores using the comparable CFPS scores as a bridge. For provinces which did not participate in PISA, CFPS scores were scaled relative to CFPS performance in provinces which *did* participate in PISA, calculating each score as follows[61]:

$$PISA_i = \frac{\text{CFPS}_i - \text{Mean}_{\text{CFPS in B-J-S-Z}}}{\text{St. Dev.}_{\text{CFPS in B-J-S-Z}}} \times \text{St. Dev.}_{\text{PISA}} + \text{Mean}_{\text{PISA}}$$

The intuition of this transformation is to use the relationship of CFPS scores outside PISA provinces to CFPS scores inside PISA provinces as a calibration to conjecture plausible PISA scores where we do not have them. These CFPS-imputed PISA scores were then combined with the original PISA data to approximate a national distribution. Because CFPS & PISA sample weights have

[61] The process is essentially the same as one described in Sarah Gust, Eric Hanushek and Ludger Woessmann, "Global Universal Basic Skills: Current Deficits and Implications for World Development," Journal of Development Economics, October 2023: 10, https://www.nber.org/papers/w30566.

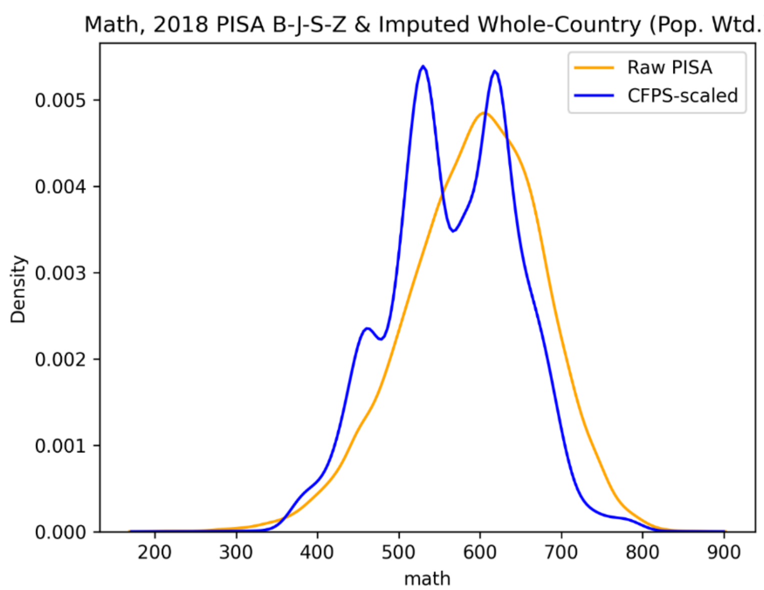
different bases & methodologies, for calculations after this imputation each observation was weighted in proportion to population in its province:

$$w_i = \frac{\text{Population}_{\text{Province j}}}{\text{Observations}_{\text{Province j}}}$$

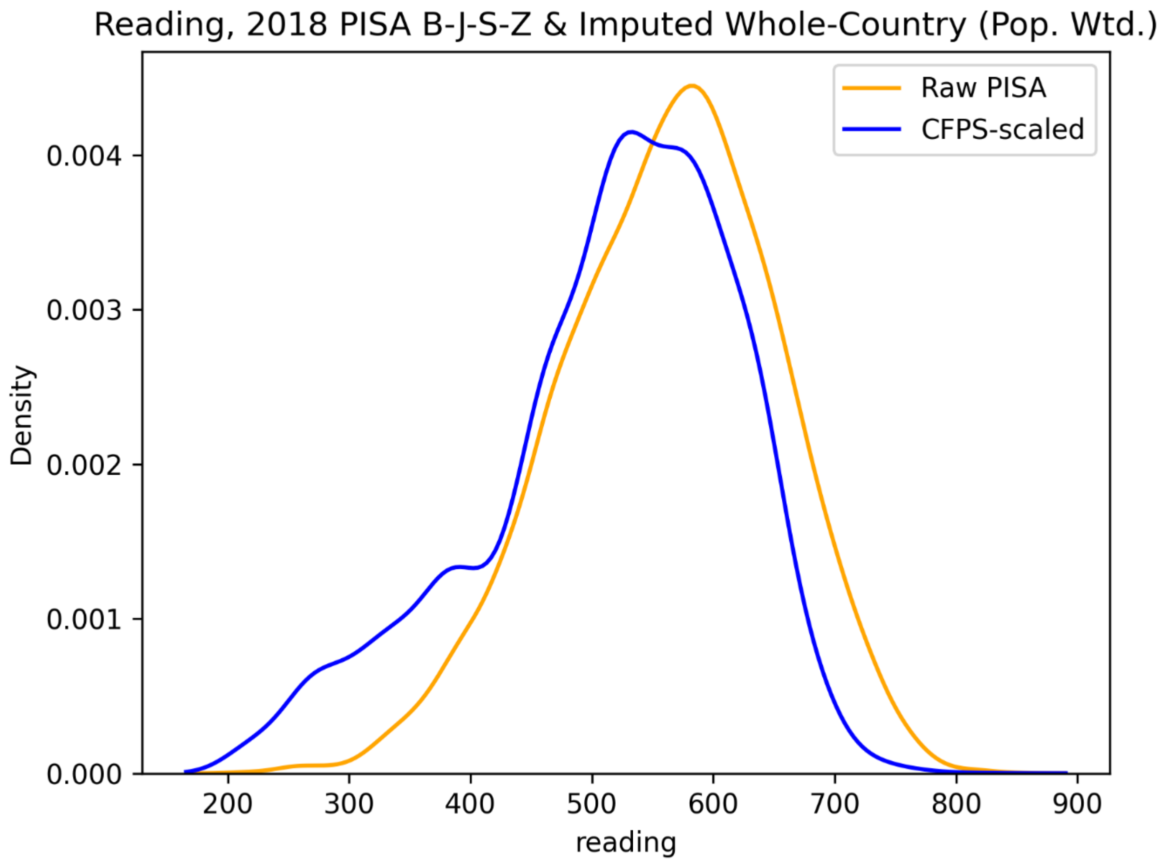
In other words, observations were weighted such that the sum of sample weights from each province is equal to that province's 2018 population. More complicated schemes could be deployed, but this simple construction has straightforward logic.

Since some provinces (notably, provinces commonsense would predict to have lower achievement) were excluded from CFPS as well, some simple inferences for these provinces were made to illustrate where the national distribution really is. For each province missing from CFPS, a "donor" province was chosen from the CFPS data, and all observations from that donor province were duplicated and weighted to represent the missing province according to the population scheme described above. A simplistic pairing method was used: if the missing province was in bottom half of 2010 GDP per capita, Guizhou's data were used (lowest CFPS provincial mean); if it was in the top half, Guangxi's data were assigned instead (second lowest CFPS provincial mean).

Slide 214



Slide 215



Slides 214 and 215 show the imputed math and reading scores distributions as they compare against the original, 4-province PISA distributions (back on a PISA-score scale, unlike the prior charts). The inference from these imputed PISA distributions tempers the impression of China's high-achieving pupil population to some degree: for math, China slips from far and away first to comparable with Singapore in second place. And in reading, China's imputed national mean falls near the mean in the Republic of Korea. The adjusted distribution had similar changes for medians.

These results are bit more favorable toward China's PISA performance than other findings in this series of work (in particular, scores modeled on several developmental factors, predicting drops well over 100 points in some specifications). One question which looms large and bears directly on the imputation here is the extent to which PISA test takers were cultivated for the exam, as discussed in our section considering the prevalence of cheating. The imputation from CFPS is sensitive to the degree of similarity (or dissimilarity as the case may be) between CFPS test-takers in B-S-J-Z and PISA test-takers in B-S-J-Z. There are reasons to suspect that the PISA examinees were selected and drilled, resulting in scores leaps and bounds above other nations', while documentation for the CFPS seems to describe good faith, best practice attempts at representative sampling that fielded a typical slate of pupils for the exam.[62] Nonetheless, this analysis takes up the available data, and establishes what might best be considered an upper bound China's on national PISA performance.

Slide 216

Math	Raw PISA	CFPS-Scaled PISA	CFPS-Scaled PISA w. inferred data*	CFPS-Scaled % Diff	W. Inferred % Diff*
Wtd. Mean	591.13	573.48	566.86	-3.0%	-4.1%
Wtd. Median	595.14	581.54	570.91	-2.3%	-4.1%
Wtd. Std. Dev.	80.87	76	77.63	-6.0%	-4.0%
Min	207.03	207.03	207.03	0.0%	0.0%
Max	863.75	863.75	863.75	0.0%	0.0%
Wtd. Obs.	992,302	1,327,350,000	1,571,760,000	-	-
N	12,058	15,204	16,192	26.1%	34.3%
10th	485.46	462.17	462.17	-4.8%	-4.8%
25th	539.68	530.38	513.33	-1.7%	-4.9%
75th	648.26	632.7	615.65	-2.3%	-4.1%
90th	690.09	666.81	666.81	-2.4%	-5.0%

* Include the provinces for which there are not original CFPS data and which received donor data.

[62] China Family Panel Studies User's Manual, 3rd Edition, July 30, 2017, pp. 17-19, https://www.isss.pku.edu.cn/cfps/docs/20220302153803194600.pdf?CSRFT=8I1K-H3Q5-FAAG-TH4J-E9OM-1WWE-D7GH-NR2W.

Slide 217

Reading	Raw PISA	CFPS-Scaled PISA	CFPS-Scaled PISA w. inferred data*	CFPS-Scaled % Diff	W. Inferred % Diff*
Wtd. Mean	555.25	526.46	514.44	-5.2%	-7.3%
Wtd. Median	558.49	537.98	532.98	-3.7%	-4.6%
Wtd. Std. Dev.	87.28	97.72	104.14	12.0%	19.3%
Min	208.22	208.22	208.22	0.0%	0.0%
Max	847.85	847.85	847.85	0.0%	0.0%
Wtd. Obs.	992,302	1,327,350,000	1,571,760,000	-	-
N	12,058	15,204	16,192	26.1%	34.3%
10th	441.58	379.65	356.49	-14.0%	-19.3%
25th	496.91	475.56	463.29	-4.3%	-6.8%
50th	558.49	537.98	532.98	-3.7%	-4.6%
75th	616.55	594.87	588.74	-3.5%	-4.5%

* Include the provinces for which there are not original CFPS data and which received donor data.

1-B-iii) CFPS Conclusion

As we have noted at various points, there are snags to these deployments of CFPS exam data for inferences from obviously different exam data. Ages, samples and purposes vary. But having tempered the findings with those caveats, the CFPS data, even without imputations and transformations to other datasets, show a starkly more diverse talent pool than more recent (and more publicized) data offer. While PISA exams give the impression that most of China's students fall within 15% of its astronomic average, the spread of CFPS scores offers a valuable and pointed *au contraire*. Further, the broadened portion seems to be weighted toward lower achievement. And for all their quirks, the adjustments to *gaokao* and PISA results offer reasonable datapoints from which to triangulate the relatively inscrutable question of true academic ability among China's youth.

1-C) The China Education Panel Survey (CEPS)[63]

1-C-i) Data Description and Challenges

The China Education Panel Survey (CEPS) is ostensibly a nationally representative sample of close to 20,000 Chinese 7th and 9th graders across 28 unidentified counties and 112 individual schools, designed as a "panel study" to track students over time. The baseline year of the panel survey is the 2013-2014 school year, with reported academic achievement scores in mathematics, Chinese, and English, as well as a measure of cognitive ability. The survey is based at the National Survey Research Center of Renmin University in Beijing, but it has also received funding from the US National Science Foundation, and one of its Co-Principal Investigators is a tenured and well-published professor of sociology from the University of Pennsylvania, Emily Hannum. In principle, this survey should give us an indication of both the cognitive ability (reflected in the cognitive scores) and academic ability (reflected in the academic achievement scores) of pupils across China. But a closer look reveals problems, giving us reason to believe that the CEPS sample may not be nationally representative.

The main variables of interest are the cognitive scores and the academic achievement scores. Of these, only the cognitive tests are administered independently by CEPS. The cognitive test consists of 20 multiple choice questions, intended to test students' ability to think logically and solve general problems. The test is designed to assess students along three dimensions: Language (including phrase analogy and linguistic reasoning); Graphics and Space (including graphical pattern analysis, origami questions, and geometry applications); and Calculation and Logic (including math applications, operation rules, sequence application, abstract law analysis, probability, and numerical size thinking). The survey data reports the students' "raw cognitive score" out of 20, and also standardizes the score based on the 3PL model. Within the sample, the

[63] This section of the paper refers to Slides 236-242 in the Online Appendix

standardized cognitive scores range from -2.03 to 2.71, with a mean of 0.01 and a standard deviation of 0.86. The academic achievement scores, in contrast, are not administered by CEPS. Instead, CEPS simply collects and reports the students' midterm scores in Math, Chinese, and English. These scores are collected directly from the individual schools, and it is not clear whether scores are directly comparable across schools or even across classrooms. In addition to the raw midterm scores, CEPS also reports standardized academic achievement scores, with the standardization performed so that the mean is 70 and the standard deviation is 10. Despite this standardization, it is still not clear whether these scores are directly comparable across students, meaning that the cognitive scores give us the best measure of ability.

According to the CEPS website,[64] the "nationally representative sample" is drawn from 28 unlabeled counties, as designated in the 2010 Census. The counties are reportedly unlabeled for privacy reasons, giving us little to work with.[65] What we do know is that there are three waves of samples drawn, called "Sampling Frames" by the survey administrators. Sampling Frame 1 is the core sample of 15 counties, drawn at random from anywhere in China. Sampling Frame 2 is a supplemental sample of 3 counties drawn from the province of Shanghai "in order to fully reflect the special circumstances of Shanghai, a megacity" [66]. And Sampling Frame 3 is a supplemental sample of 10 counties drawn at random from anywhere in China, with the stipulation that the counties have a large migrant population "in order to allow more migrant children to enter the sample, reflecting the impact of the special attributes of the floating population on the educational process and educational inequality". Specifically, Sampling Frame 3 consists of 10 counties randomly selected from a subset of 120 out of China's 2,870 counties designated

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[64] CEPS: China Education Panel Survey , http://ceps.ruc.edu.cn/English/Home.htm

[65] Thus CEPS on the officially proffered reasons for keeping the CEPS counties surveyed un-named and unidientified: *CEPS keeps in strict confidence the names, geographical locations and zip codes of schools and counties (districts) they belong to as required by relevant laws and scientific research ethics. This information will not be disclosed to any individual or institution so as to protect the privacy of the subjects and prevent any behavior that might cause damage to them…*
CEPS hereby announces that, according to the basic scientific research ethics, it decides not to disclose the names, geographic locations and zip codes of provinces, cities and counties (districts) and schools so as to protect the privacy of its subjects, guard them from any potential damage and prevent legal wrangles and other kinds of disputes.
"Announcement on CEPS Geo Data Disclosure". CEPS: China Education Panel Survey, http://ceps.ruc.edu.cn/info/1029/1044.htm.

[66] CEPS Sampling Design, http://ceps.ruc.edu.cn/English/Documentation/Sampling_Design.htm

as having a "large floating population", defined as counties "with an inflow of more than 196,000 people from across the province or more than 222,000 people from within the province"[67].

To summarize, of the 28 counties in the CEPS sample, 15 are randomly selected from anywhere in China, 3 are randomly selected from anywhere in Shanghai, and the remaining 10 are randomly selected from any county with a large migrant population. Of these, only the counties drawn from Shanghai have any sort of label as to which specific province they come from. Accordingly, we know that the three counties from Sampling Frame 2, as well as one large migrant county from Sampling Frame 3, come from Shanghai, giving us a total of 1,916 Shanghai students in the sample.

We also know that a total of 3,684 students in the sample come from "municipalities" (Beijing, Tianjin, and Shanghai), 3,815 come from "provincial capital city urban areas", 2,866 come from "prefecture level cities", and 9,122 come from "county-level cities". Other than the municipalities, these labels give us little to go on. The 'sampling design' document also specifies that six of the 28 counties in the sample come from municipalities. This gives us something to work with, as we have already identified four counties as coming from Shanghai. This means that there are an additional two counties coming from Beijing/Tianjin. In principle, since we know that 3,684 students come from municipalities, and 1,916 come from Shanghai, we should be able to find two additional counties in the sample with total observations totaling 3,684-1,916 = 1,768 students. Unfortunately, we find that this is not possible. There are, in fact, no two counties whose total number of observations equals 1,768. We conclude that something in the documentation is either wrong or mislabeled, remaining agnostic as to whether this error is intentional or not. Unfortunately, though this may give us insight into the motives and/or competence of those designing the survey, it does not help us get any further in identifying which provinces the CEPS observations come from, other than the labeled Shanghai observations.

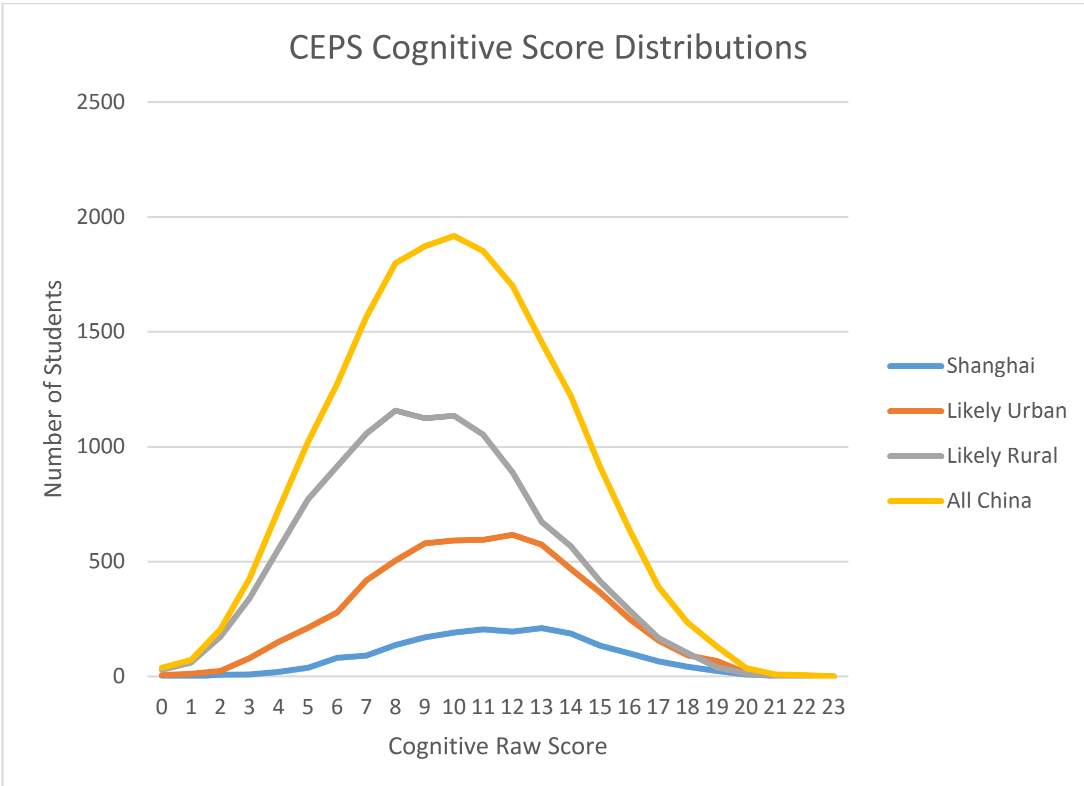
Along with the "Sampling Frame" designation, the counties in the CEPS sample are divided along another dimension which may give us insight: Each county is

[67] See the CEPS 2013-2014 User Guide

labeled as coming from one of three “Subsamples”. Subsample 1 is a “sub-sample of the nation (except Shanghai and districts with large influxes of migrants)”; Subsample 2 is a “sub-sample of Shanghai”; Subsample 3 is a “sub-sample of districts with large influxes of migrants in the country (except Shanghai)”. We interpret this language as indicating that while counties in Subsample 2 definitely come from Shanghai, counties in Subsample 1 likely come from rural areas, and counties in Subsample 3 likely come from urban areas outside of Shanghai. Dividing our sample along these lines gives us the four aforementioned counties from Shanghai, 14 counties from likely rural areas, and nine counties from urban areas outside of Shanghai.

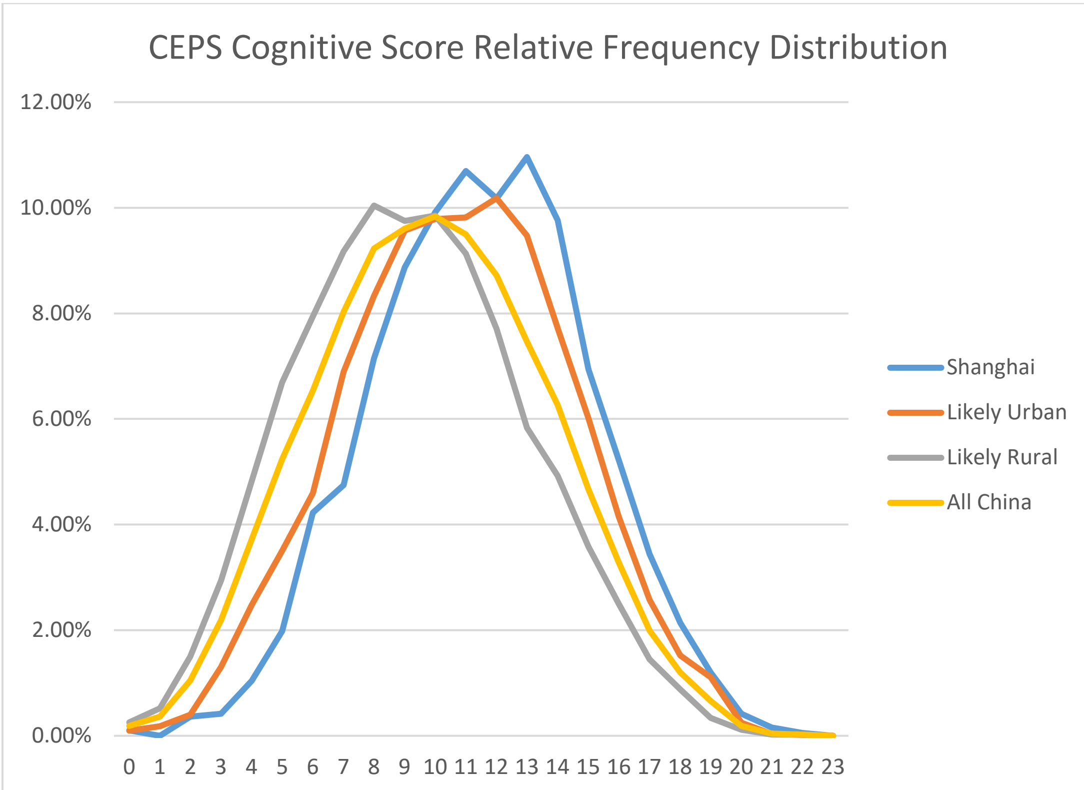
A primary question that arises from this analysis is: Is CEPS actually nationally representative? Or does it skew towards urban, more highly educated areas? Our guess that 14 sample counties come from rural areas comes only from the fact that they are labeled as a “sub-sample of the nation (except Shanghai and districts with large influxes of migrants)”. In other words, can we expect that the vast urban/rural divide observed in Chinese education is fully captured by a division based on whether a county can be classified as a having a “large migrant” population? We posit that the answer is no for several reasons. First, the aforementioned division of counties in the sampling design manual (“municipalities”, “provincial capital city urban areas”, “prefecture-level cities”, “county level cities”) does not inspire confidence that impoverished rural communities are well represented in the CEPS. Second, the observed differences between students known to be from Shanghai and students from the “likely rural” category are much smaller than we would expect based on work done by others in assessing urban-rural cognitive and educational gaps in China.

Slide 236

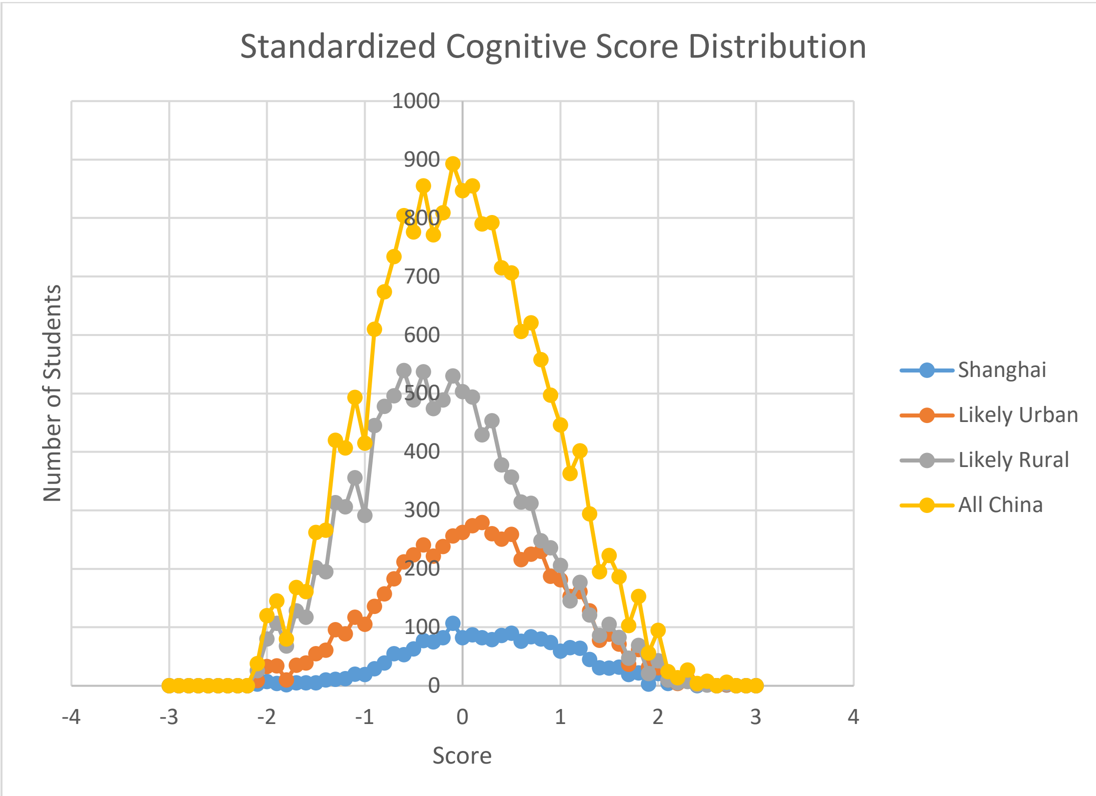


CEPS Cognitive Score Distributions
Number of Students
2500
2000
1500
1000
500
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
Cognitive Raw Score
Shanghai
Likely Urban
Likely Rural
All China

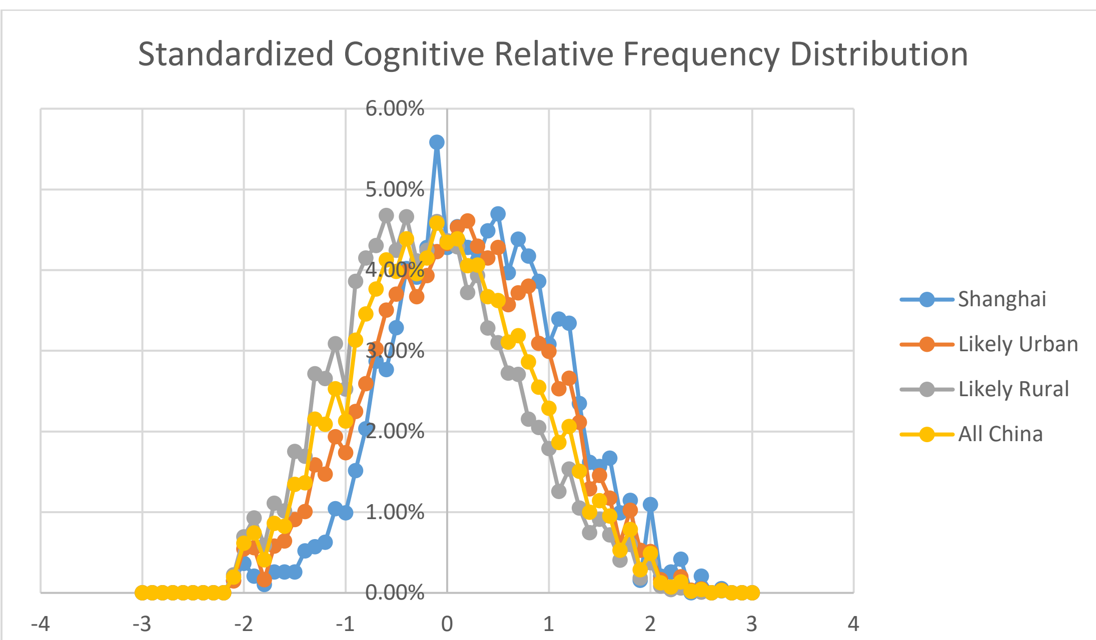
Slide 237


CEPS Cognitive Score Relative Frequency Distribution
12.00%
10.00%
8.00%
6.00%
4.00%
2.00%
0.00%
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
Shanghai
Likely Urban
Likely Rural
All China

Slide 238

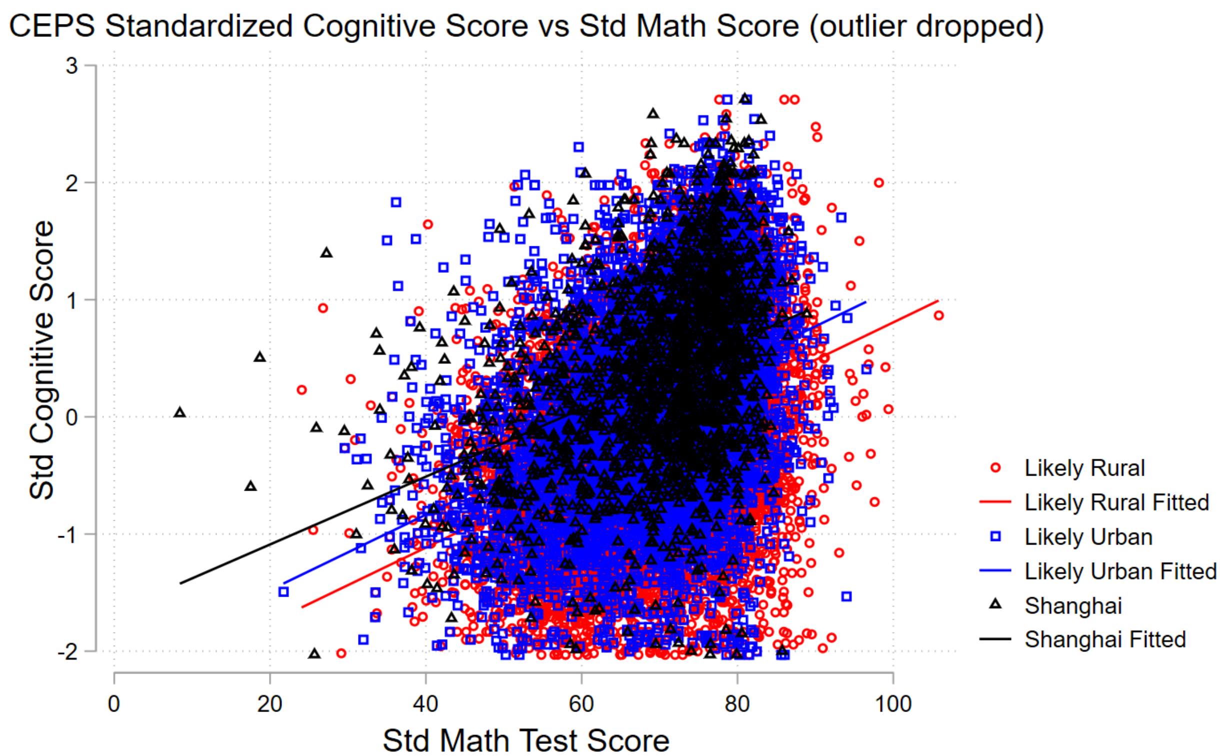


Slide 239



Would we really expect, for instance, that rural students and Shanghai students are less than one standard deviation apart in cognitive ability? If so then either the REAP Program vastly underestimates rural education levels or PISA vastly overestimates Shanghai's education levels. In any case, the CEPS data seems inconsistent with our prior assumptions on the distribution of "knowledge capital" across urban and rural parts of China.

Slide 240



And lest it be thought that cognitive score is a potentially misleading indicator of the urban/rural achievement gap reflected in education, the above chart shows the relationship between standardized math scores and standardized cognitive scores across the three different regions. Though all three regression lines have different intercepts, their slopes are nearly identical. For a given math score, Shanghai cognitive scores are generally higher than urban scores, which in turn are higher than rural scores, but an additional unit of achievement on the math test yields a comparable increase in cognitive score regardless of whether one is in Shanghai, a city, or the country. The stability of the relationship between test score and cognitive score is striking, and the lack of any apparent tendency for Shanghai students to outperform rural students in either dimension of the above figure provides further evidence that the CEPS survey is not quite representative of China as a whole.

1-C-ii) Linking CEPS and PISA

The aforementioned challenges and skepticism notwithstanding, we can still use the CEPS data to get a rough indication of what the true level of knowledge in China looks like. *If* CEPS is truly representative of China, and *if* the 2018 PISA scores are truly representative of knowledge in Shanghai, we can use the CEPS data labeled as coming from Shanghai to create an "illustrative distribution" of all-China PISA scores in 2018. Because of the two big "ifs" that are required for us to validly perform this linking, our preferred interpretation of the results is that they represent an "extreme upper bound" of true all-China PISA scores.

To perform this transformation we follow Gust, Hanushek, and Woessmann (2023)[68] and use Shanghai as the "linking province" between the PISA data and the CEPS data. Because Shanghai participates in both PISA and CEPS, and (just as importantly) because Shanghai data is labeled in CEPS, we can use our assumption that the distribution of scores from the two different tests are two samples of academic achievement from an underlying achievement distribution for Shanghai. *If* the scores follow a normal distribution, we can convert a non-Shanghai CEPS score ($\mathrm{CEPS_i}$) to a PISA scale ($S_{\mathrm{PISA\_i}}$) in the following way:

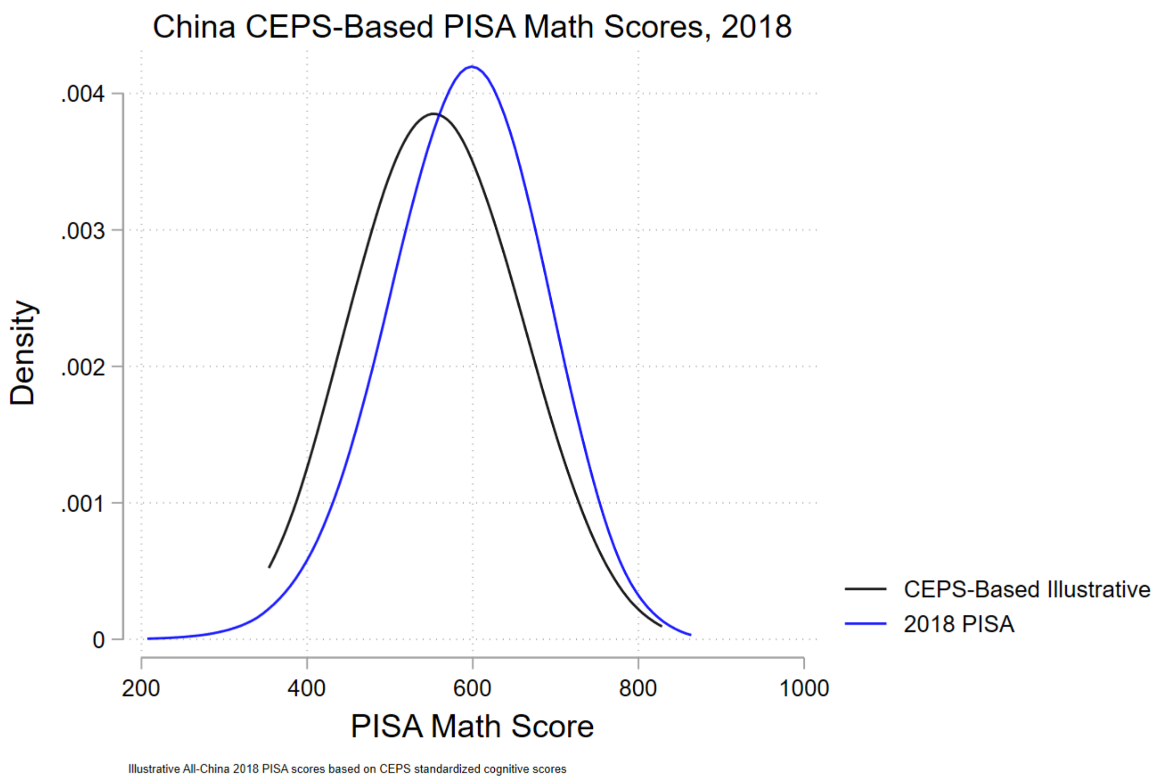
$$S_{\mathrm{PISA\_i}} = \frac{\mathrm{CEPS}_i - \mathrm{Mean}_{\textit{CEPS-Shanghai}}}{\mathrm{Std.\ Dev.}_{\textit{CEPS-Shanghai}}} \times \mathrm{Std.\ Dev.}_{\mathrm{PISA\text{-}Shanghai}} + \mathrm{Mean}_{\mathrm{PISA\text{-}Shanghai}}$$

Where $\mathrm{Mean}_{\textit{CEPS-Shanghai}}$ and $\mathrm{Std.\ Dev.}_{\textit{CEPS-Shanghai}}$ are the estimated mean and standard deviation, respectively, of the Shanghai cognitive scores in the CEPS data and $\mathrm{Mean}_{\mathrm{PISA\text{-}Shanghai}}$ and $\mathrm{Std.\ Dev.}_{\mathrm{PISA\text{-}Shanghai}}$ are, respectively, the estimated mean and standard deviation of Shanghai PISA scores.

[68] Sarah Gust, Eric A. Hanushek, and Ludger Woessman, "Global Universal Basic Skills: Current Deficits and Implications for World Development" *Journal of Development Economics*, 166 (January), 103205. 2024

In other words, we convert an individual CEPS score to a standard normal score and then re-express as a normally distributed score on the PISA scale, giving us an illustrative "all-China" PISA distribution using the CEPS dataset.

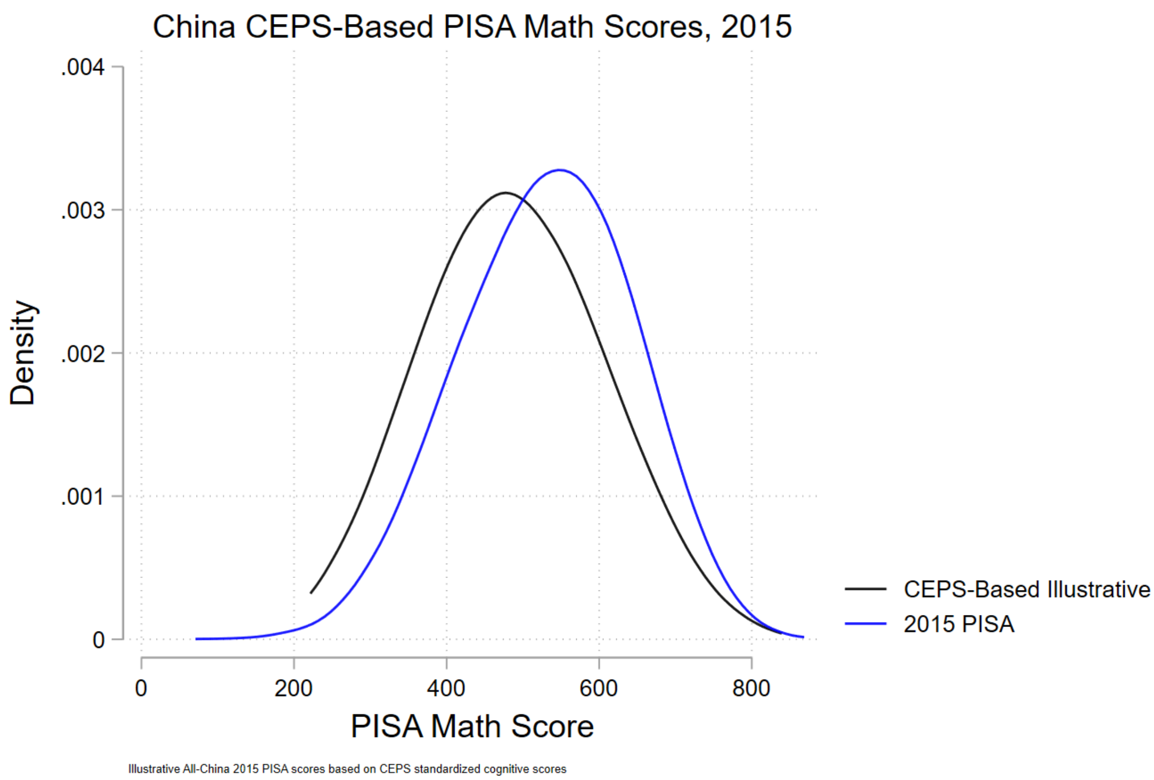
Slide 241



The above figure shows the illustrative distribution calculated in this way (in black), along with the true reported 2018 PISA distribution (in blue). A few caveats are in order: First, as noted, this assumes that CEPS is indeed nationally representative – if it skews towards richer, more highly educated areas (as we suspect) then this illustrative distribution will still tend to overestimate the distribution of knowledge in China. Unfortunately, we do not have any quantitative indication of by how much we are overestimating. Second, this analysis uses CEPS data from 2013-14 compared to 2018 PISA scores. This implicitly assumes that the distribution of knowledge capital has not changed significantly from 2013 to 2018. Finally, since CEPS only labels scores as coming from Shanghai, we treat the 2018 PISA scores as if they only come from Shanghai, which we know to be untrue.

Suffice it to say that this is a very rough first-order approximation of what a true all-China PISA distribution would look like in 2018. Similar exercises can be done for 2015 as well.

Slide 242



1-D) *Gaokao*[69]

By far the greatest trove of data on test performance for pre-college boys and girls in China comes from *gaokao,* the national college entrance test administered annually by the Ministry of Education. For reasons we shall explain, however, this trove is not an unalloyed treasure for outsiders interested in gleaning information about all-China academic achievement for the PRC's pre-college population.

The formal name for the *gaokao*[70] test translates as "National Unified Examination for Admissions to General Colleges and Universities", sometimes also referred to

[69] This section of the paper refers to Slides 292-307 in the Online Appendix.

[70] For background on the *gaokao* test, see inter alia: Adam Sichta, "Challenges of Reform: Ending the Dominance of the Gaokao, Beginning the Future of Chinese Education", VDM Verlag, May 15, 2009, https://www.amazon.com/Challenges-Reform-Dominance-Beginning-

as the “National College Entrance Examination”; *Gaokao* means “higher exam”. That is the same term that was used historically for the annual imperial examination. The imperial examination process was also referred to as “exam hell”.[71] Although the modern test is vastly different from its predecessor, it continues in the ancient tradition in the respect of remaining a grueling ordeal for examinees—that at least is its reputation.

Just as with the imperial exam, the modern test is a two day affair in June, taking up nine full hours. It is said to be extraordinarily demanding, setting a level of intellectual rigor and requirements in knowledge and skills that are essentially beyond reach. It is said that no student has ever earned a perfect score on the modern PRC *gaokao* exam. To put that fact in perspective: over the past two decades over 200 million young Chinese, including virtually all of the nation’s most talented young men and women from their birth years, have competed in the *gaokao* process.

Chinese and math are compulsory subjects for every examinee, but there is a range of optional subjects as well, including foreign languages, history, and the sciences. Conventionally, the test is taken by 12th graders, and so we notionally here assume the age of the testees to be 17—but in practice some of these 12th graders may be older or younger than 17, and *gaokao* rules allow others to take the test as well (under age 20 for those serving in the military, for example). In theory, the *gaokao*

Education/dp/3639154339/ref=sr_1_17?crid=3QFB75TJF041A&keywords=gaokao&qid=1706537381&sprefix=gaokao%2Caps%2C110&sr=8-17; Zachary Howlett, “Meritocracy and Its Discontents: Anxiety and the National College Entrance Exam in China”, Cornell University Press, April 15, 2021, https://www.amazon.com/Meritocracy-Its-Discontents-National-Entrance/dp/1501754467/ref=sr_1_11?crid=3QFB75TJF041A&keywords=gaokao&qid=1706537545&sprefix=gaokao%2Caps%2C110&sr=8-11; Yong Zhao, “Who’s Afraid of the Big Bad Dragon?: Why China Has the Best (and Worst) Education System in the World”, September 5, 2014, https://www.amazon.com/Whos-Afraid-Big-Bad-Dragon/dp/1118487133/ref=sr_1_7?crid=1ZIHPWIQGDWE0&keywords=gaokao+history&qid=1706537737&sprefix=gaokao+history%2Caps%2C66&sr=8-7; Chris Hamnett, Shen Hua, and Liang Bingjie, “The reproduction of regional inequality through university access: the *Gaokao* in China”, Area Development and Policy, Volume 4, 2019 – Issue 3, https://www.tandfonline.com/doi/abs/10.1080/23792949.2018.1559703; for a memoir on how the the “*gaokao* mentality” crosses the ocean to influence educational attitudes of Chinese Americans in the US, see Yanna Gong, “Gaokao: A Personal Journey Behind China’s Examintion Culture”, October 10, 2013, https://www.amazon.com/Gaokao-Personal-Journey-Examination-Culture/dp/0835100626/ref=sr_1_1?crid=3QFB75TJF041A&keywords=gaokao&qid=1706537545&sprefix=gaokao%2Caps%2C110&sr=8-1 . As it happens, one of the best succinct descriptions of the modern PRC *gaokao* system can be found in Wikipedia: https://en.wikipedia.org/wiki/Gaokao . We draw on these sources in this section of the paper.

[71]For a superb history, see Ichisada Miyazaki, *China’s Examination Hell* (New Haven: Yale University Press, 1981), https://yalebooks.yale.edu/book/9780300026399/chinas-examination-hell/.

is graded on a scale with up to 750 points—but as we shall see, local procedures can be different in practice.

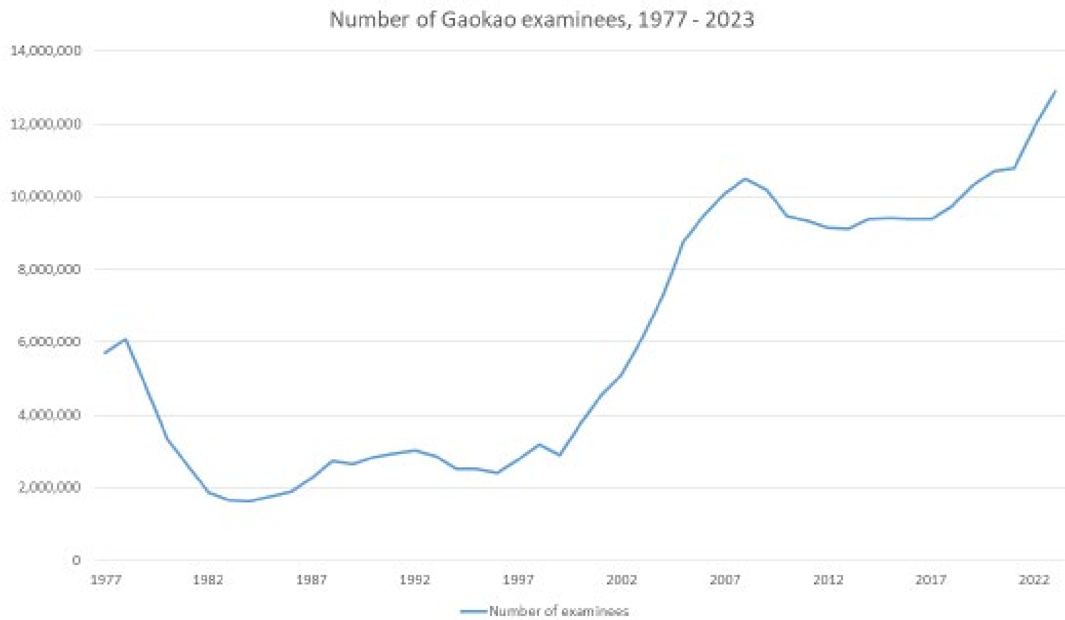
A brief history of the modern *gaokao* is in order. The modern PRC college entrance exam was established after the consolidation of Communist power in 1952, three years after the Liberation. The test was set to the needs of a mass industrial economy, with exams in Chinese, math etc. (There was of course an ideological element to the testing and test taking as well, including a review of the class background of aspirants). The universities were suspended during the upheaval of the Cultural Revolution, and *gaokao* tests were not conducted for years. After Mao's death the *gaokao* was finally reintroduced in 1977 (Deng Xiaoping was reportedly instrumental here, even though he had not yet risen to supreme power.) The modern *gaokao* test system dates from the early 1980s.

In the 1980s and early 1990s the *gaokao* was an elite test. Only a small fraction of each cohort of 17 year olds took it. (In part because high school education was still limited in China in those years). Of those who applied, only a small share passed. But with the official decision in the late 1990s to expand higher education massively, all this changed. The number of applicants exploded, and the pass rates soared.

With available slots in higher education soaring, the *gaokao* was transformed from an elite test to a mass testing practice, almost a formality for what has become in effect something like an open admissions process for higher education in the PRC today. But the ruthless competition and endless preparation that still attends each new annual cycle of the *gaokao* test is no longer because college berths are scarce. Rather, this is now a competition for getting into the best schools: not unlike the SAT competition in the USA. (In fact, the proportion of Chinese youth who take *gaokao* nowadays is around 40%: not that much lower than the proportion of their US counterparts who take SAT.)

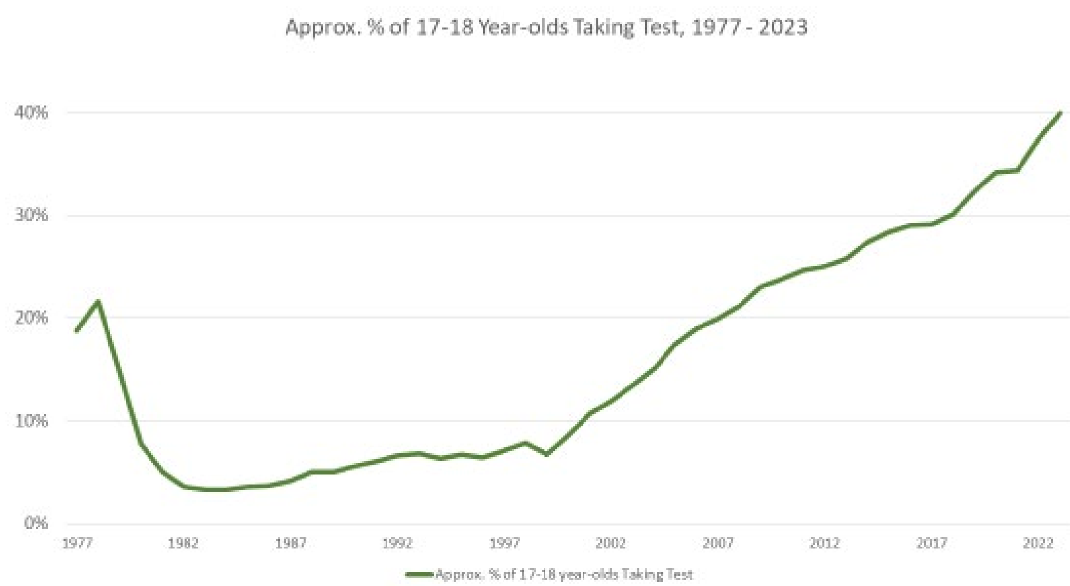
Between 1999 and 2023, the number of *gaokao* examinees leapt from under 3 million to almost 13 million, more than quadrupling. Over those same years the fraction of test takers from the cohort of 17-18 year old Chinese catapulted from 7% to 40%. The pass rate, which had been just 17% in 1982, was 55% by 1999—but it was 85% in 2022 (the latest available year), after exceeding 90% in both 2020 and 2021. By 2021 and 2022, slightly over 10 million examinees were passing the *gaokao* test each year.

Slide 292



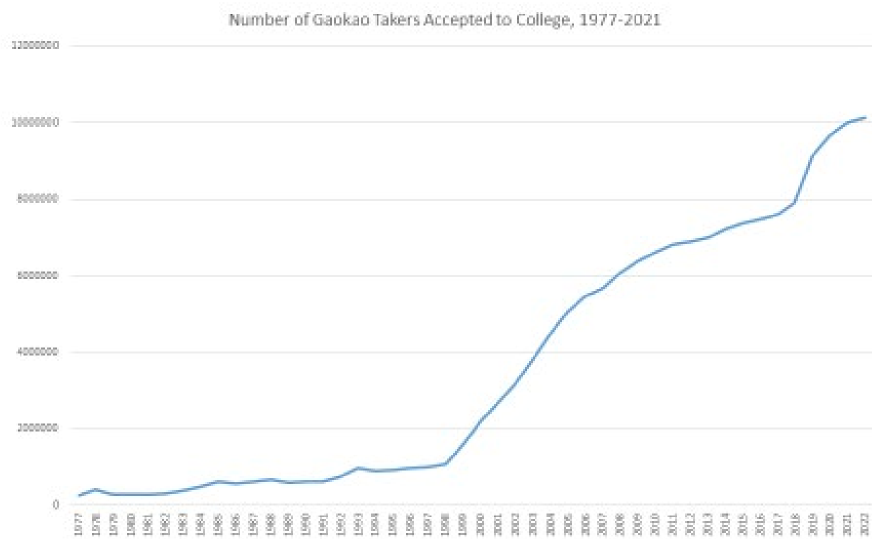
Source: https://www.eol.cn/e_html/gk/gkrs/index.shtml via https://en.wikipedia.org/wiki/Gaokao.

Slide 293



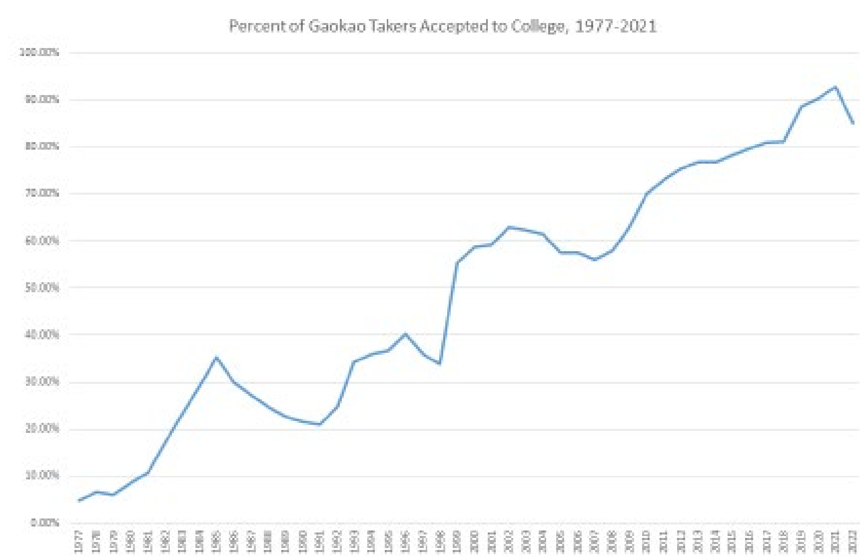
Sources: https://www.eol.cn/e_html/gk/gkrs/index.shtml via https://en.wikipedia.org/wiki/Gaokao;
United Nations, Department of Economic and Social Affairs, Population Division (2022). World Population Prospects 2022, Online Edition, https://population.un.org/wpp/Download/Standard/MostUsed/.

Slide 294



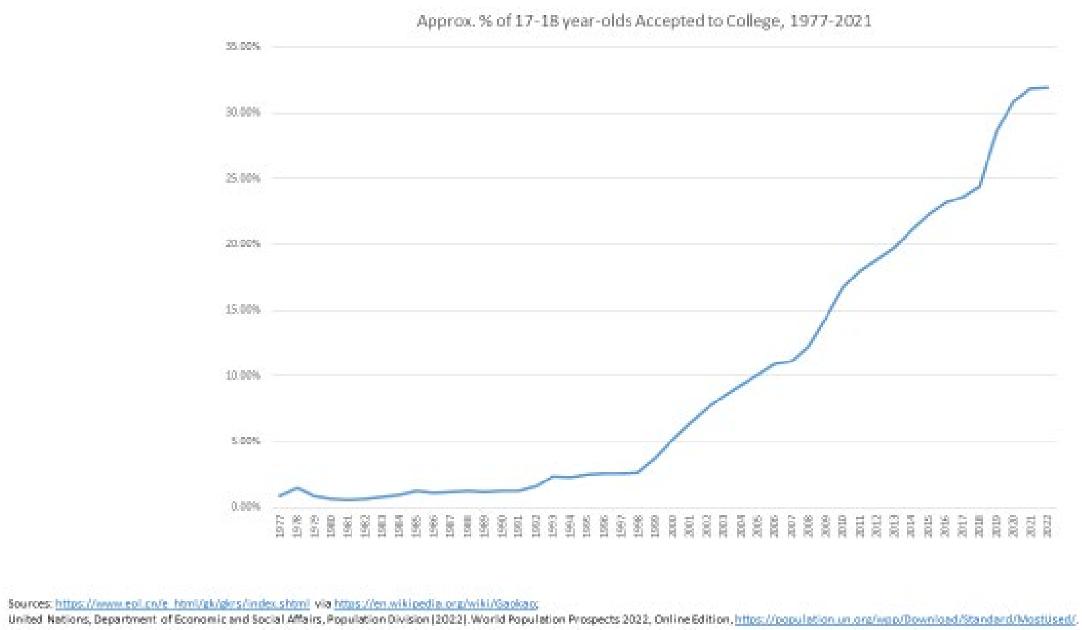
Source: Gaokao - Wikipedia

Slide 295



Source: Gaokao - Wikipedia

Slide 296



Just to be clear: although the *gaokao* is now in effect a mass exam, a distinct majority of young Chinese still do not compete in it. The 60 percent who do not take it in all likelihood substantially over-represent rural students (or young men and women who did not stay in school to 12th grade). But *gaokao* data are compiled and published on a provincial basis, with no indication of urban vs. rural status. In any case, it is a reasonable surmise that the 40% of Chinese 12th graders taking the *gaokao* rank in the top half of "knowledge capital" for their birth cohort. *Gaokao* is *not* testing a random sample of PRC youth. The majority who do not compete certainly would earn lower scores on average—though just how much lower is an unanswerable question to those outside the confines of the MOE's NAEQ, where confidential data might just provide an informed impression.

Yet even though current *gaokao* testing excludes the great majority of rising youth in China today, the enormous increase in college admissions over the past generation has fundamentally transformed the test in one critical respect: it is no longer tenable to maintain a single national grading standard for the millions and millions of young applicants from such a stratified population and such an inegalitarian society.

Until the 1990s, there was a single all-China grading standard for the *gaokao*. It would be highly informative to assess those all-China data—but they are no longer available to outsiders, if they ever were.

In the 21st Century, with the burgeoning test-taking population, examiners in the MOE faced a dilemma. If they kept an elite, all-country system of evaluation of *gaokao* results, test scores for most of the examinees would be embarrassingly low by comparison with results from say Beijing and Shanghai. The political repercussions of revealing just how poor the preparations were for even good students from the localities were deemed to be unacceptable. Consequently, at some point in the early 21st Century (the exact date is not clear to us), *gaokao* protocols were fundamentally changed. Since then, *gaokao* scores have in effect been graded on a curve—with separate curves for each province in the PRC.

The curves for the provinces, of course, are set by the central authorities—after all this is China. We do not know the method used—even in China this is unknown, outside a tiny select circle of educational policymakers. The process, though, is necessarily political and inescapably arbitrary. We believe, though we cannot prove, that the curve is re-set for China as a whole, and for every province in China, anew each year, in the face of the new corpus of *gaokao* test results.

As enforced, the curve conceals the vast differences across China in examinees academic achievement, while still leaving some hints as to the difference in academic prowess across the country's provinces. (Within each province, we have no reason to believe that rankings of student scores on the *gaokao* is officially altered or adjusted.) In effect, the *gaokao* exam scores reported for China represent something like "affirmative action with Chinese characteristics".

We were able to obtain detailed *gaokao* statistics for China by province for recent years because these are currently available online. Their availability reflects their domestic importance—to students and parents; a statistical blackout of *gaokao* scores would be a *very* unpopular move by the regime. Students and parents are intensely concerned about the integrity of the national examination—not least because of the honored and well-practiced tradition of test cheating in China[72], about which more later.

[72] "How is Chinese Gaokao assessment done?" *China Global Television Network*, July 2020, https://news.cgtn.com/news/2020-07-17/How-is-Chinese-Gaokao-assessement-done--Sc2qgs3qPm/index.html

Great attention by ordinary people in the PRC—perhaps this can be called "public attention", even under a dictatorship?—is devoted to detecting any signs of malfeasance by local *gaokao* examiners, or tolerance of (other) students' tricks for beating the test. But the locals do not seem to be aware (or concerned) that even faithfully conducted and graded tests will then be sent to the center, where the points assigned to the test taker will no longer necessarily have any correspondence to the number of correct answers he or she achieved on the test.

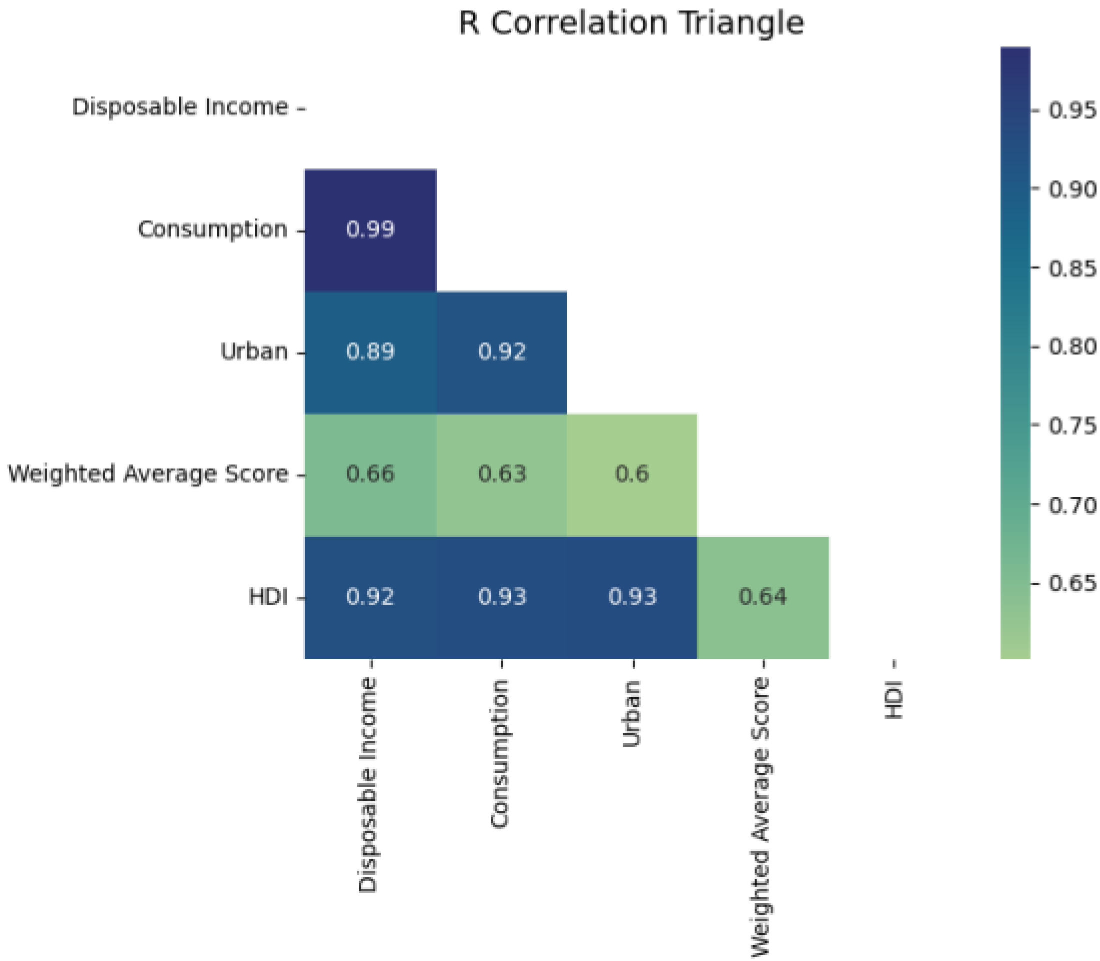
We selected just two years of *gaokao* scores—2019 and 2021, i.e. just before and just after the COVID-19 pandemic outbreak—for our statistical analysis for this paper.

We have subjected these data to almost excruciating interrogation in our statistical effort to separate "signal" from the "noise" in which this is, by official intent, disguised. The voluminous results of our statistical exploration are appended to this paper in Slides 297-348 of the Online Appendix, and the curious are invited to explore them at leisure.

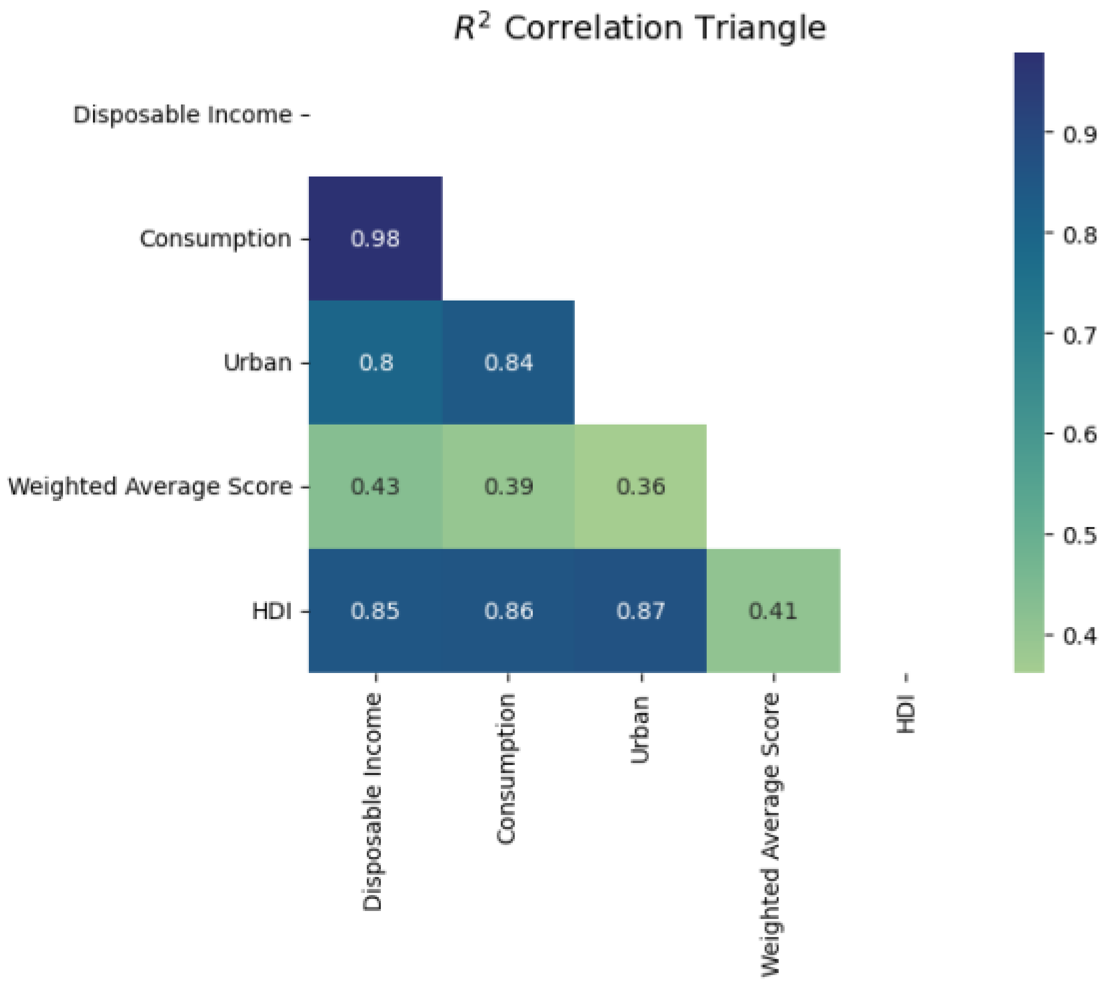
Two findings from our evaluation are worth presenting below. The first is the "correlation triangle" for mean *gaokao* scores and development indicators by province. We show the correlations from 2019, but they are very similar in 2021, and presumably all recent years.

The development indicators (i.e. urbanization rates, levels of per capita income, and Human Development Index scores) correlate very closely between China's provinces. On the other hand, mean provincial *gaokao* scores have a much lower correlation with all these indicators. This is not evidence that China's educational system is so successful that it has overcome interprovincial development differences in student achievement. It is evidence that the *gaokao* test scores have been systematically altered on a province by province basis by the hand of the central authorities

Slide 297


R Correlation Triangle
Disposable Income
Consumption
Urban
Weighted Average Score
HDI
0.99
0.89
0.92
0.66
0.63
0.6
0.92
0.93
0.93
0.64
Disposable Income
Consumption
Urban
Weighted Average Score
HDI
0.95
0.90
0.85
0.80
0.75
0.70
0.65

Slide 298

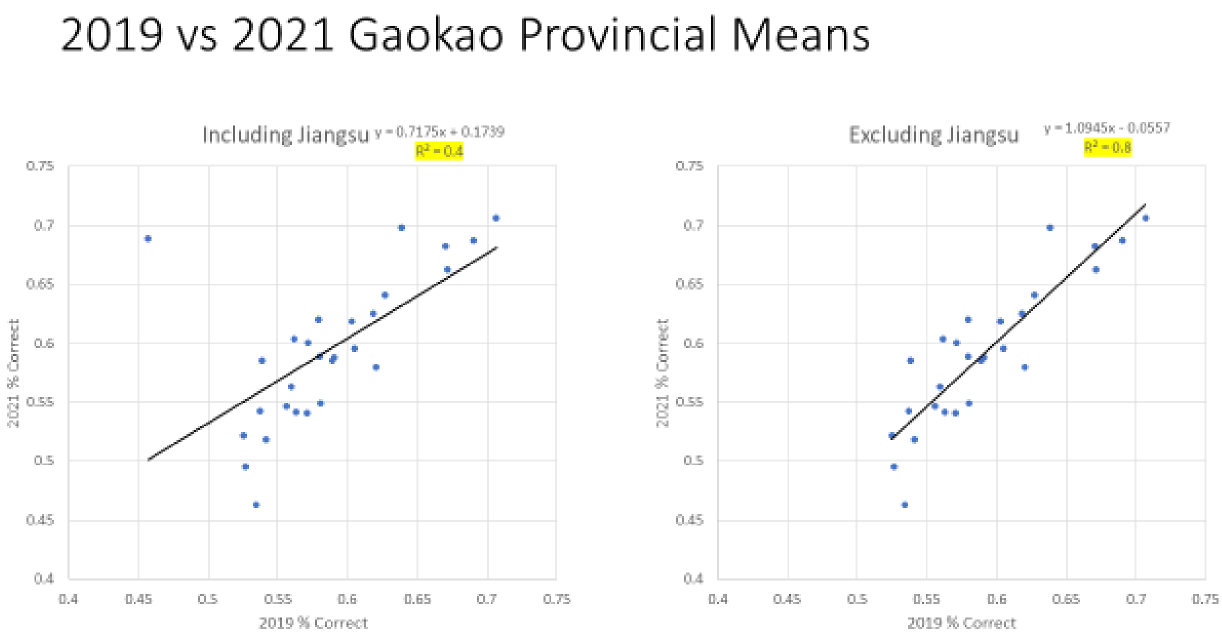


We can also demonstrate that the correspondence in mean test scores from one year to the next by province in China is suspiciously weak. We see this in the figures below, which compare mean scores by province for 2019 and 2021. For the SAT, we would see a near perfect correlation between test scores by state from one year to the next. (And as we have already seen in our discussion on PISA test results, scores for Massachusetts, and the US as a whole, are quite stable over time for an international examination.) The surprisingly attenuated correspondence between 2019 and 2021 scores by province—and the jolt in the scores for Jiangsu province—attest to a political process of "curving" reported test grades from one year to the next. Even if we exclude the Jiangsu observations (possibly due to a provincial shift toward a different scoring system[73]), the 2019 and 2021 mean test

[73] By one report, both Jiangsu and Shanghai have changed—and lowered—their score range: instead of a 750 point scale, Shanghai's top possible scores is now said to be 680, with the dramatically lower ceiling of 480 for adjacent

scores by province are a telltale sign that the *gaokao* results are "customized"—or put less charitably, doctored—on a province-by-province basis, each and every year.

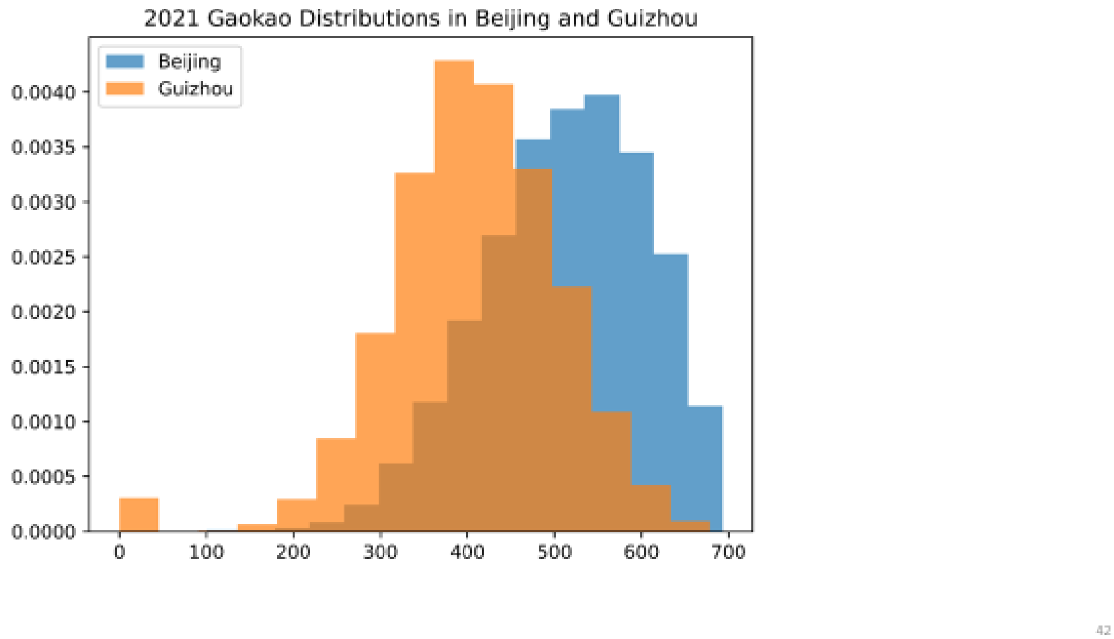
Slide 299



Despite the opaque, systematic, and (we believe) deep revisions to raw student scores in the *gaokao* grading process, apparently on a year-by-year basis, there is nevertheless some "signal" remaining in the numbers that the PRC allows to be published. We can see this, for example, in a comparison of distributions of scores for elite Beijing and backward Guizhou, below:

Jiangsu. Nuffic, "China: Grades and Study Results" https://www.nuffic.nl/en/education-systems/china/grades-and-study-results Note that this report does not explain why or how some provincial top *gaokao* scores in 2021 exceeded the notional 750 point maximum: in Hainan, for example, at least one examinee in 2021 scored a 903.

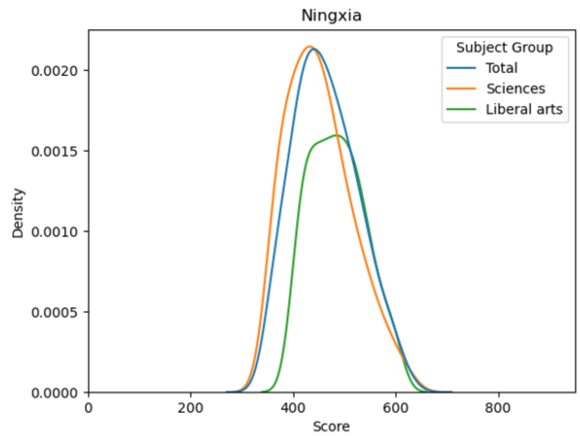
Slide 300

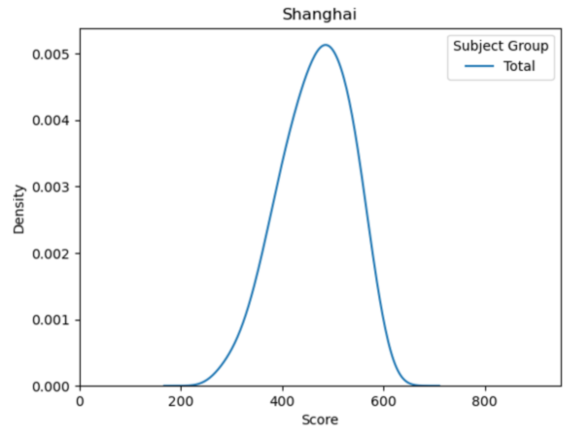


Possibly, this reported comparison will not fail the laugh-out-loud-test for some viewers, since it shows Beijing outperforming Guizhou. But to anyone at all familiar with the particulars under consideration, the purported "close-ness" of the test score distributions for the two provinces is preposterous on its very face. (Remember: Guizhou is where Scott Rozelle and colleagues determined that rural children were reading at a level of proficiency lower than that of any country in the PIRLS international database.)

But there are results here that do fail the laugh-out-loud-test, too. Thus, the remote Ningxia region—largely disadvantaged Hui people of Muslim heritage; 17th in per capita GDP out of China's 31 provinces (Shanghai is 2nd)—comes in as basically equal to Shanghai in reported mean *gaokao* scores for 2021 (mean score of 463 vs Shanghai's 466).

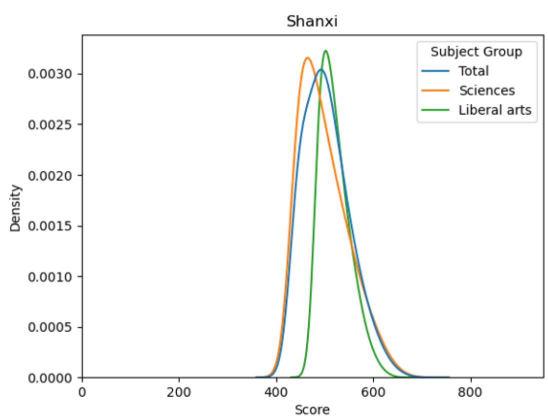
Slides 301 and 302

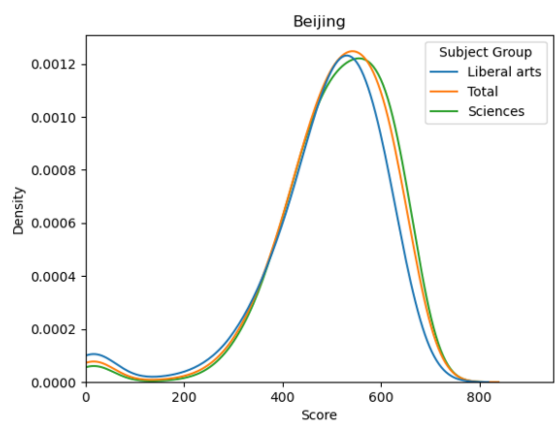




Ningxia's rating, and ranking, appear to have been aided by the fact that it was impossible to score below a 350 on the test if you lived in the province that year. But the deities of the curve smiled even more that year on Shanxi province, another below average income province where Rozelle and colleagues conducted some of their rural reading tests for schoolkids. In 2021 it was apparently impossible to score below a 410 in Shanxi: this in a test with a reported all-China mean that year of 438! So Shanxi reportedly ranked 2nd in all China: ahead of Beijing that year, behind only PISA-included Zhejiang.

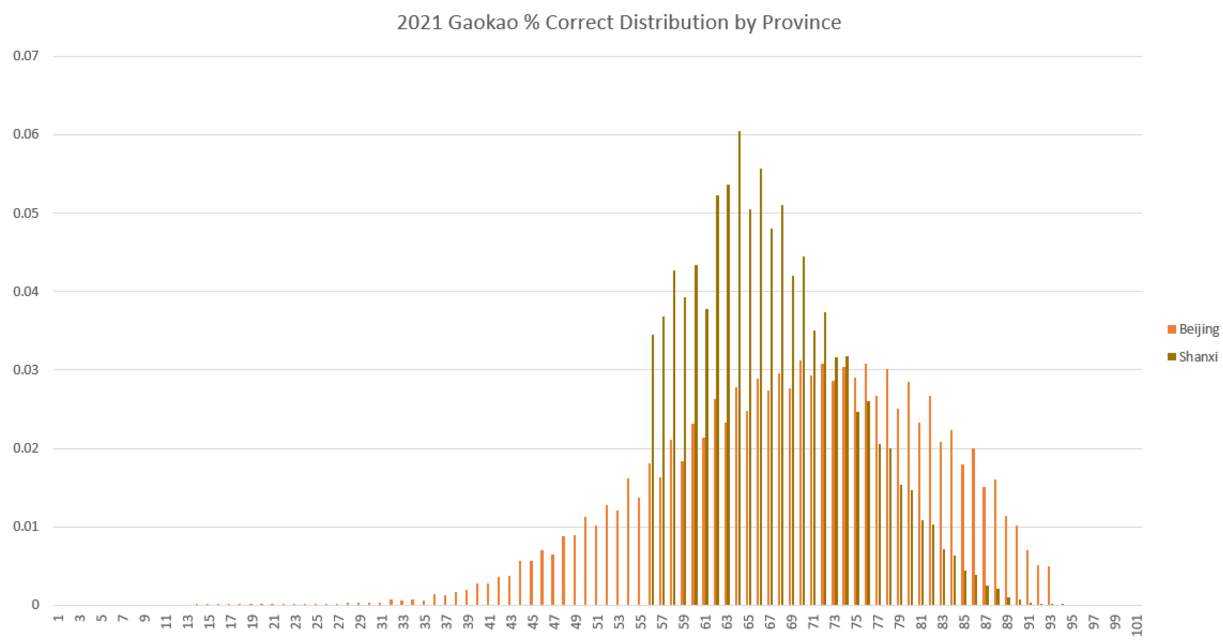
Slides 303 and 304





We even have reason to believe that the "affirmative action adjustment" runs deeper than a simple realignment of the scale used for each province. For the 2021 data we know the maximum possible score by province, allowing us to calculate a 'percent correct' distribution by province. Even if the raw scale of the test varies by province, we would expect that the 'percent correct' distribution should be fairly stable from province to province. As seen in the figure below, however, this is clearly not the case. While students from Beijing got anywhere from 13-93% of their questions correct, it appears that no one from Shanxi was allowed to get below a 56%. What happens, we are forced to wonder, if a student from Shanxi turns in a blank exam? The sharp cutoff in the distribution below forces us to suspect that a score of zero would not be reported.

Slide 305



In sum: unlike the methodologically rigorous results on test scores from internationally standardized achievement tests like PISA, and from the achievement results from domestic CEPS and CFPS panel studies, there is no methodologically coherent, consistent and transparent all-China standardization to the *gaokao* scores.

Even with all the impenetrable and opaque corrections to aspiring test-taker's raw scores across the whole of China, variations in tested *gaokao* achievement are still visible—and these variations may be informative, despite the intentions of Beijing's test-grading authorities.

Statistically speaking wider variation of scores tracks with lower mean scores in any test sample of similarly scored results. As a simple arithmetic proposition a high mean test score result can only occur if there is a low variation in results (the population must be consistently high performing), and conversely a wide distribution of result means that average score levels cannot be high. Possibly there is something to learn from the dispersion of *gaokao* scores, in and of itself?

We entertain this proposition by comparing "coefficients of variation" from a variety of tests, and tested populations, in our project.

"Coefficient of variation", as mentioned above, is the dispersion within a sample of observations (as measured by its Standard Deviation) divided by the mean of the sample. It is a measure that allows us to compare dispersion within sample—in this case, within samples of test scores—in an apples-to-apples basis, regardless of, for example, the set numerical range for possible scores for a given test.

Below are the COVs for the sample populations and the relevant tests examined in this paper, ranked from highest to lowest:

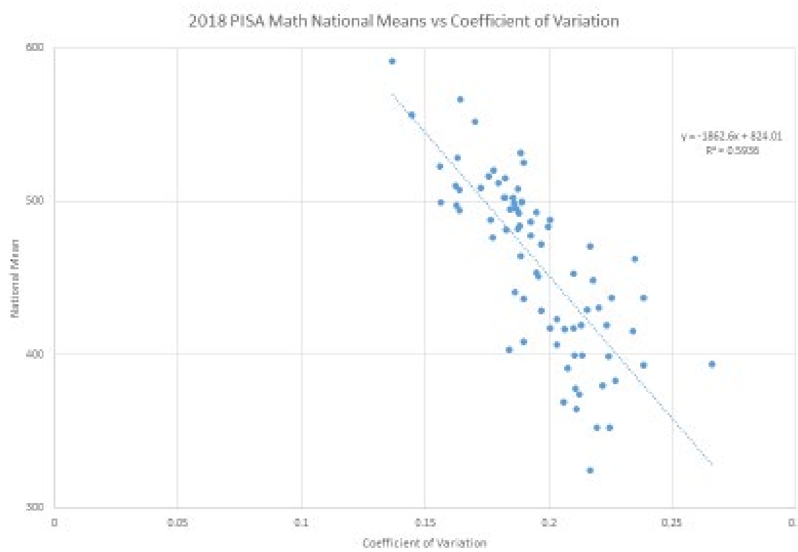
Slide 306

Country	Dataset	Coefficient of Variation
China	CEPS Math	40.6%
China	CFPS Math	39.3%
China	CEPS English	38.2%
China	CEPS Cognitive	37.6%
China	CFPS Reading	33.1%
China	Gaokao (overall), CFPS-Adjusted	26.8%
China	CEPS Chinese	24.9%
China	Gaokao (overall)	24.3%
Brazil	PISA Reading	24.2%
Brazil	PISA Math	22.7%
Brazil	PISA Science	22.5%
USA	PISA Reading	21.4%
China	PISA Reading, CFPS Adjusted w. inferred provinces	20.2%
USA	PISA Science	19.5%
Russian Federation	PISA Reading	19.4%
USA	PISA Math	19.2%
China	PISA Reading, CFPS Adjusted	18.6%
Russian Federation	PISA Science	17.7%
Russian Federation	PISA Math	17.6%
China	PISA Reading	15.7%
Vietnam	PISA Math	15.0%
Vietnam	PISA Reading	14.6%
Vietnam	PISA Science	13.9%
China	PISA Math, CFPS Adjusted w. inferred provinces	13.7%
China	PISA Math	13.7%
China	PISA Math, CFPS Adjusted	13.3%

We note that even with all the pains MOE authorities take to reduce the reported dispersion of scores for *gaokao*, the COV for overall *gaokao* 2021 results, for example, was 24.2%--nearly twice as high as the COV of 13.7% reported in PISA for math tests in “China”. The overall *gaokao* COV was higher than PISA COVs for Vietnam, Russia, and the United States. Perhaps tellingly, it was also higher than the measure for the dispersion of PISA test scores *for Brazil*.

Within PISA itself, there is a correlation between mean national test score levels on the one hand and COV on the other. We see this for example in PISA 2018 math results:

Slide 307



The correlation is not high—its R-Squared is just under 0.6—but it is strong enough to excuse a bit more exploration.

IF a population in PISA 2018 exhibited a COV of 24.3%, as the *gaokao* 2021 scores did, its estimated mean test level in the above figure would be **371.** As we recall, the mean overall score for “China” in PISA 2018 was **579.**

Yes—there are any number of good statistical reasons to call foul on this comparison. And we ourselves cannot stand by it. The calculation, however, is both revealing and informative—in that it is suggestive of how very different a true and faithful measure of “knowledge capital” for China might look from the profile presented for “China” in PISA 2018.

1-D-i) International *Gaokao* Usage by Non-PRC Universities and Testing Services

While the National College Entrance Exam (NCEE/*gaokao*) is the national test for college entrance in China, several subsets of colleges and universities outside of China have begun accepting *gaokao* scores for admission in recent years. Most notably, universities in Hong Kong, Singapore, and Australia all accept *gaokao* scores either in addition to, or instead of, alternative national entrance examinations. According to *The Economist,* as of 2018 several western universities accepted the *gaokao*, primarily located in Australia and Canada[74]. A handful of universities in the United Kingdom, France, Italy, and even the United States (including the University of Delaware, the University of New Hampshire, and the University of San Francisco) also accept *gaokao* scores for college admission.

While many western universities, particularly those in Australia, have become sympathetic to the *gaokao*, most remain skeptical. The chief issue is that there is no systematic/algorithmic way to evaluate *gaokao* scores. Instead, universities willing to accept *gaokao* scores must decide how to compare students across provinces, most setting an arbitrary "cutoff" for student performance in relation to their peers in order to qualify for admission. For example, The University of Western Australia sets cutoff targets for the 'percentage correct' in a given year. Because the *gaokao* tests are province-specific, the maximum possible score and even the difficulty of the test can vary greatly from province to province and from year to year, making direct comparisons impossible. As such, the University of Western Australia and others use an idiosyncratic approach to evaluate scores, assessing percentiles based on what they think seems to match their admission requirements.

To take another example, the University of Cambridge says the following about accepting *gaokao* scores:

> The Gaokao is regarded as suitable preparation for Cambridge, but competitive applicants studying for the Gaokao are typically able to demonstrate additional academic achievement. This might include, for example, relevant science Olympiads or College Board Advanced Placement Tests (APs).

[74] The Economist, "Some Western universities see merit in China's flawed exam", June 28, 2018

> Normally, any offer made to a successful applicant based upon achievement in the Gaokao will be conditional on achieving a rank within the top 0.1% of those taking the Gaokao in their province, but individual circumstances will be carefully considered by the admitting College when confirming an offer with particular attention paid to relevant subjects examined. It is therefore recommended that you contact the College you wish to apply to for further advice and guidance[75].

To summarize at a high level, most universities outside of China either discredit *gaokao* scores entirely, or accept them in an idiosyncratic way with a large grain of salt, typically supplementing with other measures of academic achievement. Though many see *gaokao* as a rough measure of student achievement in China, few (if any) see it as a methodologically rigorous measure of aptitude that makes for easy comparisons with traditional American metrics of achievement such as the SAT or ACT. A few recent studies have attempted to bridge the gap between *gaokao* scores and more traditional western measures of achievement, but none have been able to do so in a satisfactory way without access to proprietary data[76].

A conversation with a lead researcher at the US-based Educational Testing Service (ETS) revealed that little work is being done to compare *gaokao* scores to more traditional standardized testing scores for the exact reasons we have identified: Because *gaokao* tests, scores, and scales are all province-specific and kept private, it is just not possible to conduct the kind of research necessary to independently evaluate the validity of the test. The researcher we spoke to said that her team had looked at *gaokao* before, but they were unable to make any systematic comparisons because of a lack of access to data. She indicated that though *gaokao* scores used to be standardized, over the past 20 years the system has become "a mess", making international comparisons, and even inter-provincial comparisons, impossible. Should the CCP relax their control over data on *gaokao* scores and testing, and should the exam become standardized across China, perhaps fruitful

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[75] University of Cambridge, Undergraduate Study International Entry Requirements, https://www.undergraduate.study.cam.ac.uk/international-students/international-entry-requirements

[76] See in particular Farley, Alan, and Yang, Hong (2019) "Comparison of Chinese Gaokao and Western university undergraduate admission criteria: Australian ATAR as an example. Higher Education Research and Development". ISSN 0729-4360; and Hao Zhang, and Jie Wang "Effective Predictions of Gaokao Admission Scores for College Applications in Mainland China" September 2018, 1809.06362.pdf (arxiv.org)

research can be done in the future. But for the time being, *gaokao* is not an internationally respected test akin to the ACT or SAT because of policy choices inside China. Any universities outside China which decide to accept *gaokao* scores for admission must also decide whether they will accept scores from all provinces (as opposed to only the wealthy ones) and how to set cutoffs.

It is for these reasons that we can separate little signal from the noise in *gaokao* scores – perhaps comparisons within a given province in a given year are valid, but anything beyond that is mere speculation.

One additional point made by the researcher we spoke to at ETS was of interest: she indicated that several years ago she published a paper with a professor at Stanford and a few Chinese colleagues. The work was critical of the levels of education and skills and knowledge in China, and this got her Chinese colleagues into trouble with the CCP. As a result, they are no longer able to collaborate on issues like *gaokao* evaluation unless the Chinese government becomes explicitly less hostile to criticism and international input.

2. **Cheating**[77]

Any investigation of test performance in China that does not acknowledge the extraordinary role of *test-cheating*, *test-beating*, and *gaming the test system* in China, past and present, is missing a huge—and glaringly obvious—aspect of "meritocracy with Chinese characteristics".

Under China's absolutist rule, it is an interest of state to identify (and then employ and promote) top talent among the ordinary people. Academic tests have been an essential state vehicle for identifying such talent for so long that we may be tempted to describe this as an essential characteristic of the Chinese ethos. But China's "rule by law"—this is not "rule of law"—has incentivized efforts to "beat the test"—and have brought out creativity and remarkable ingenuity in doing so for basically as long as China's testing regimen has been in place. Thus the dialectic: the absolutist system that requires honest information on merit and talent from "little people" also encourages "little people" to gain whatever life-altering advantages they can in the face of "the test".

2-i) The Historical Tradition of Test Cheating in China

Cheating on tests would appear to be such a well-established and pervasive tradition in China that one is tempted to describe it as fused into Chinese civilization itself. Integral to and inseparable from China's long and hallowed tradition of merit-based examinations is the tradition of ingenious and widespread efforts to game tests for personal advantage.[78] No other country may have so perfected the art of test cheating, or so elevated its importance in social life.

Wide scale cheating on the Imperial examination for civil servants goes back centuries. In fact cheating was carefully organized and remarkably commercialized, with publishers regularly selling booklets summarizing the most common questions and answers from previous exams all around the country.[79] Tiny "cheat sheets" with tens of thousands of characters have survived from the

[77] This section of the paper refers to Slides 350-351 in the Online Appendix.
[78] Cf. Ichisada Miyazaki, *China's Examination Hell* (New Haven: Yale University Press, 1981), https://yalebooks.yale.edu/book/9780300026399/chinas-examination-hell/.
[79] Miyazaki, *China's Examination Hell*, p. 17

Ming and Qing dynasties.[80] And cheating also included bribing officials and evaluators, substituting "ringers" for ostensible candidates in the examination hall, and a whole panoply of ingenious tricks to beat the system. From at least the Song dynasty (960-1279 AD), which saw the advent of moveable type printing and thus something like mass publishing in China, Imperial examiners were necessarily preoccupied with preventing cheating, catching cheats, and assuring the integrity of the nation's merit-based examination system[81]--but the frequency with which Imperial edits prohibited publication of the exam-beating booklet attests to the enduring and inexhaustible demand for them.

Historically, cheating on tests was almost an exquisite art in China. Nowhere else in the world, it would appear, was such cheating so widely practiced--and appreciated—as in China. Widespread test cheating remains a fact of life in China today—practiced around the country by enterprising students on an individual basis, but also sometimes involving large rings involving teachers and higher officials. Cheating on tests regularly features in Chinese news reports—and we only read about the cheats who were caught.

Even when not technically engaging in illegal cheating on tests, "gaming the system" to perform better on tests has been worked out to a fine science in contemporary China, and appears to be at times not only tolerated but admired. China is far from the only country in which test cheating is widespread: India and Indonesia are also among the numerous national competitors for this distinction.[82] The point here is not that China is unique (although it may be) but rather that an inquiry drawing inferences about knowledge capital from exam results in China will be hopelessly naïve if it ignores the cheating factor.

[80] Sarah Zheng, "How students cheated in exams to get into China's imperial civil service," *South China Morning Post*, April 18, 2017, https://www.scmp.com/news/china/society/article/2088423/how-students-cheated-exams-get-chinas-imperial-civil-service.

[81] Command of the material for the imperial exam implied rote memorization of the classics, whose total contents exceeded 400,000 Chinese characters—though to the modern eye, such content may seem a questionable qualification for high office, such a test will have favored those with very high cognitive abilities!

[82] Yue-Yi Hwa, "Combatting Cheating: Ideas from India and Indonesia," Research on Improving Systems of Education (RISE) Programme, October 23, 2022, https://riseprogramme.org/blog/combatting-cheating-ideas-india-indonesia lists some papers studying test cheating in those countries.

2-ii) “Test-Beating”: the Implications for PISA Scores for Test-Takers from ‘China’

While the usual dialectic in test-cheating in China is for students to cheat and administrators to try to prevent cheating or to catch cheats, there are times when test administrators may also benefits from the inflating of score results by test-takers.[83] The PISA test is an example of an academic exercise in which officials in China appear to have been highly incentivized to “over-perform”—not to break the rules of PISA testing outright, but to bend them as far as possible without actually violating formal rules in the service of harvesting much higher scores than students would have earned under “normal” circumstances. We might call this “test-beating”, a practice distinct from “test-cheating”, insofar as “test-cheating” invites punishment if caught.

Anecdotal reports about local schools that bent the rules so that students might ‘do their best’ in the aforementioned PISA exam have been shared with the principal investigator, for example.

In the “small world department”, the principal investigator was told about a 15 year old girl in Shanghai, the daughter of the friend of a friend, who was one of the students who happened to be selected for testing by PISA.

As recounted: the school, aware of this upcoming event, spent weeks drilling the students and “teaching to the test”.

The classroom was even favored with a visit from Shanghai’s Deputy Mayor, who emphasized to the children how important this test was, and how diligently they needed to study for it; the official further told the children they should let their parents know that no distractions should be allowed, and no household chores imposed, that might interfere with their giving this exam their utmost.

Another reliable source, familiar with the mechanics of PISA testing in China, reported to the principal investigator that the only way China would concede to participate in PISA at all was (with or without the knowledge of PISA and OECD) by being allowed to selectively report scores from not only the *provinces* they desired – but also whichever *schools*, *classrooms*, and possibly even *students* they

[83] There is an obvious analogy here with performance reporting in a centrally planned economy, when local officials gain from misreported and overly favorable claims to the center about performance under their aegis.

wished. This is “selection bias” at its most extreme: the reported 2018 ‘all-China’ PISA results reportedly come not from a “random sample” within the most affluent provinces, but from a sample of students explicitly chosen to paint the country in the best possible light.

How much does this systematic “test-beating” intrigue affect reported PISA scores for China? One need not be a terribly clever detective to suspect that such machinations had much to do with the remarkable (impossible?) leap in reported test scores in “China” between 2015 and 2018—a phenomenon that we have already described in some skeptical detail.

In 2015, as we saw, mean PISA math scores for “China” were about 10 points lower than in Massachusetts, the top-achieving state in America. Very creditable to be sure—but not yet off the charts. Then, just three years later, “China” had soared over 60 points above Massachusetts! Even *if* we take “China” to just be affluent Shanghai in both years, this fact is astounding. How did they manage to do it?

Consider this American analogy. Say Massachusetts administrators were allowed to choose any schools from which to draw their “random sample” of test takers: Phillips Academy Andover; Groton; Boston Latin; Belmont High; and as many others as needed to fill the roster. Then say the administrators picked the most impressive home-rooms or classes of 15 year olds within each of these elite secondary schools – those with the most high-performing honors students. And then let’s say the teachers in these schools were allowed to “teach to the test” for PISA for weeks on end—at the expense of the regular curriculum. How much do we think this might change reported test scores for “Massachusetts”?

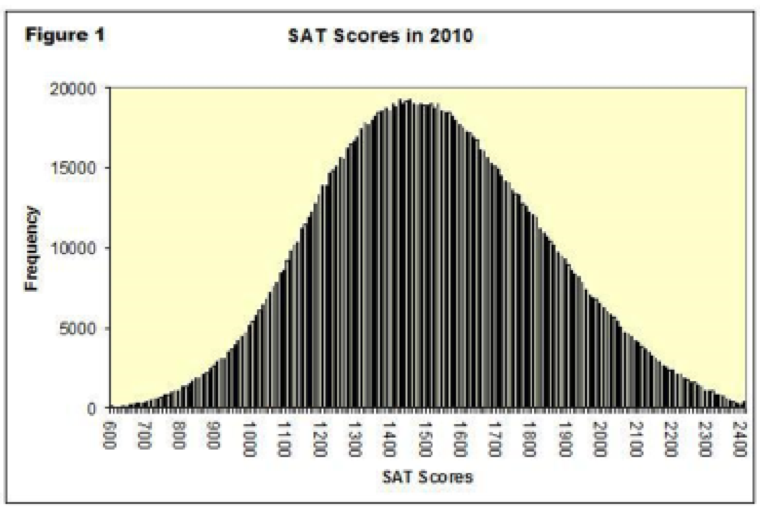
This is a question we cannot answer in precise quantitative terms. However, there are quantitative clues and hints that can help us appreciate the general magnitude of the answer.

Consider in particular the everyday realities of the “test prep” industry in the United States—the tutors who help rising high school juniors and seniors earn better scores in the SAT tests for college admission. The SAT is administered by the Educational Testing Service (ETS), a gold-standard organization with the highest reputation for both academic rigor and academic integrity. Test-prep tutors do not know what will be on the upcoming rounds of SAT tests—but they have mastered the art of SAT test-taking, and they earn their pay from students because they can *reliably raise a tutees' score.*

The Kaplan Test Prep company, for example, reportedly promises that it can raise a paying tutees' total SAT score by 160 points.[84] Since the SAT 3-part test has a combined range of possible scores spanning a scale from 600 to 2400, such a promise would be highly meaningful. Although the actual distribution of SAT scores changes somewhat from year to year, in one recent year (2016), the Standard Deviation was reportedly 117 points for Critical Reading, 121 points for Math, and 115 points for Writing.[85]

The figure below may help the reader get a visual sense of what a jump of 160 point in SAT score might mean to an aspiring collegiate.

Slide 350



Source: probability - What is the best way to interpolate over the 25th and 75th percentile of SAT scores? - Mathematics Stack Exchange

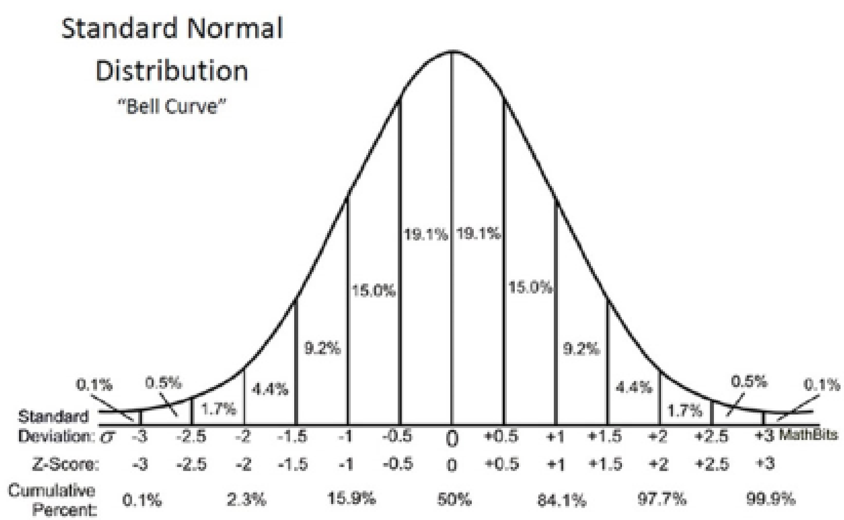
Statistically speaking, a 160 point increase in combined SAT scores in 2016 would have amounted to an increase in test performance of almost half a Standard

[84] Evelyn Jones, "Kaplan SAT Prep Review 2023: Is It Worth It?" July 19, 2023, https://2400expert.com/kaplan-sat-prep-review/

[85] CollegeBoard "2016 College-Bound Seniors Total Group Profile Report", Table 1: Overall Mean Scores, https://reports.collegeboard.org/media/pdf/2016-total-group-sat-suite-assessments-annual-report.pdf

Deviation (0.45 S.D.). In a normally distributed "bell curve" of results, half a Standard Distribution is enough to take one up from the 50th percentile to about the 70th percentile; or from about the 85th percentile to about the 93rd percentile; or from about the 93rd percentile to roughly the 98th percentile. It constitutes a very considerable advantage—which is why so many students and their families are willing to pay for this service.

Slide 351



Source: https://mathbitsnotebook.com/Algebra2/Statistics/STstandardNormalDistribution.html

Given that PISA is scaled for a Standard Deviation of 100 points, the Kaplan Test Prep would appear to be worth the equivalent of almost 50 PISA points.

In other words, "test prep" by itself might account for almost 50 of the roughly 60 point difference in PISA scores between Massachusetts 2015 and "China" 2018.

If we were to select special top-performing schools with which to represent the Massachusetts "random sample"—and then look for the best home rooms within this "universe" of testable students, we could no doubt manufacture further increases in reported scores for Massachusetts.

All of this is to indicate that "test-beating" is all we need to presume to account for the stellar PISA scores emanating from "China". It could well be that, absent test-beating, China's students from its most privileged provinces might perform capably, even creditably, against counterparts from OECD countries. They might even match, or for that matter outperform, places like Massachusetts; certainly this would seem to be at the very least possible for PISA's "China"-tested cities, Beijing and Shanghai. At the moment, unfortunately, there is no way for us to correct statistically for pervasive "test-beating" in these PISA 2018 results for "China".

2-iii) Human Capital Attainment, Education and Exam Cheating In China

It is worth attempting to place the phenomenon of exam cheating in China in a greater context of "knowledge capital" augmentation, given realities facing China's pupils and students.

For this broader assessment, we asked our colleague Mr. Radek Sabatka, a recent master's graduate from the London School of Economics, to help us better understand these realities given his own direct experience in education in China. The following is the memo he prepared on these questions for this project:

> [This] is a reflection on factors that ought to be considered while examining higher education and major bottlenecks in human capital attainment in the People's Republic of China (China). As China is undergoing a transformation of its economic system from an investment-driven economy to an economy based on higher-value jobs and services, what skills and abilities are needed in such a new economy? Any consideration of human capital attainment in China cannot be disconnected from a conceptualization of what type of economy and country China will be in the upcoming decades. While it is challenging to design an ideal framework for human capital attainment through education, certain criteria must be upheld in order to equip students with the best possible outcomes for a country's economy. Human Capital no longer entails an individual's ability to perform well in academic assignments, but more importantly, to be an individual capable of navigating modern economies and global trends while staying well both in terms of mental and physical health.

1) Exam Cheating

Since the Chinese education system leans heavily towards test-taking and would not be an exaggeration to state that the college entrance examination (*gaokao*) is an apogee in Chinese people's lives, it would be natural to expect a widespread tendency to contrive the exams and cheat. Overall, given the importance of *gaokao* exams for Chinese society, there is a growing importance for the CCP to eliminate cheating. However, it would be quite fallacious to conclude, simply on the basis of the CCP's interest in preventing such cheating, that the Chinese public trust in *gaokao* remains firm—or that the Chinese government made it near impossible to cheat during exams in China either through 'anti-corruption' campaign or any other means.

Cheating during *gaokao* exams has a long history in China given the broader social and economic importance of achieving good results. In 2017, the South China Morning Post reported about a small book on public display in Changsha; "the book is about the size of a pack of cigarettes and its characters are the size of grains of rice."[86] The book contains Chinese classic texts such as those of Confucius and other thinkers, and it was used by students during Civil service examinations in imperial China for over 1300 years. These books are to be found all across China. Chinese students have always tried to improve their scores even through the deployment of illegal means.

There is a widespread tendency to cheat on *gaokao* exams in China. As of June 2022, Chinese authorities investigated over 129 alleged cheating on exams and around 200 suspects were caught in one province only. Also, quite controversially, the Chinese National Education Examinations Authority released a statement saying that "no test papers were leaked before this year's national college entrance exam" in 2022.[87] What one could reasonably infer from this statement is that it is a common practice that exam papers, or a few questions at least, get leaked before the actual exam.

[86] Zheng, Sarah. "How Students Cheated in China's Imperial Civil Service Exams." *South China Morning Post*, April 18, 2017. https://www.scmp.com/news/china/society/article/2088423/how-students-cheated-exams-get-chinas-imperial-civil-service.

[87] Yang, Zekun, and Xinying Zhao. "No Papers Leaked before College Entrance Exams." China Daily, June 9, 2022. https://www.chinadaily.com.cn/a/202206/09/WS62a14f0aa310fd2b29e6199d.html.

Cheating at college entrance exams, the so-called *gaokao*, seems to be so prevalent that Chinese authorities had to adopt strict measures and punishments. For those organizing cheating, the Chinese state raises to seven years of imprisonment as well as for students who resort to cheating.[88] Some high schools even banned girls from wearing bras that included metal clasps for suspicions that they included devices helping them to cheat.[89]

a) Technology

Due to technological advancements in the 21st century, it is perhaps easier than ever to resort to cheating during *gaokao*. And it would be reasonable to state that the market for cheating equipment is booming in China; in a way, it is a never-ending cycle between a cat and a mouse: does the Chinese state dispose of enough enforcement power to stop creative ways of using technology to cheat? When one looks up copying robots (a device that can mimic handwriting) on Taobao (the Chinese version of Amazon), one finds plentiful offerings ranging from as low as 200 yuan to more than 1000 yuan.[90]

And Beijing is aware of these practices as Chinese authorities have been tightening their anti-cheating measures each year. Chinese officials have been trying to curb exam cheating using sophisticated technological means such as drones and radio surveillance trucks.[91] In 2018, the Chinese government launched a campaign to curb cheating across five provinces (Liaoning, Shandong, Hubei, Guangdong, and Sichuan), resulting in the discovery of at least 12 groups that facilitated cheating and seized equipment over hundreds

[88] Bearak, Max. "In China, Cheating on an Exam Will Get Students Detention — in Prison." Washington Post, December 1, 2021. https://www.washingtonpost.com/news/worldviews/wp/2016/06/09/in-china-cheating-on-an-exam-will-get-students-detention-in-prison/.

[89] Press Trust of India. "Ban on Bra for Exams in China Draws Flack." Business Standard India, June 9, 2013. https://www.business-standard.com/article/pti-stories/ban-on-bra-for-exams-in-china-draws-flack-113060900302_1.html.

[90] Zhang, Phoebe. "Chinese Schoolgirl Shamed for Using Robot to Write Homework. Now Everybody Wants One." South China Morning Post, February 19, 2019. https://www.scmp.com/news/china/society/article/2186780/chinese-schoolgirl-shamed-using-robot-write-homework-now.

[91] Associated Press in Beijing. "China Deploys Drones to Stamp out Cheating in College Entrance Exams." The Guardian, June 8, 2015, sec. World news. https://www.theguardian.com/world/2015/jun/08/china-deploys-drones-stamp-out-cheating-college-entrance-exams.

of millions of yuan.[92] It is not only advancements in hardware, but also in increasing the hacking capacity of Chinese individuals. Numerous groups claim to provide that year's exam questions for large sums of money; it would be reasonable to believe that the Chinese elites have access to those either by having the ability to bear such costs or through other means.

b) Hiring Others to Take the Test

This is perhaps where the Chinese authorities were successful in the elimination of cheating by resorting to advanced technology. One of the ways through which one could cheat on *gaokao* Exams in the past has been hiring a 'test-taker' who would sit and write the exam on behalf of a student. For instance, in 2014, the Higher Education Admission Office of the central Henan province reported at least 127 test-takers who were hired to sit the *gaokao* exam instead of the right student.[93] Cheating groups have found ways to obtain forged ids as to prove their identity of being the student. However, these practices have become practically impossible as the Chinese state is deploying facial recognition systems to identify the test-takers[94]; and in Inner Mongolia, education authorities adopted a finger vein recognition system instead of the traditional methods of fingerprint detection since Chinese test-taker have found a way to bypass the archaic technology[95].

But it is not only cheating on the Chinese mainland, Chinese cheating practices have also affected international exams where students pay for test takers to take their GRE, GMAT, or SAT (the Atlantic). For instance, the chairwoman of the GRE board commented that in 2002, average GRE scores rose significantly in China, Korea, and other 2 countries following the public's access to current and past GRE scores online. Some professors at US universities even noted that it is a common practice that they get a few students

[92] Shaikh, Zeeshan. "Chinese Police Seize One Lakh Wireless Devices in Crackdown against Cheaters in Gaokao Exams." The Indian Express (blog), June 7, 2018. https://indianexpress.com/article/world/chinese-police-seize-one-lakh-wireless-devices-in-crackdown-against-cheaters-in-gaokao-exams-5207561/.
[93] Press Trust of India. "China Says 127 Students Hired Others for Exam." The Indian Express (blog), June 18, 2014. https://indianexpress.com/article/world/asia/china-says-127-students-hired-others-for-exam/.
[94] Reuters Staff. "Chinese Exam Authorities Use Facial Recognition, Drones to Catch Cheats." Reuters, June 8, 2017, sec. World News. https://www.reuters.com/article/uk-china-exam-idUKKBN18Z14G.
[95] Zuo, Mandy. "How China's Innovative Cheats Tap Hi-Tech to Beat Exams." South China Morning Post, June 7, 2018. https://www.scmp.com/news/china/society/article/2149613/fake-fingerprints-and-electronic-erasers-how-chinas-innovative.

each year who scored high scores on GRE verbal section but spoke very little English during actual classes (AP). Based on my experience in China, it is common to hear that students have the option to cheat on graduate exams for US universities – either by hiring a test taker or obtaining the questions beforehand.

2) Laying Flat (tang ping 躺平)

"Lie Flat" and "Let It Rot" are both terms that became popular in China, predominantly among the urban youth who feel a sense of disappointment, burnout, and discontent with school, work, and other social developments, caused by a lack of social mobility and increased uncertainty in today's China. The term 'lie down' implies a reactionary movement of passive response by not participating in rigorous academic work and the labor market (traditional values highly regarded by the generations of their parents) since any action by the youth is perceived as meaningless. The movement also derives from the widely believed increasing income and wealth inequality, which needless to say have been on a decline since 2012. It is estimated that more than 15 million Gen Z youth are jobless and lowering their employment expectation way beyond their attained education and professional experience,[96] and the proportion of unemployed Chinese youth is predicted to rise to 19.3% in 2023 and beyond.[97]

The importance of the laying-down movement is underscored by the reaction of the Chinese government to censor any slogans and social media content relating to the movement. The 'Lie Down' movement is among the topmost-censored words on the Chinese internet in 2022, among words like 'Support Xinjiang People'.[98] The Chinese government realizes that the "lie down' movement has the potential to grow into wider social unrest at the grass root

[96] Ngai, Catherine. "China's Gen Z Is Dejected, Underemployed and Slowing the Economy." Taipei Times, July 28, 2022. https://www.taipeitimes.com/News/feat/archives/2022/07/28/2003782548.
[97] Lau, Yvonne. "China's Gen Z and Millennials Have a Word for Their Disaffection with the Economy and Life in General. Evolution Is Dead, Meet 'Involution.'" Fortune, July 31, 2022. https://fortune.com/2022/07/31/china-gen-z-millennials-unemployment-jobs-college-graduates-white-collar-education/.
[98] Beach, Sophie. "Minitrue: Remove Products Touting 'Lying Down' and 'Involution.'" China Digital Times (CDT), June 25, 2021. http://chinadigitaltimes.net/2021/06/minitrue-remove-products-touting-lying-down-and-involution/.

level and possibly challenge the regime's benefit to the country.[99] And it is even exacerbated by the Chinese government's persistence in sticking to the draconian measures adopted under the 'dynamic zero Covid' policies, where uncertainty defines people's everyday lives; finding a job in big cities like Shanghai is nearly impossible, and living costs and rents are skyrocketing, there is no other way than to give up. If the Chinese youth finds themselves increasingly disconnected from the Chinese economy and underemployed, then the "Lie Flat" movement might be one the most significant pieces of evidence of Chinese slowing economy.

Factors essential in considering Human Capital Attainment in China

1) Individual Critical Thinkers

Critical Thinking skills are essential in preparing students for a modern workforce where constant changes are expected. Given the test-oriented Chinese education system, which focuses on respecting authority and a collective mindset, it would be interesting to look at the metrics/scores (if there are any) of how Chinese students perform on critical thinking assignments – although I am aware of how complicated it is to define the term critical thinking. Chinese citizens are indeed aware of the failures of their educational system in achieving so. In 2022, a verified blogger on Chinese Twitter Weibo wrote that "I think that the best education, or the best higher education in the world, should be to create individuals with personalities and pay attention to the cultivation of individual thoughts. But our domestic education does not pay attention to the cultivation of individual thoughts and only seeks a correct answer."

2) Formalisms

Xi Jinping is only the second Chinese leader (after Mao) whose thought has been made as part of the country's constitution. The logical conclusion is that Xi's thoughts must be taught at all levels of Chinese education. But Xi's thought goes well beyond a simple political indoctrination, it is in fact, a comprehensive set of ideals and 'facts' aimed at politics, economics, foreign

[99] Yu, Jianrong. "China's 'Angry Youth' May Fuel Social Unrest, Researcher Writes" Bloomberg News. Accessed November 10, 2022

relations, education, family life, and so on. The members of CCP even have a phone app called 'Xuexi Qiangguo', where they have to complete tasks each day about the thought of Xi Jinping and get reward points.

But even though Xi's thought has made its way to every level of the Chinese education system, the mandatory sessions about Marxism, the Chinese nation, and Xi's thought are the sole exercise of formalisms, at least in urban centers. I attended a few in Shanghai, where all students were constantly on their phones and social media, knowing that the 3-hour workshop is a necessity but nothing to be remembered. At the end of the session, all students (about a hundred) were asked to submit an essay on the topic related to Xi's thoughts. In practice, only a single student writes the essay, and others copy-paste and submit the same one. Those teaching the course did not seem to mind. And this is the *formalism* present in all socialist countries where things are done only to satisfy a certain requirement, but it is well-known what the nature and meaning of such assignments are.

4) Oversupply

The expansion of Chinese university enrollments has contributed to the decline in educational quality. In 1999, China began the process of drastically expanding enrollment in Chinese colleges. Since then, their gross enrollment rate increased from less than 10 percent to over 50 percent, achieving what Martin Trow calls the transformation from elite to universal higher education. Unlike Western countries that underwent this process, however, China did not achieve this transformation from an incremental market-oriented approach, but from a distinctly government-led approach. From the enrollment rate of 7 percent to 50 percent, the United States took seventy years. China only took twenty-two (Zhou 5). The achievement significantly improved Chinese economic performance but, at the same time, contributed to the decline of educational quality. First, the expansion of enrollment led many Chinese students to go to college just for the sake of a diploma, not to enhance their knowledge, thus decreasing the quality of student bodies. Second, the enrollment expansion meant that resources for students became insufficient. Among the top 100 universities in the world in terms of faculty-student ratio, according to Times Higher Education, only three are Chinese. Third, the expansion of enrollment calls for an increase in the number of universities, which added bad-quality institutions into the mix. In order to get thousands of new institutions, even many specialized secondary schools suddenly became colleges without much sophistication in academics (Wen HSWH.org).

6) Private Tutoring

As a major step to avoid the poor quality of public schools and increase chances of their kids at being accepted into one of the top national universities, or perhaps even abroad, Chinese middle- and higher-income families spend a considerable amount of money on private tutoring right up to 2021, when private tutoring was officially banned by the Chinese government. Seven months after the introduction of the ban on private tutoring, offline primary and secondary school tutoring institutions in China dropped from over 124,000 to 9,728, while 87 percent of those online completely shut down.[100] The question is, of course, whether these institutions now operate in a grey zone as the demand for private tutoring could not have been eliminated, and my evidence suggests that it is so.

However, it has been argued that private tutoring, especially in cases where teachers employed at public institutions demanded money for additional training, served as a major force in the unequal distribution of educational outcomes. In certain instances, public teachers purposely lowered their teaching quality to incentivize students to pay for their private lessons. Chinese, especially in middle to higher-income families, are determined to give their children the best education possible, and accessing private tutoring is essential in giving their children a comparative advantage when applying to top Chinese universities and abroad.

7) Apps and Social Media

This is perhaps one of the spaces where Chinese society is doing well. When the government fails, the Chinese people stood up through the socialization of reskilling. Social media apps have become a common way to share industry

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[100] Zou, Shuo. "Number of Tutoring Institutions Plummets." China Daily, March 1, 2022. https://global.chinadaily.com.cn/a/202203/01/WS621d6b5da310cdd39bc8968e.html.

knowledge and train incoming employees for their roles. Platforms such as Douyin (TikTok) and Weibo provide community-level training where companies and the state fails. The same could be said of higher education; TikTok serves as a learning platform in the same way as YouTube does outside of China.

8) Overeducation

Especially in the top-tier cities of Beijing, Shanghai, Chongqing, and Tianjin, children face immense pressure to achieve near-perfect academic results. I recall one instance during my tutoring work in a small English library in Shanghai. During one interview with one of the students on Sunday, I asked him what his weekend was like and what he did during his *free time*. His answer shocked me for his reply did not consist of a list of activities, but he asked me what free time meant. Indeed, there might have been a language barrier at play, but his reaction illustrates a common pattern among Chinese children from middle-income families: they are extremely overworked. A typical schedule consists of waking up before 7 am and going to bed around 11 pm. Every hour of each day is filled with various activities, and when there is no school on the weekend, Chinese children usually catch up on homework. Indeed, Chinese 'Tiger Mothers' have improved their micro-management of their children – to the extent that parenting apps and technology market accounted for billions in 2019.

9) Mental Health

This is a factor yet to be explored. Although studies show that the Chinese education systems have steadily improved the provision of high-quality counseling, there are still reasons to consider the mental health of Chinese youth as a hindering factor in human capital attainment in China. Suicides are vastly common due to the academic pressure and low quality of living conditions in Chinese dormitories (usually 6 students per room, electricity turns off at 10 pm every day forcing students to rely on power banks to finish their assignments).

Summary

There are indeed many other factors that determine human capital attainment in China and there are many underlying questions to be looked at such as the impacts of those studying abroad and then returning to China. Nonetheless, the main lesson seems obvious, the educational system in the People's Republic of China faces many challenges in raising a new generation of well-educated citizens.

3. Cognitive Ability and "Knowledge Capital" for China's Children and Youth[101]

Research on "psychometrics" and its implications is in bad odor in much of the academy, and in many policy circles as well. This is not a new trend—severe criticism of cognitive testing and research has been a feature of US academic discourse for at least two generations—but the intensity of the emotion and vitriol on the topic is arguably new. Speakers on this general topic now have to be prepared for the possibility that they will be shouted down or canceled on US campuses. Not so long ago AEI's Dr. Charles Murray, the scholar known for his bestselling "Bell Curve" research, was shut down and swarmed by a menacing masked mob at Middlebury College, and the professor who invited him there to speak was sent to the hospital. The chilling effect of such episodes is apparent, and of course deliberate

Why the often furious controversy over "psychometrics" and its implications? The reasons passions are inflamed may mainly be ideological. Marxism reaches widely and deeply on campus nowadays—further than many students or professors realize themselves. To update Keynes' aphorism, intellectuals who believe themselves to be freethinkers are usually slaves of some defunct economist—an economist by name of Karl Marx. Even if they have never cracked *The Communist Manifesto* or *Das Kapital*, they have been infected with the now omnipresent and ethereal Marxian notion that human beings are inherently mutable, and their given characteristics can be changed under the influence of a new environment (or a New

[101] This section of the paper refers to Slides 353-358 in the Online Appendix.

World). To suggest otherwise—as cognitive testing tends to do—is quintessentially politically incorrect.

Furthermore, in suggesting there are measurable cognitive differences between different population groups, and that these can include differences by nationality and ethnicity, psychometrics opens itself up to automatic accusations of "racism"—an intellectual third rail in academic and policy circles today. It does not help matters here, we should emphasize, that both psychometrics and anthropometrics *were* put to some pretty ugly uses in the now-distant past: including what now look like obviously pseudo-scientific claims in the international "eugenics" movement in the late 19th and early 20th Centuries.

Apart from such deeply emotional—but intellectually superficial—reservations, there are more substantive questions and objections that can, and are, lodged with psychometrics *per se*, as well as with the inferences that are drawn from it for social science research. A brief summary of these would include the following: 1) measured results on cognitive potential for given individuals can change over time; 2) measurement of cognitive potential for given population groups can change over time; 3) cognitive measurement perforce overlooks personal traits, attitudes, habits and motivations that have direct and often considerable impact on individual or group performance; 4) social science researchers may draw unwarranted inferences from cognitive data.

We recognize and accept these substantive research caveats concerning psychometric data. But we also believe this field of research offers genuine promise for evaluation of "knowledge capital" in China—valuable insights that would be sacrificed if we were to ignore this body of work out of ideology—or current fashion.

A century and more of cognitive testing has established that there is a deep and powerful statistical correspondence between test-takers' cognitive measures (such as IQ) and their measured achievement in standardized academic tests. Measured cognitive ability, indeed, is arguably a uniquely powerful predictor of measured academic achievement—more powerful than social or family background, socioeconomic status, or other important factors in explaining scholastic results. Evidence on the cognitive potential of China's pre-college age population—infants, children and youths—consequently appears to be central to any overall assessment of the terrain of knowledge and skills in China, and indispensable to evaluating reported claims of test scores, including internationally standardized test scores, for China's pupils.

3-i) Cognitive Measures, "Knowledge Capital", and National Economic Performance

To date relatively little attention has been devoted to any possible relationship between cognitive ability and knowledge capital—a lacuna perhaps largely due to the fact that Hanushek et al.'s formal conceptualization of "knowledge capital" and its influence on economic performance is still less than a decade old. Before this research, economists and social scientists had done at least some work on the possible correspondence between measured cognitive ability and national economic performance. Lynn and Vanhanen's 2002 treatise *IQ and the Wealth of Nations*[102] established the strong and in statistical terms[103] significant relationship between available data on measured cognitive ability by country and contemporary levels of measured national economic performance (using such metrics as per capita income or GDP). On its face, it seems reasonable to conjecture that there might be a positive relationship between those two quantities, if only because greater cognitive ability could make retention of knowledge easier and acquisition of skills quicker and less onerous. But as economists will surely observe, neither of those authors are economists: Lynn is a psychologist and Vanhanen is a political scientist.

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[102] Richard Lynn and Tau Vanhanen, "IQ and the Wealth of Nations", Praeger, February 2002, https://www.amazon.com/IQ-Wealth-Nations-Richard-Lynn/dp/027597510X/ref=sr_1_1?crid=3GRM8L5MDPUXC&keywords=iq+and+the+wealth+of+nations+vanhanen&qid=1706028866&sprefix=iq+and+the+wealth+of+nations+vanhanen%2Caps%2C64&sr=8-1

[103] "Statistically significant" is a term of art often misinterpreted by non-statisticians (and sometimes even by some from within the guild). It does *not* mean "important", much less "a big deal we should all pay attention to". It simply means "an observed association which is unlikely to have occurred simply on the basis of chance". That is why statistical significance measures offer probability calculations: i.e. the odds that results observed might have been completely random.

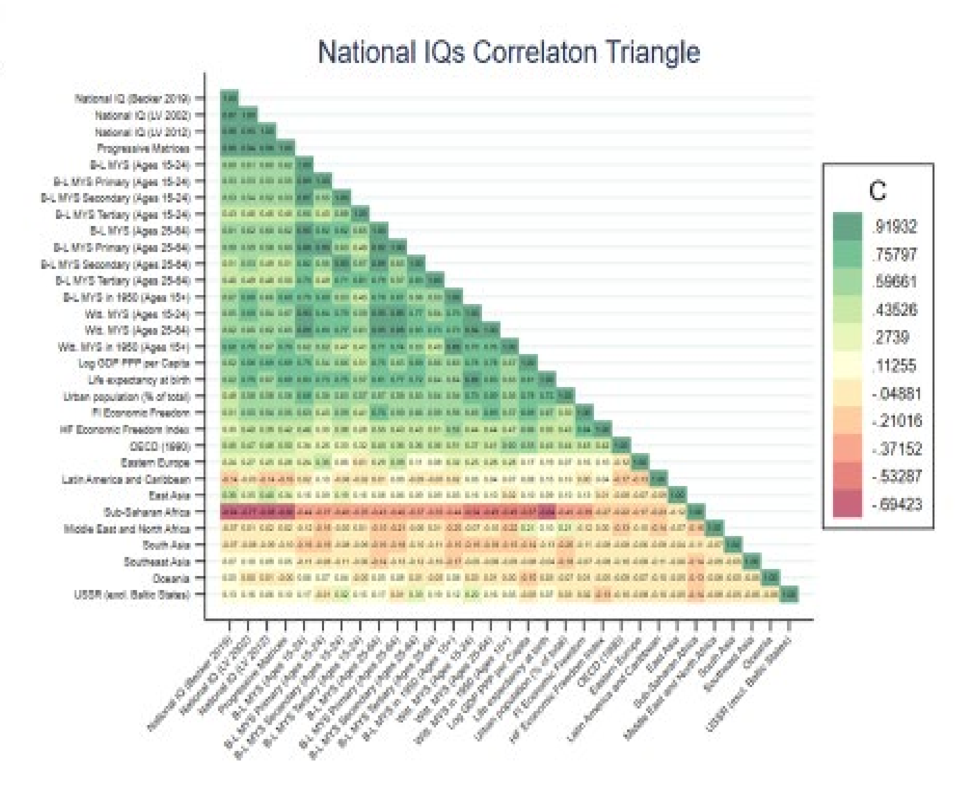
One economist who has dared to tread into the territory of IQ and national economic performance is Garrett Jones (of George Mason University) with his study *Hive Mind: How Your Nation's IQ Matters So Much More Than Your Own*[104]. Jones also accepts the data suggesting a positive correspondence between a country's measured cognitive ability and its level of national economic performance as genuine and meaningful, rather than an artefact of statistics. His explanation for the strong relationship focuses on the collective, not the individual, potentialities for higher average levels of cognitive ability in large populations. Jones argues that national economic performance depends crucially on often complex processes of cooperation: among both individuals and specialized groups. Consequently, in his telling, populations with higher overall levels of cognitive ability are better able to explore economic cooperation's potentialities, and to grasp them.

In a previous phase of this project, our own statistical analysis pointed to a relationship at the national level between cognitive ability (as measured by IQ testing) and tested levels of both academic achievement (a proxy for "knowledge capital") and national economic performance. We see this in the 'correlation triangle' below, utilizing four separate datasets that present reported data on national IQ levels.

[104] Garett Jones, "Hive Mind: How Your Nation's IQ Matters So Much More Than Your Own", Stanford University Press, November 2015, https://www.amazon.com/Hive-Mind-Your-Nations-Matters/dp/150360067X/ref=sr_1_1?crid=26S7IAAX4JOX0&keywords=hive+mind+garrett&qid=1706029313&sprefix=hive+mind+garrett%2Caps%2C67&sr=8-1

Slide 353

Figure 10b.



Our simple statistical analysis demonstrates that information on IQ adds no additional “predictive power” to models attempting to estimate international economic performance at the national level if standardized student achievement scores are already in the mix, and that the addition of IQ into these models results in a “stepwise” variable with no meaningful weight or reliability of its own. This is what one would expect to find with two indicators as closely correlated as IQ and student achievement scores—it is a problem known to statisticians as “collinearity”.

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Table 6.

Secondary Math Score, MYS Witt (Raven's Progressive Matrices estimates)

	(1)	(2)	(3)	(4)	(5)	(6)
Raven's Progressive Matrices estimates'	7.221***	7.225***	7.287***	7.249***	6.651***	6.651***
	(21.60)	(20.72)	(20.57)	(20.78)	(17.99)	(18.14)
Witt. MYS (Ages 25-64)	0.426	-0.0399	-0.613	-0.782	-2.938*	-2.657*
	(0.39)	(-0.04)	(-0.54)	(-0.71)	(-2.40)	(-2.30)
Life expectancy at birth, total (years)	-0.0923	-0.156	-0.197	-0.250	1.492**	1.421**
	(-0.27)	(-0.41)	(-0.51)	(-0.65)	(2.80)	(2.75)
Urban population (% of total population)	-0.112	-0.167	-0.200*	-0.180*	-0.0479	-0.0497
	(-1.30)	(-1.89)	(-2.31)	(-2.08)	(-0.40)	(-0.50)
Natural log of GDP PPP per capita-2017)		2.280	1.219	1.526	1.288	1.922
		(0.71)	(0.36)	(0.46)	(0.35)	(0.55)
Economic Freedom Index			0.298		0.284	
			(1.66)		(1.52)	
Economic Freedom of the World Index (o-)				4.252		2.518
				(1.92)		(1.23)
Eastern Europe					6.478	5.874
					(1.60)	(1.40)
Latin America and Caribbean					-14.66**	-12.97**
					(-3.02)	(-2.65)
East Asia					26.38***	25.81***
					(4.40)	(4.75)
Sub-Saharan Africa					33.54*	34.42*
					(2.49)	(2.57)
Middle East and North Africa					-12.31**	-10.83*
					(-2.71)	(-2.32)
South Asia					23.95**	25.41***
					(3.29)	(3.43)
Southeast Asia					-2.041	-0.680
					(-0.31)	(-0.10)
Oceania					0	0
					(.)	(.)
USSR (excl. Baltic States)					21.32**	21.13**
					(2.69)	(2.63)
Constant	-184.9***	-194.5***	-197.5***	-204.0***	-252.0***	-258.2***
	(-7.10)	(-7.48)	(-7.67)	(-8.12)	(-5.41)	(-5.46)
Observations	447	443	442	443	442	443
R-Squared	0.868	0.868	0.869	0.869	0.898	0.898

t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001

But we also found (perhaps not surprisingly given the previous result) that reported national IQ was an extremely powerful predictor of academic performance as measured by international standardized testing in the World Bank HLO database.

Slide 355

Table 6.

Secondary Math Score, MYS Witt (Raven's Progressive Matrices estimates)

	(1)	(2)	(3)	(4)	(5)	(6)
Raven's Progressive Matrices estimates'	7.221***	7.225***	7.287***	7.249***	6.651***	6.651***
	(21.60)	(20.72)	(20.57)	(20.78)	(17.99)	(18.14)
Witt. MYS (Ages 25-64)	0.426	-0.0399	-0.613	-0.782	-2.938*	-2.657*
	(0.39)	(-0.04)	(-0.54)	(-0.71)	(-2.40)	(-2.30)
Life expectancy at birth, total (years)	-0.0923	-0.156	-0.197	-0.250	1.492**	1.421**
	(-0.27)	(-0.41)	(-0.51)	(-0.65)	(2.80)	(2.75)
Urban population (% of total population)	-0.112	-0.167	-0.200*	-0.180*	-0.0479	-0.0497
	(-1.30)	(-1.89)	(-2.31)	(-2.08)	(-0.40)	(-0.58)
Natural log of GDP PPP per capita-2017)		2.280	1.219	1.526	1.266	1.922
		(0.71)	(0.36)	(0.46)	(0.35)	(0.55)
Economic Freedom Index			0.298		0.264	
			(1.66)		(1.52)	
Economic Freedom of the World Index (p-)				4.252		2.518
				(1.92)		(1.23)
Eastern Europe					6.678	5.874
					(1.60)	(1.48)
Latin America and Caribbean					-14.66**	-12.97**
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					(4.60)	(4.78)
Sub-Saharan Africa					33.54*	34.42*
					(2.49)	(2.57)
Middle East and North Africa					-12.31**	-10.83*
					(-2.71)	(-2.32)
South Asia					23.95**	25.41***
					(3.29)	(3.43)
Southeast Asia					-1.041	-0.680
					(-0.31)	(-0.18)
Oceania					0	0
					(.)	(.)
USSR (excl. Baltic States)					21.32**	21.13**
					(2.69)	(2.63)
Constant	-184.9***	-194.5***	-197.5***	-204.0***	-252.0***	-258.2***
	(-7.10)	(-7.48)	(-7.67)	(-8.12)	(-6.41)	(-6.46)
Observations	447	443	442	443	442	443
R-Squared	0.868	0.868	0.869	0.869	0.890	0.890

t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001

Our simple models showed that developmental indicators *plus* geographical dummy variables for regions (indicators we took to reflect "culture of learning" for East Asia, Latin America, etc.) *plus* IQ variables would "predict" (or to a statistician: "correspond with") up to 90 percent of inter-country variations in reported test performance for pupils tested.

And since our simple models also suggested that "knowledge capital" as proxied by reported scores on internationally standardized tests has a strong association with per capita GDP *a decade in the future*[105] (per the table below—testing on math scores for pupils from all around the world in the HLO dataset), this strongly suggests an association between IQ and national economic performance from our own findings. We emphasize that our statistical analysis is only simple econometrics: more sophisticated forays could illuminate additional facets of the interplay here. But it is clear enough from this work that IQ and measured test scores track very closely—and that measured test scores track very strongly with future economic performance.

[105] Strictly speaking, this is a 'lagged variable'—we are showing how test scores etc. from ten years ago track with economic performance today.

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Table 3.

lnGDP/CAP = Math Primary scores * MYS BarroLee

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
HLO Primary School Math Mean Score # ~	0.000412***	0.000197***	0.000195***	0.000188***	0.000174***	0.000118***	0.000125***
	(12.85)	(6.36)	(8.84)	(7.91)	(7.86)	(4.63)	(4.87)
Life expectancy at birth, total (years)		0.0657***	0.0269***	0.0242***	0.0259***	0.0286***	0.0309***
		(10.00)	(4.17)	(3.69)	(4.21)	(4.42)	(5.01)
Urban population (% of total population)			0.0236***	0.0237***	0.0218***	0.0269***	0.0248***
			(7.65)	(7.62)	(6.97)	(9.77)	(8.90)
FI Index				0.0909*		0.137**	
				(2.22)		(3.31)	
HF Score					0.0125***		0.0125***
					(4.07)		(4.08)
East Asia						-0.490***	-0.495***
						(-5.85)	(-6.48)
Latin America and Caribbean						-0.553***	-0.508***
						(-6.17)	(-5.55)
Constant	7.984***	4.121***	5.381***	4.976***	4.853***	4.525***	4.629***
	(43.13)	(10.13)	(14.96)	(14.62)	(16.14)	(13.91)	(15.67)
No of Observations	202	202	202	202	202	202	202
R-Squared	0.514	0.737	0.816	0.818	0.823	0.853	0.855
F Statistics	165.1	197.6	219.3	190.1	240.5	144.8	170.3

t statistics in parentheses
* $p<0.05$, ** $p<0.01$, *** $p<0.001$

3-ii) The "Stylized Facts" and "Received Wisdom" about Cognitive Ability in China

"Stylized facts" from still-limited psychometric testing hold that mean IQ in China is consistently reported through standardized testing as higher than international averages[106]. The map of tested global IQ results below reflects the "received wisdom" in the psychometric field. A variety of datasets for reported international mean IQ are available at this point, and all of them hold China to be a "high performer" in terms of measured IQ.

[106] Richard Lynn, *IQ and the Wealth of Nations* (Westport: Praeger Publishers, 2002), https://www.amazon.com/IQ-Wealth-Nations-Richard-Lynn/dp/027597510X.

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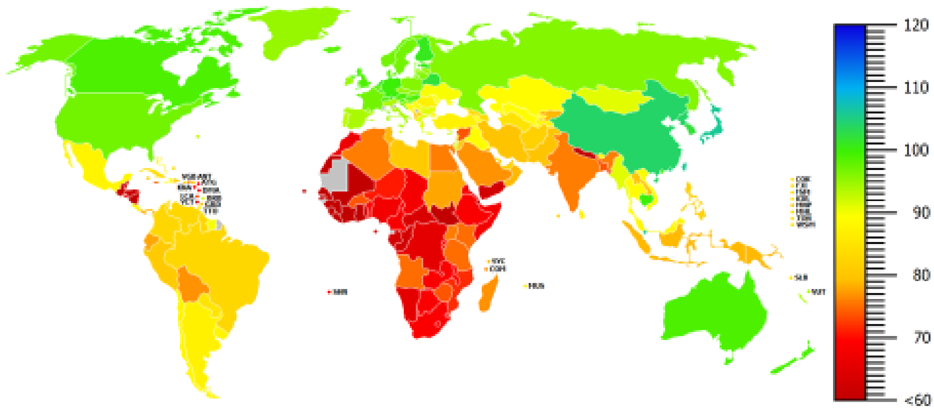

Source: Becker, D. (2019). The NIQ-dataset (V1.3.3). Chemnitz, Germany.
https://www.researchgate.net/project/Worlds-IQ

This particular dataset suggests that average IQ levels in China are about 7.5 points higher than the internationally normed score of 100. (It is one of the very highest readings for China, but all available readings at this writing are above 100 and taken to be "statistically significantly" higher than 100.)

Since IQ tests have normally distributed results and are set with a Standard Deviation of 15 points, a 7.5 point differential would mean that China's national IQ would be one half a standard deviation higher than the international norm. To those who allow themselves to look at psychometric measures and their possible implications, this would appear to be a highly consequential finding, full of potentially important real world implications.

If this result did actually reflect the facts on the ground, it would imply benefits that might be advantageous for the development of knowledge capital (and, by extension, for national economic performance).

Assuming a normal distribution of ability, a half standard deviation difference would mean, for example, that about 70% of such a population would be in the "top half" for ability; that ability of the "ordinary" population's top 2.5% would be possessed by about 7% of the “special” population—almost three times the prevalence in the “ordinary” distribution—and so on.

Other datasets with slightly lower reads for IQ in China would imply correspondingly lower levels of potential advantage: but these presage potential advantage nonetheless. Thus, “stylized facts” and “received wisdom” in psychometrics is consonant with the proposition that China’s pupils would be high performers on international standardized testing and that these rising generations would have advantageous endowments of “knowledge capital”. (By extension, and all else equal, “stylized facts” and “received wisdom” in the psychometric field would seem to encourage the proposition that China is favorably placed for national economic performance and future economic growth.)

But we refer to China’s purportedly high IQ levels as a “stylized fact” for three reasons.

First: there have been perhaps surprisingly few serious studies attempting to measure this quantity in China’s huge population.[107] The happenstance of China’s history may help to explain this. In the tradition of the Marx-Engels worldview and Soviet Lenin/Stalin governance, psychometrics was a suspect intellectual endeavor; with Mao in power, this field of inquiry was not only disfavored but positively dangerous for practitioners for decades. Before Communism, further, there was no great interest in IQ testing either. Japan had a strong influence in China for decades before Communist “Liberation”. But Japan took many of its intellectual leads, including styles, tastes and fads, from Prussia rather than Britain. The British (and Americans) were pioneers in psychometrics; the Germans, by contrast, favored physical measurement of the human form and physiognomy (anthropometrics). Thus, just as in Prussia, Japanese administrators throughout the Home Islands, the colonies, and the conquered territories recorded heights and weights of schoolchildren—but did not test for cognitive abilities.

In 2022, PRC psychologists and clinicians published a paper on a “China-Brief Cognitive Test (C-BCT)” they had developed and begun using to test cognitive

[107] Louise T. Higgins and Gao Xiang, “The Development and Use of Intelligence Tests in China,” *Psychology and Developing Societies*, Vol. 21, No. 2, https://doi.org/10.1177/097133360902100205.

capabilities of Chinese patients.[108] With this domestically developed and vetted test for Chinese citizens by Chinese researchers, it is possible that psychometric testing will become more acceptable, and widespread, in the PRC. Whether that proves to be the case, however, remains to be seen.

Second: given the "Flynn effect" of secular increases in measured IQs for national populations all around the world over the postwar era[109], IQ in China may well be a moving target.[110] The "Flynn effect chart below underscores these particular uncertainties.

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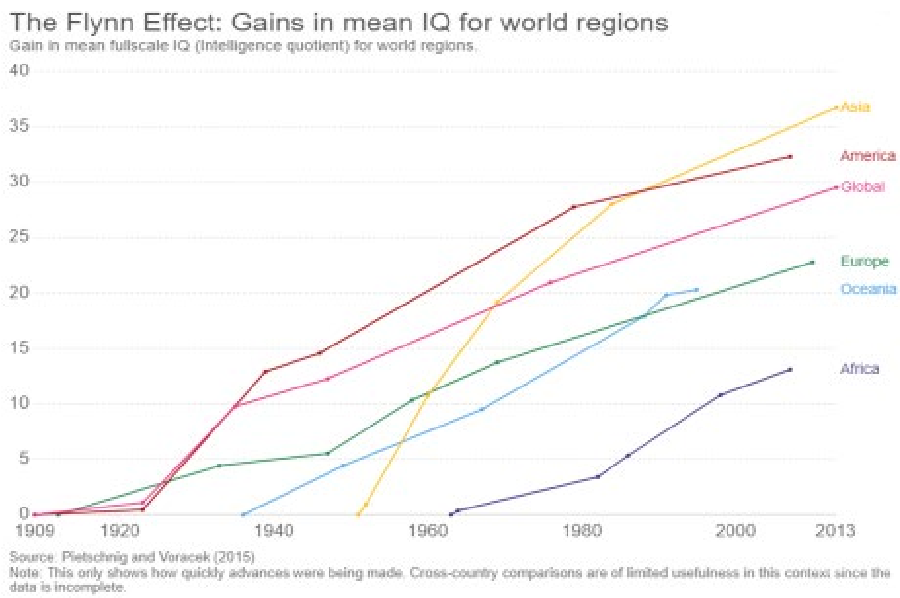

Source: https://practicalpie.com/the-flynn-effect/

Third: IQ testing in China appears to have been generally conducted in urban rather than rural settings—and this means most IQ testing to date has overlooked

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[108] Shuling Ye, et al., "The Chinese Brief Cognitive Test: Normative Data Stratified by Gender, Age and Education", *Frontiers in Psychiatry*, Volume 13, 2022, https://doi.org/10.3389/fpsyt.2022.933642

[109] Lisa Trahan, Karla K. Stuebing, Merril K. Hiscock, and Jack M. Fletcher, "The Flynn Effect: A Meta-analysis," *Psychol Bull*, Vol. 140, No. 4, June 30, 2014, https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4152423/; and Jakob Pietschnig, and Martin Voracek, "One Century of Global IQ Gains: A Formal Meta-Analysis of the Flynn Effect (1909-2013)" *Perspectives on Psychological Science*, 2015, Vol. 10(3) 282-306, DOI: 10.1177/1745691615577701

[110] Mingrui Wang and Richard Lynn, "Intelligence in the People's Republic of China," *Personality and Individual Differences*, Vol. 134, No. 1, November 2018, https://www.sciencedirect.com/science/article/abs/pii/S0191886918303301.

possible urban-rural divergences in measured cognitive abilities in China due to developmental or other differences.

Urban-rural differences in measured cognitive abilities in China would not stand as a conspicuous exception in the corpus of psychometric results to date. There are strong hints of the similar findings from elsewhere in the world.

One of the intriguing findings concerning rural-urban differences in cognitive abilities comes from the General Social Survey (GSS), an instrument developed and administered by the National Opinion Research Center (NORC) at the University of Chicago. GSS, which by now has been in use for over half a century, and utilized to sample populations in over 40 countries, is a staple of academic social science research nowadays.[111] GSS offers an enormous database for examining patterns within and across countries in very rich detail over a time frame now spanning generations.

One of the GSS' many variables is WORDSUM: an indicator drawn from the questionnaire's findings about respondents' vocabulary competence. Researchers have argued that WORDSUM has predictive properties as an indicator of cognitive capabilities. That proposition has been challenged, but increasingly seems to be accepted in the field.[112] WORDSUM data have been used to show a statistically meaningful differential in developed Western countries between urban and rural populations. While part of the differential can likely be explained by acculturation, WORDSUM is seen to retain predictive psychometric properties nonetheless. A research consensus seems to have developed: as one recent paper on WORDSUM's psychometric properties concludes, "despite being culturally loaded, WORDSUM shows no evidence of culture bias".[113] But this is a consensus only within a niche of scholars examining this still academically arcane question.

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[111] The General Social Survey, https://gss.norc.org/

[112] Neil Malhotra, Jon Krosnick, and Edward Haertel, "The Psychometric Properties of the GSS Wordsum Vocabulary Test", August, 2007, https://gss.norc.org/Documents/reports/methodological-reports/MR111%20The%20Psychometric%20Properties%20of%20the%20GSS%20Wordsum%20Vocabulary%20Test.pdf

[113] Meng Hu, "On the Validity of the GSS Vocabulary Test Across Groups", *OpenPsych*, June 2023, https://www.researchgate.net/profile/Meng-Hu-25/publication/371782410_On_The_Validity_of_The_GSS_Vocabulary_Test_Across_Groups/links/64948a7fc41fb852dd2706cf/On-The-Validity-of-The-GSS-Vocabulary-Test-Across-Groups.pdf

GSS is almost ideally placed for providing data on a possible urban-rural cognitive divide. That WORDUM seems to offer evidence for just such a differential ought perhaps to occasion much more research on the matter.[114] But for our purposes here it serves as an invitation to examine, to the extent possible, cognitive performance in China from the under-examined, rural, areas.

3-iii) Cognitive Testing in China: What New Data Reveal for Rural China

Over the past decade, remarkable new research within China has focused on conditions for rural boys and girls in China: infants, children, youth, the range of the pre-college population. Much of this new academic and public policy research has been generated by a collaboration between US and PRC scholars, and is especially associated with Stanford University's REAP (Rural Education Action Program) and its indefatigable Professor Scott Rozelle, the prolific author of among other studies *Invisible China: How the Urban-Rural Divide Threatens China's Rise*.[115]

The illuminating new research on rural China and its young includes psychometric testing. And the findings here are simply stunning.[116] These studies consistently report below-average measured IQs for rural Chinese children. The result also reportedly holds for a range of rural children, from toddlers to 8th graders to migrant children from rural areas currently in urban schools.

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[114] Charles Murray has also found indications of rural-urban differentials of this sort—in prewar Norway. See Charles Murray, "Human Diversity: The Biology of Gender, Race, and Class", January 2020, p. 320, https://www.amazon.com/Human-Diversity-Biology-Gender-Class/dp/1538744015?asin=1538744015&revisionId=&format=4&depth=1 p. 320.

[115] Scott Rozelle and Natalie Hell, "Invisible China: How the Urban-Rural Divide Threatens China's Rise" The University of Chicago Press, 2020, https://press.uchicago.edu/ucp/books/book/chicago/I/bo61544815.html

[116] See for example Xinyue He, Huan Wang, Fang Chang, Sarah-Eve Dill, Han Liu, Bin Tang, Yaojiang Shi, "IQ, grit, and academic achievement: Evidence from rural China," *International Journal of Educational Development*, Vol. 80, 2021, https://fsi9-prod.s3.us-west-1.amazonaws.com/s3fs-public/iq_grit_and_academic_achievement-_evidence_from_rural_china.pdf; Ai Yue, Yaojiang Shi, Renfu Luo, Jamie Chen, James Garth, Jimmy Zhang, Sarah Kotb, Alexis Medina and Scott Rozelle, "China's Invisible Crisis: Cognitive Delays among Rural Toddlers and the Absence of Modern Parenting," Rural Education Action Program, https://sccei.fsi.stanford.edu/reap/publication/chinas-invisible-crisis-cognitive-delays-among-rural-toddlers-and-absence-modern; Dennis Normile, "One in three Chinese children faces an education apocalypse. An ambitious experiment hopes to save them," *Science*, September 21, 2017, https://www.science.org/content/article/one-three-chinese-children-faces-education-apocalypse-ambitious-experiment-hopes-save.

Far from testing above international averages, the rural Chinese children and youth in these studies perform at the other end of the "bell curve distribution". Although the sampling is perforce limited, the findings are very consistent regardless of age or testing instrument. In short: the mean for these rural children in IQ testing come in at around 88 or 89—11 or 12 points below international averages, i.e. close to a Standard Deviation *lower* than that notional global average.

Even more serious cognitive shortfalls are reported for a large fraction of these rural young. Defining "low IQ" as any below 85—i.e. a full Standard Deviation below the norm—these studies identify over a third of rural testees as "low IQ". This work led REAP researchers and Rozelle to warn in 2017 that fully a third of young China is facing the risk of "cognitive deficits".

Needless to say, these research forays into "Invisible China" are depicting a rather different reality for the country as a whole than the "stylized facts" from the "received wisdom" in the field. On the basis of these still preliminary explorations, it would seem likely that current assumptions about average levels of cognitive testing for China's children are biased upward—perhaps quite seriously. The implications for *true* levels of all-China pupil achievement, and for "knowledge capital", should be self-evident.

3-iv) The CEPS: Does its Read on Cognitive for China's Young Make Sense?

One possible treasure trove for information on cognitive capabilities among pupils in China might seem to be CEPS. This panel survey has the promising feature of testing students not only for achievement, but also for cognitive ability. We were excited by the potentialities of the dataset when we learned about this. But after extensive analysis of the data, excitement transformed to disappointment. Rather than a treasure trove, what we found looks more like fool's gold.

Our findings are presented in Slides 360-373 of the Online Appendix. (Also see Slides 244-290 for more detail on our investigation into this dataset.)

Briefly, below the reasons for our disappointment—and indeed skepticism—about what we found:

First: self-evident carelessness in the dataset. We found several inconsistencies and impossibilities in the dataset – numbers not adding up to their reported values, extreme outliers, etc. For example, CEPS includes data on students' academic achievement by asking teachers of sampled classrooms to report how their students did on their midterm exams. While not standardized across schools or across classrooms, the upper bound for CEPS' math score appears to be 150 points. Only one student out of nearly 20,000 sampled had a math score over 150 – with one student reportedly scoring 400 on their midterm. While not *strictly* impossible since scores are not standardized, having one extreme outlier certainly draws skepticism. It seems to us most likely an impossible error, miscoding perhaps. But it was obviously neither identified by the panel study authorities nor corrected, leading us to wonder what else was erroneously incorporated.

Second: the correlations between cognitive scores and math scores in CEPS are curiously low. In the literature on student academic achievement and cognitive ability it is common to find simple correlations of 0.6, 0.7, or 0.8—implying a simple R-squared (or coefficient of determination) ranging from about 36% to well over 60%. In the CEPS dataset, by contrast, we find R-squared values of 25% or less—some of these being as low as just 13%. Why this highly uncharacteristic lack of relationship between cognitive and achievement? We are left without answers to the question.

Third, and perhaps most puzzling, is the correspondence between measured cognitive and student achievement in identified Shanghai counties and the rest. There was only a weak correlation between cognitive and achievement within the Shanghai-identified students, an R-squared of just 14%. The R-squared for the rest was likewise weak, just 13%.

But bizarrely, the regression models for the two groups, with a total of almost 20,000 students between them, cranked out almost identical slopes: a coefficient of 4.411648 for Shanghai-identified, and 4.39992 for the other. That works out to a difference of less than a third of a percentage point between the two—these are almost parallel lines. What was the last time a result like that was seen in a genuinely random panel survey on cognitive ability and student achievement?

To make matters even more outlandish, the "constant" for the simple regression model is lower for Shanghai-identified pupils than the others. In plain English: at any given level of tested cognitive ability, the students from privileged Shanghai were likely to earn *lower* scores than the others! This finding does not pass the laugh-out-loud test. There is no obvious way to explain-or excuse—it. And further attempts at disaggregation show the constant term for "likely rural" to be higher than for "likely urban", which in turn is higher than for identified Shanghai. All three of these populations, incidentally, report almost identical R-squared values, and slopes, for their relationships between cognitive scores and math scores.

What "universe" is CEPS surveying and sampling? From all other evidence presented we have seen in this paper, we would be hard pressed to make the case that these numbers describe any version of the real "China". We are even led to wonder: whether any genuinely rural children are included in this highly vaunted international collaboration on pupil achievement in China? The unapologetic lack of transparency in the CEPS makes it impossible to resolve any such doubts—or to answer other basic questions about the peculiar quirks of this seeming academic resource.

3-v) Rethinking Cognitive and "Knowledge Capital" for Young China

We lack anything resembling comprehensive and accurate data on cognitive ability for China's far-flung and highly differentiated populace. Yet the initial, still exploratory evidence presented above should be sufficient to challenge received wisdom and stylized facts about overall cognitive ability on the Mainland—and invite a much more careful and comprehensive examination of those contours by Chinese and international researchers. Now that PRC specialists have developed their own "China-Brief Cognitive Test", maybe we can hope for more "truth from facts" here. The other alternative, of course is continuing official opacity—more unsatisfying research efforts like CEPS.

Initial evidence on cognitive performance among rural children in China is only suggestive of general patterns for the huge numbers of children in China's hinterlands. We cannot extrapolate from these initial soundings with any precision. Yet it is worth noting that these initial cognitive testing results do seem to track with the strikingly poor academic achievement results recorded by the REAP collaborators in poor rural China, results already discussed. Those standardized achievement readings do not look like anything PISA presents on "China"—they are more akin to the very low results from impoverished Third and Fourth World settings.

It is true of course that PRC is overwhelmingly ethnically Han Chinese—like the Singapore, Hong Kong and Taiwan that rank so high in PISA and other tests of scholarly achievement, and likewise test well in cognitive ability tests . But Singapore and Hong Kong are totally urban populations—and Taiwan's population is also overwhelmingly urban. By contrast, a very large fraction of China's under-18 population is rural. Depending on how one counts "rural"—as we shall see in the following section—the majority, or even the great majority, of primary and secondary boys and girls today in China could be classified as "rural", despite Beijing's urbanization drive and the rapid increase in the share of the country being described as "urban" in national statistics.

We in this project do not believe we are yet in a position to talk confidently about what a comprehensive and accurate cognitive survey would reveal for China or its infants, children and youth. There seems good reason, however, to question existing presumptions. We do not know how much a national estimate of cognitive ability, such as mean IQ, should be adjusted downward for all-China. But it clearly should be adjusted downward—and with those adjustments likewise expectations about overall "knowledge capital" for China's young population.

There are other important issues to raise here as well.

One possibility suggested by the REAP-associated studies might be that China is characterized not by a single normal distribution of cognitive ability but instead several different but overlapping distributions.[117] A "bimodal" distribution of ability, for example, would be an overall population in which both "tails" were "fatter" than one would expect in a normal distribution—with more high ability and more low ability persons than ordinarily expected at one and the same time.

Another implication would concern the workforce today and tomorrow, given the importance of migration from the countryside to urban growth. Is IQ mutable in some measure during a given lifetime? The question arises because of the arithmetic significance of former rural men and women in the contemporary urban population and workforce. If the answer is no, we would expect such a constraint to have a bearing on the development of knowledge capital in the Chinese workforce in the years ahead. If the answer is yes, understanding the mechanisms

[117] Depending on the "malleability" of any such differences in the face of early childhood interventions, other policy measures, family behavior, etc.

through which such effects occur would be key, because Chinese policy could possibly accelerate the influence and thereby potentially augment urban workforce knowledge and skills.

4. **Demography, Family Structure, and "Knowledge Capital" in Pre-College China**[118]

In the broadest sense, demographic trends proscribe the realm of the possible for human populations and the individuals who comprise them. We cannot hope to glean an informed understanding of the realm of knowledge and skills for China's pre-college-age boys and girls unless we also examine the demographic factors that bear on their prospects for achievement: where they live; with whom they live; and what sorts of communities, families, and resources they are being raised with.

An exhaustive treatise could be written on the demographic influences on knowledge capital in China. The overview in this section, instead, will be brief, only touching on key points.

The prospects for knowledge and skill development for China's youth are critically shaped by the geography of disadvantage. Given the yawning urban-rural gap in China, rural background is likely to be a disadvantage for boys and girls—for many of them, a serious disadvantage, as we have already seen. Getting a better sense of the odds of "growing up rural" is therefore necessary for any informed impression of educational quality for China's children today.

Then there is the phenomenon of migration—which relates to the geography of disadvantage, but contributes its own distinctive influence to child prospects. China has undergone momentous urbanization over the past three decades—a phenomenon we have analyzed in detail elsewhere.[119] But police-state social controls on the population have shaped—distorted—the migration process in the PRC, especially since the early 1990s. With notorious *hukou* residence identity cards still in force, permission to migrate is only provisional—and usually only granted to wage earners, not their family members. So the rise of migration in contemporary China has also led to the rise of "left behind children" – children left

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[118] This section of the paper refers to Slides 387-395 in the Online Appendix.

[119] Nicholas Eberstadt and Alex Coblin, with Cecilia Joy-Perez and Kangyu Mark Wang, "Urbanization with Chinese Characteristics: Domestic Migration and Urban Growth in Contemporary China", AEI Press, October 2019, https://www.aei.org/wp-content/uploads/2019/10/Urbanization-with-Chinese-Characteristics.pdf?x91208

to grow up in rural villages while their parents migrate to work in urban areas and send remittances home. Any serious examination of knowledge capital for China's children and youth must recognize this unfortunate fact of life for scores of millions of boys and girls in China today.

Then there is family structure. The influence of family structure on youth outcomes, including educational outcomes and scholastic achievement—is a sensitive topic in many quarters nowadays. Increasingly, it is considered impolite to discuss, at least in elite circles in the US. But this inconvenient topic has generated a mountain of social science research establishing a correspondence—indeed, a contribution—of family structure to youth outcomes. With its huge increases in adult survival chances and its extraordinary—at times coerced—declines in fertility, China has undergone a veritable revolution in family structure in recent decades. We examine some of these basic trends and their possible bearing on the outlook for youth knowledge and skills.

Finally there is the question of wealth transfers and population structure. China's population aging has implications for children, since the very young and the very old may both be claimants for a limited pool of resources. We include some research that may help place this dilemma in perspective.

4-i) Urbanization and the Rural Remnant Population

China has undergone colossal urbanization since the death of Mao and the "opening and reform" that commenced under Deng Xiaoping in the late 1970s. Just how many people live in rural areas today is not quite clear. According to official data, PRC's urbanization ratio leapt from about 21% in 1982 to 64% in 2020. As the world's most populous country (at least until 2020), this means China's urbanization over the past four decades was the most massive urban resettlement of population in human history.

But statistics on urbanization in China are less straightforward than some might assume. For one thing, big redefinitions of "urban" took place in successive population counts—sometimes lowering "urban" totals, sometimes raising them. Consequently, official figures on "urban" and "rural" population are not referring to the same thing over time. For another, the inescapably political nature of Chinese statistics means that politics can take a role in the designation of an area as "urban"—if a place has a military or national security significance, for example.

Under the current PRC statistical approach, if the government moved a nuclear missile base into Yellowstone Park, that immense and almost completely uninhabited spot could be defined as “urban”.

Irrespective of definitions—China’s youth population has been declining since the advent of the One Child Policy in 1979/81. This fact holds regardless of what age range one sets for the youth group. If we look at those who might be in school, the drop is major and ongoing. (We use UN Population Division reconstructions.) For the 6-15 age group—roughly speaking, the years for which the 9 years of compulsory schooling would ordinarily be expected—estimated totals drop by almost a third over the 1980-2020 period, from just under 260 million to just under 180 million. For the 6-18 age group—i.e. the entire precollege school-age population—the decline is also about a third—from just under 330 million to about 220 million.

Although China’s population is rapidly urbanizing—by official PRC statistics based upon residence, a matter to which we will return—the share of young people reportedly living in urban areas is falling. But the proportion of children in rural areas is consistently higher than the national proportion. This abiding distortion on rural population structure has much to do with the PRC’s notorious *hukou* restrictions. To be clear: urban populations tend to attract working age men and women in search of opportunity all around the world. In free societies, the share of working people generally tends to be lower in the countryside than the city, too. *Hukou* restrictions exaggerate this tendency, though.

A large share of China’s infants, children and youth still live in the Chinese countryside according to official PRC residence statistics. In the 2010 Census, over half of China’s under 18 population was counted as residing in rural areas; in the 2020 population count, roughly 3 out of 8 were still counted as rural by that definition of the term.

But there are uncertainties here. Note that, based upon our calculations, the CFPS survey for 2010 indicated a somewhat higher share of rural children as a whole, and also of compulsory school age rural children than the 2010 Census reports. But others working with the CFPS derive much higher rural youth fractions from that dataset. For example, a team including Dali Yang from University of Chicago, a widely regarded political scientist and China authority, conclude that almost three quarters of China's children were rural in 2010 according to the CFPS.[120] And noted authorities on China use even higher figures to represent the share of China's children in rural areas. The renowned Scott Rozelle, for example, routinely uses the number 70 percent to describe the share of China's rural children.[121] And Eric Hanushek, in a recent 2023 paper, uses the figure of 65 percent as the share of China children who are rural.[122]

How to resolve these discrepancies? The written record does not allow us to do this to complete satisfaction. One partial explanation, however, may be the difference between actual residence and *hukou* identity card assigned residence. With the urbanization drive of the 2010s, children were, increasingly, granted provisional—temporary—permission to move to urban areas (formally classified as townships and cities) with their parents. In consequence, we would get very different numbers for "rural" children depending on whether we go by location at the time of a population count on the one hand, or reading their *hukou* papers on the other. Thus, whereas the 2020 census counts 40 percent of China's 6-15 compulsory-school-age children as "rural", 64 percent of the 5-9 age group in China that year had rural *hukou*, as did 68 percent of the 10-14 group. From the standpoint of *hukou data*—the overwhelming majority of boys and girls in China are still considered to be "rural".[123]

[120] Chen, L.J., Yang, D.L., & Ren, Q. (2015). Report on the State of Children in China. Chicago: Chapin Hall at the University of Chicago, Table 1.1 https://www.chapinhall.org/wp-content/uploads/Chapin_CFPSReport2016_ENGLISH_FNLweb-1.pdf
[121] Invisible China pp 8
[122] Gust Hanushek and Woessmann October 13 2023, pp. 21, 67.
[123] We are indebted to Profs. Yong Cai of UNC and Wang Feng of UC Irvine for helping us understand this wrinkle in Chinese demographic data. Private correspondence January 2024.

Slide 387

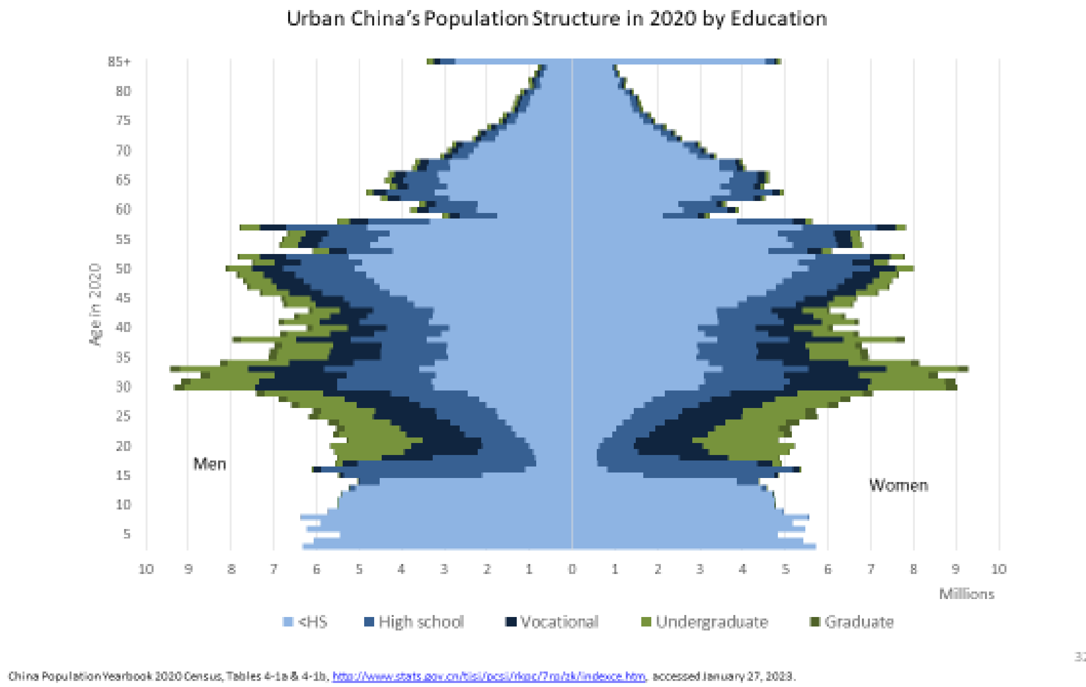
What Percentage Of China's Children Are Rural?

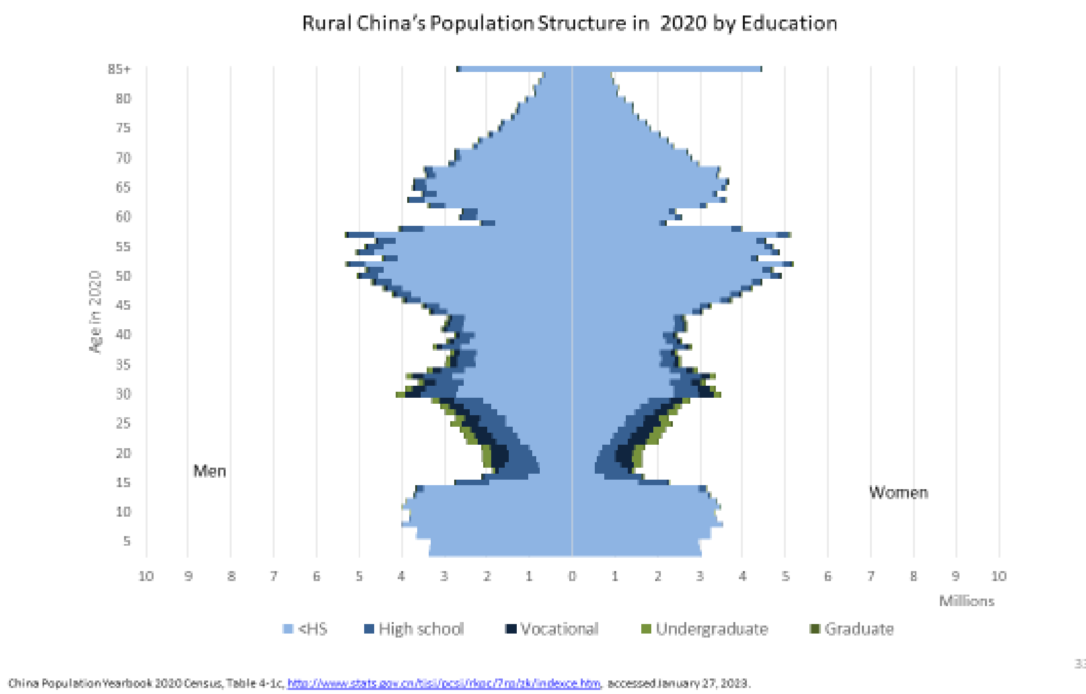
Source	Year Covered	Age range	Percent	Method
CPS NBS	2010	Under 15	57.4	Census--by location
CFPS	2010	Under 15	58.2	Our calculations
CPFS	2010	Under 15	73.1	Dali Yang et al calculation
Rozelle and Hell (2017)	?	?	70	Hukou(?)
Hanushek et al (2023)	?	?	65	Hukou(?)
CPS NBS	2020	Under 15	38.8	Census—by location
CPS NBS	2020	0-4 5-9 10-14	61 64 68	Census-by hukou

We cannot completely resolve these data uncertainties ourselves. But whatever the exact numbers on China's rural population of children may be, there is no uncertainty about the implications of being a rural child in China today: this is a serious mark of socioeconomic disadvantage.

We have already discussed some of the facets of disadvantage for these children. These tendencies include lower family incomes; poorer educational quality; apparently also lower tested cognitive ability. But the fact of the matter is that China's rural children also *are growing up in an environment where adults with even a high school education are scarce—even today*. We may see this in Slide 389. Despite the explosion of higher education in China over the past two decades, scarcely any adults with higher education lived in the countryside in the 2020 census—and the great majority of working age men and women did not even have a high school education in rural China.

Slide 388 and 389

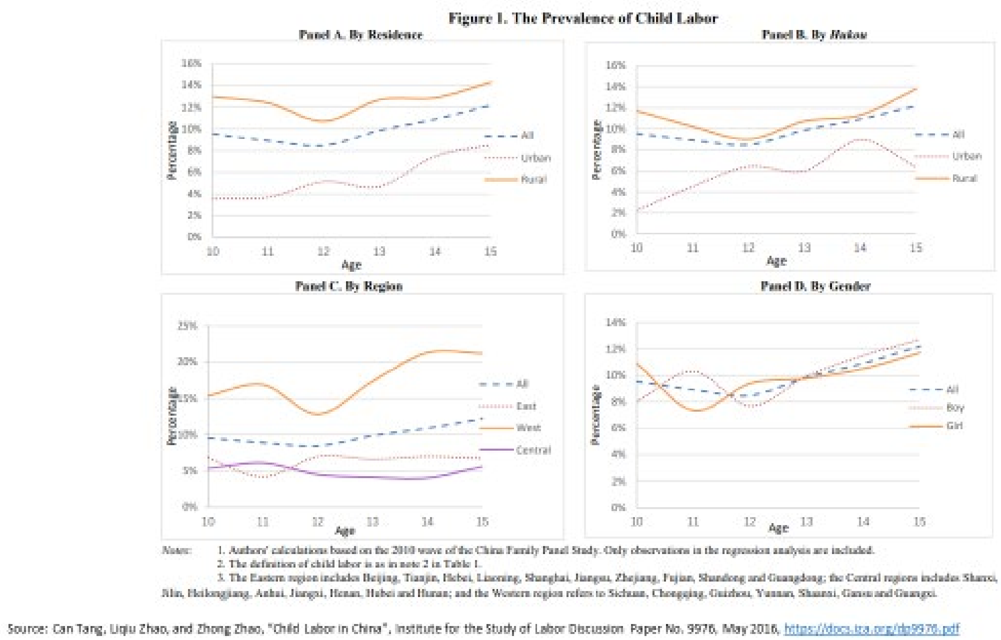




The absence of caregivers with even modest levels of formal educational attainment is yet another disadvantage in the environment for rural children. We cannot say precisely how this affects their prospects for knowledge capital—but it is most surely disadvantageous.

One direct factor influencing the gap between urban and rural knowledge capital is the implicit cost of schooling. Though education at the early levels is mandatory and paid for by the state, there is reason to think that the opportunity cost of time in education may be higher for rural children than for urban ones. One comprehensive study on child-labor in China using the CFPS dataset finds that in 2010, 7.74% of children aged 10-15 in China were engaged in labor in some form.[124] They estimate that 10-15% of rural children participate in the labor force, as opposed to 3-8% of urban children

Slide 390



Surprisingly, while the study finds that children engaged in child-labor are more likely to drop out of school, they found that around 90% of children engaged in the workforce remain in school and divide their time between education and work. This means that many children in China, particularly the rural areas, are accumulating nominal "years of education" despite likely not gaining as much as their nonworking peers – even after controlling for the education quality of a given school. The study finds that on average, "child laborers spend about 2.0 hours a day on study, compared with 8.4 hours for children who are not laborers". This alarming gap of 6.4 hours *per day* clearly has bearing on the knowledge capital of

[124] Can Tang, Liqiu Zhao, and Zhong Zhao, "Child Labor in China", Institute for the Study of Labor Discussion Paper No. 9976, May 2016, https://docs.iza.org/dp9976.pdf

children in China, with rural children coming from poor households the most disproportionately affected.

4-ii) The "Left Behind Children" Phenomenon in China

The dimensions and changing geography of the "left behind child" phenomenon merits coverage in the terrain of knowledge capital for China's pupils. But the socioeconomic condition of these children is not straightforward.

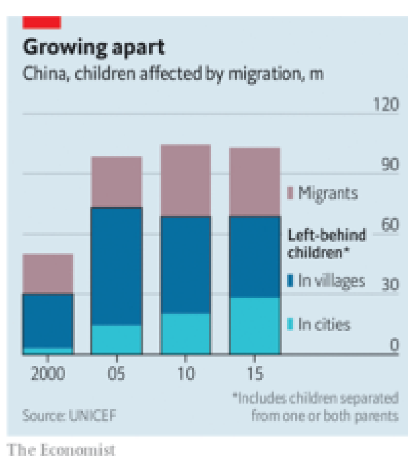
On the one hand, they are being raised without one or both parents—all else equal, a negative. But everything else is not equal: for the most part, their parents are sending remittances home—presumably leaving the children with more resources than they would have had had they remained in the village. The question becomes whether parental time or income is more important for the development of human capital.

Though the evidence is ambiguous, children may lack for their want of parenting. Rural grandparents are just about the least educated and least skilled people in China today—and this may matter. These varying influences deserve examination, and there is considerable research from both Chinse and international scholars to help us parse this out.

Slide 391

The "Left Behind Children" Phenomenon

According to official data for 2015
One third of rural children
And one sixth of urban children
Are "left behind children"



The Economist, "The plight of China's 'left-behind' children", April 8, 2021

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According to UNICEF, in 2015 (the most recent year for which we could obtain such data), 103 million children under 18 years of age were "affected by migration".[125] Of these, 34 million were migrants themselves (disproportionately 15-17 years of age—in other words, job seekers finished with schooling) while 69 million were urban or rural children (41 million of the former, 28 million of the latter) who had one or both parents out of the house seeking work elsewhere. By our rough calculations, this means fully a third of the under-18s in the countryside were "left behinds" in 2015, with one sixth of urban children also counting as "left behinds".

Anecdotal accounts of the "plight of left behind children", as *the Economist* calls it[126], can be heart-rending: tales of suicide, abuse, and tragedy can be found in the Chinese press despite the strict censorship. The question for our purposes: should we regard "left behind" status as a "risk factor" with respect to knowledge capital accumulation? In other words: does being a left behind systematically affect knowledge and skills for Chinese children?

Evidence here seems to be mixed. As we have already seen in our discussion on "cognitive factors", IQ testing for rural left-behind children shows cognitive performance well below international norms. But those same tests, conducted by collaborators with Stanford's REAP program, also show that IQ is well below international norms for all rural children, not just those who have been left-behind. In fact, IQ scores for left behind children were only slightly lower than for children living with parents in rural areas—only a bit over one point separated them. And other research, drawing from the 2010 CFPS, indicates that test scores in math and vocabulary for left behind children 10-15 years of age were virtually identical to those from intact rural families. According to the CFPS, children 10-15 were no less likely to be in compulsory schooling, or to aspire for college, from left behind families than rural intact ones.

There are, however, some signs that left behind children may not receive the nurturing of other children. Left behinds in CFPS were less likely than any other children from any other family type (rural intact, urban intact, migrant) to be read to, or have books bought for them, or to have their homework supervised, or to be

[125] UNICEF, "Children Affected by Migration", Figures 10.1 and 10.2, https://www.unicef.cn/sites/unicef.org.china/files/2019-06/10EN-Children%20affected%20by%20migration%20Atlas%202018.pdf

[126] "The plight of China's 'left-behind children'", *The Economist*, April 2021, https://www.economist.com/china/2021/04/08/the-plight-of-chinas-left-behind-children

offered tutoring—though interestingly that lack of attention did not apparently translate into poorer test performance than for children from intact rural families.

Left behind children may also be more likely to bear certain kinds of emotional or psychological burdens than others: to lack good personal relations; to have lower self-esteem; to have fewer good friends. But this is in comparison to urban intact and to migrant, not just intact rural—there, the differences were very small.

One recent study has found that the "left-behind risk factor" may be systematically different for girls versus boys. This study finds that girls are more likely to be left behind than boys, and that girls receive significantly less than boys in remittances, with estimates of the differential between boys and girls ranging from 9-17%. That is, girls suffer both in terms of parental time *and* income. This work also takes a full life-cycle view of the left behind risk factor and finds that while migrant workers "earn much more and advance economically,…longitudinal data reveals that their children complete fewer years of schooling, remain poor, and have worse mental and physical health later in life".[127] In other words, this study finds that being left behind has a significant negative impact on human capital when viewed over an entire lifetime.

This last study is not only intriguing, but suggestive in another sense: it may be that the identification of systematic disadvantage in knowledge capital require more sophisticated methodological and quantitative approaches that disaggregate sub-groups within the larger pool of left behind children. In itself, that fact may be telling.

Left behind rural children suffer a major disadvantage with respect to prospects for academic achievement by dint of residing in the countryside. That much is clear. Just how much additional disadvantage they face is a question that can only be answered by further careful study.

4-iii) Trends in Family Structure and Kinship Networks

For better or worse, officially collected demographic data today focus only on headcount and household. This is true the world over. It is an enduring legacy of the design of the earliest population counts from the empires of antiquity two

[127] Xuwen Gao, Wenquan Liang, Ahmed Mushfiq Mobarak, and Ran Song, "Migration Restrictions Can Create Gender Inequality: The Story of China's Left-Behind Children", NBER Working Paper, February 2023, https://www.nber.org/system/files/working_papers/w30990/w30990.pdf

millennia ago, both in the Mediterranean and the Far East. That design served the purposes of state: they provided information for manpower mobilization and for taxation. The design arguably still serves the purposes of state, but those same parameters also make it difficult glean the information needed to analyze family structure and extended family networks, factors with a direct bearing on outcomes for rising young boys and girls—not least aforementioned cognitive development and prospects for scholastic achievement.

Fortunately, population *modeling* can simulate family and extended family trends and patterns by in effect generating notional families and kinship networks as "output" from demographic computer software that uses observed (or reconstructed) population information—deaths, births, age and sex totals, marriage, divorce—as the "inputs. Such modeling provides highly serviceable approximations for the evolution of family and kinship in any society since the results must mimic observed demographic trends, and that constraint is highly binding on simulated output.

Long term modeling of China's family structure and kinship networks is now available. Nicholas Eberstadt and Ashton Verdery have published this in a 2023 report.[128] Verdery also pushed initial findings in a 2019 study.[129]

Drawing upon that work, we can glean a better understanding of how China's atrophy of kinship networks, and its rapid increase in the probability that those raising children are also taking care of their own elderly parents, may affect both attention and resources for children.

Thanks to improved survival prospects, orphanhood—a great disadvantage for children through time immemorial—is now almost a thing of the past in China. As may be seen, in these simulations, less than an estimated 0.1% of all Chinese children and youth under 20 years of age had no living parents in 2020—while over 99% of those under 10, and about 96% of those 10-19, had two living parents. This does not necessarily mean both parents were living with these children, though, as we know from the discussion on "left behind children" above. But it is a

[128] Nicholas Eberstadt and Ashton Verdery, "A Huge Demographic Blind Spot with Surprises Ahead", AEI Press, February 2023, https://www.aei.org/research-products/report/chinas-revolution-in-family-structure-a-huge-demographic-blind-spot-with-surprises-ahead/ Unfortunately, modeling limitations do not allow us to simulate urban-rural differences in patterns and trends in family structure and extended kinship network, although we recognize that cleavage may be major and consequential.

[129] Ashton Verdery, "Modeling the Future of China's Changing Family Structure to 2100," in *China's Changing Family Structure: Dimensions and Implications*, ed. Nicholas Eberstadt (Washington: AEI Press, 2019), 23-78, https://www.aei.org/wp-content/uploads/2019/09/RPT-China's-Changing-Family-Structure-online.pdf.

development that leans in favor of better outcomes for children, including educational outcomes.

In addition, these simulations highlight the rise of the only-child in modern China's families. Everyone is aware of this general trend of course, but these simulations add precision to that "stylized fact". Between 1980 (around the start of the One Child Policy) and 2020, the share of boys and girls with no siblings shot up from about 20% to almost 50%; for those 10-19 years of age, the corresponding leap was from 4% to about 43%--nearly a tenfold increase.

(Projections to 2050 are offered in these charts, but those projections depend upon assumptions about future fertility that may not prove accurate.)

These projections demonstrate that even in the One Child Policy era, only children in China were not an overwhelming majority. In fact, they may not even have been a majority—at least not yet. The reason is the arithmetic of childbearing—a highly disproportionate number of births need to be only children for their share to be high in a population. If two mothers have only children and a third mother has twins, only children only make up half the kids in that "population".

Given the broad finding in research on the economics of the family that more resources are devoted per child to households with lower fertility and fewer children (resources here meaning not only financial means but also parental time and attention), the rise of the only child in China would seem, all else equal, to augur positively for child development and prospects for academic achievement.[130]

But the plunge in fertility that propelled the rise of the only child in China also had immense consequences for the extended family, and not all of these were self-evidently auspicious.

Steep and prolonged sub replacement fertility has also affected extended family networks—the traditional social safety net in China, and in practice the main social safety net for the great majority of Chinese today.

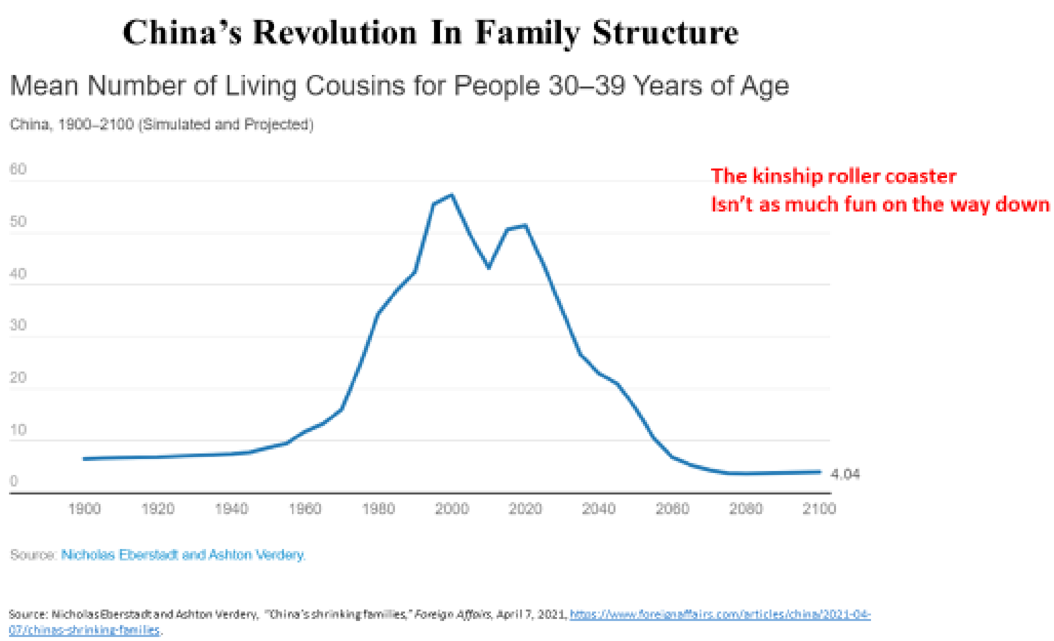
As may be seen from these simulations, there has been an upsurge in the proportion of young Chinese with no cousins, no uncles, and no aunts. The percentages are still low—just 5% of 10-19s in 2020 with no uncles or aunts, for example, but the trend is "baked into the cake" at this point and cannot be stopped

[130] Not all Western observers may agree with this received wisdom from studies on "the economics of the family", it should be noted; some of the new pro-natal activists in Europe and other low fertility regions come to mind in this regard.

for at least the next generation; by 2040 a sixth or more of China's boys and girls 10-19 stand to have no cousins, aunts or uncles.

And the "zero kin" outcome is just the extreme—the impact for family networks on the whole is better represented through average changes. Young people in China are already experiencing a stunning collapse in the extent of their family networks. Between 1990 and 2020, the mean number of cousins for children under 10 plunged by over 75% according to these simulations. Between just 2005 and 2020, they plummeted by nearly 60% for the 10-19 group. And they are on track to fall another 80% or more over the next generation—a change that is essentially unstoppable.

Slide 392



China benefitted in important respects from a "kin explosion" that took place in the postwar era. This demographic earthquake went underappreciated by demographers, economists, and policymakers—perhaps because no data were available to detail it. In retrospect it seems likely that this "kin explosion" was an unobserved variable contributing to China's remarkable economic performance in the decades after Mao. (The same may also be true for other countries that participated in the so-called "East Asian Miracle"—but modeled simulations of their family structures and kinship networks are not yet available). Now that the extended family structure is atrophying—and rapidly—the contribution of this factor to the economic equation has switched signs. Although we lack precise calculations for the magnitude of the effects, it is reasonable to expect the withering of kinship networks will be an economic negative on the macro-side (for overall economic performance) and on the micro-side (for the prospects of individuals—including children of primary and secondary school age).

4-iv) Wealth and Resource Transfers in China in a Family Context with Population Aging

All else equal, the fewer children of a sharply sub-replacement fertility regimen would be expected to receive more familial attention and investment—but are all things equal in a "4-2-1" family, and in a China set on a trajectory of rapid and pronounced population aging?

The National Transfer Accounts (NTA) project[131] helps cast light on this question NTA is an international network of population economists who have created a cross-national database to trace out patterns of lifetime earnings and consumption for dozens of countries—examining patterns across the life cycle for given generations, and also flows of resources between generations. NTA has an active research membership in China. (The NTA China team published some of their findings in a 2019 volume edited by Nicholas Eberstadt—part of what we draw on below.[132])

Empirical evidence matters in this space because results can surprise—so it is worth examining what the data show, and seem to portend, including what NTA may reveal about China's urban-rural differences. We bring other data to bear in this discussion as well.

Within countries and across countries, the inverse relationship between investment in children (on a per capita basis) and fertility levels is well established, as already noted. But the curves in these relationship are not perfectly fitted to the data—there are outliers, and on the international charts China is an important one of them.

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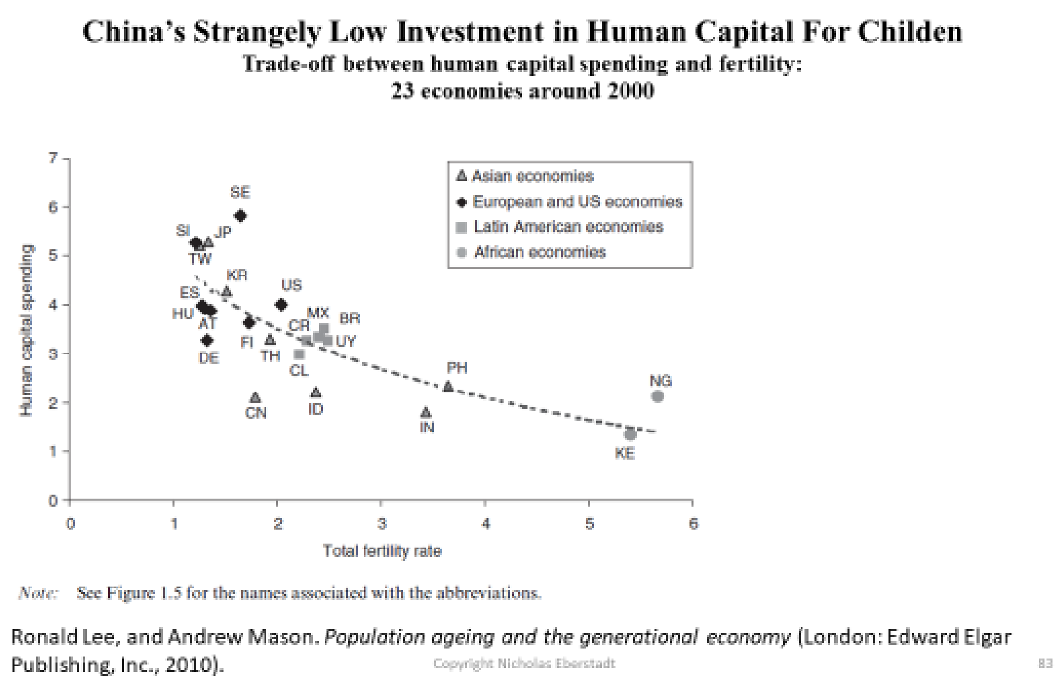
[131] National Transfer Accounts: Understanding the Generational Economy, https://www.ntaccounts.org/web/nta/show/

[132] Wang Feng, Shen Ke, and Cai Yong, "Household Change and Intergenerational Transfers in China," in *China's Changing Family Structure: Dimensions and Implications*, ed. Nicholas Eberstadt (Washington: AEI Press, 2019), 102-115, https://www.aei.org/wp-content/uploads/2019/09/RPT-China's-Changing-Family-Structure-online.pdf

According to NTA, PRC relative spending per child on human capital (health and education, private and public outlays combined) was strikingly low for a population with such low fertility. In calculations for countries around the world around the year 2000, NTA findings suggest that China's relative spending was only about a third of Sweden's—a country with then comparable sub-replacement fertility; its relative level of investment in human capital per child, indeed, was comparable with Nigeria's—where fertility was almost four times as high. Additional work by NTA again confirmed the PRC's 'outlier' status—with relative levels of per child investment lower than the Philippines, where fertility levels were over twice as high, and less than half as high as in South Korea, Taiwan, Singapore, or Japan where fertility rates were only slightly lower.

(Note, by the way, that those four East Asian countries happen to rank up at the very top of PISA's achievement scores for 15-year-old pupils around the world.)

Slide 393

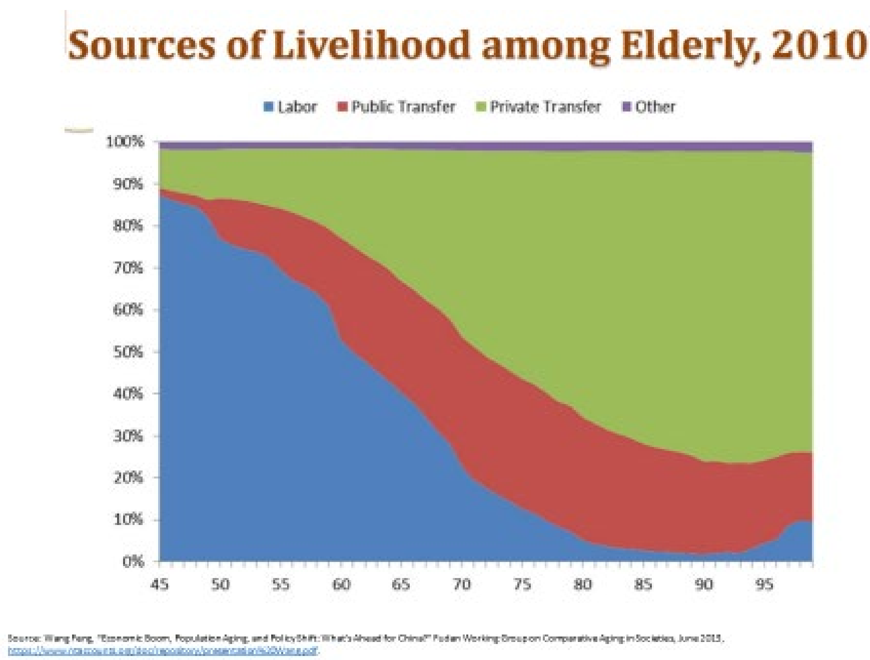


The curiosity of unexpectedly low investment in human capital for children in the PRC may derive from a variety of factors. But one possible factor that comes immediately to a demographer's mind concerns the competition between the young and the old—between two groups of contending "dependents"—for support from relatively scarce private and public resources.

Forty years ago, in a landmark address to the Population Association of America, demographer Samuel Preston of the University of Pennsylvania argued that US children were being squeezed for public resources by seniors: and attributed the imbalance to the fact that the elders were politically influential—i.e., they could vote.[133] The PRC is a rather different system, but it has politics nonetheless, and its politics may favor the elderly—or at least certain contingents of its seniors.

By age 60, according to NTA research, only half of the consumption of older Chinese was coming from the labor-earnings of the men and women in question—and after age 60 there was a sharp falloff in the share of consumption covered by the labor of PRC seniors themselves. By age 65, PRC seniors were earning just a third of what they needed to make ends meet; by 70, just a fifth; by 75, a mere tenth. Public and private transfers covered the rest.

Slide 394



By the reckoning of NTA, perhaps surprisingly, China's elderly already appear to rely quite heavily on public spending for their support. But a huge cleavage is apparent between public support for some seniors in urban areas on the one hand, and the masses of elderly in the countryside on the other. The latter have only the most penurious public guarantees at the moment. Note further that the proportion

[133] Samuel H. Preston, "Children and the Elderly: Divergent Paths for America's Dependents", *Demography,* Vol. 21, No. 4 (Nov., 1984), p. 435-457, https://www.jstor.org/stable/2060909

of elderly seniors living alone is much higher than for the country as a whole—and in 2010, none were more likely to be living alone than men and women in their early 80s.

Taking care of the needs of rural seniors is still mainly an obligation of family members. And as we just saw, the extended family networks that have been set to that task are rapidly atrophying.

Their atrophy is underway at a most inopportune moment, for China is in the midst of extraordinarily rapid population aging. The story of China's impending population aging is already fairly familiar by now, though we illustrate it in depth with our charts.

It is important to appreciate not only the speed and dimensions of China's coming population aging but 1) that this aging is taking place at a much lower level of national per capita income than for other countries that reached corresponding degrees of "graying"; and 2) that the aging is particularly acute in rural areas, where per capita incomes are far lower than in urban areas.

Data on estimated private wealth holding by country from UBS—an innovative project begun at Credit Suisse before its troubles prompted the UBS merger—also highlight the unfavorable relationship between aging and resources for China from yet another international perspective.

The coming demographic pressure of population aging may make it difficult for China to correct the surprisingly low level of human capital investment for its dwindling numbers of children. As Eberstadt-Verdery projections demonstrate, high levels of senior survival and low levels of fertility for future senior during their own childbearing years mean that a generation hence most middle aged Chinese men and women will have elderly parents and in-laws to think about in their consanguineous family units. By 2040, many men and women in their 60s may have to be thinking about supporting elderly parents or in laws—nearly three quarters of married sixty-somethings then are projected to have at least one surviving parent or in-law by then, with almost half projected to have two or more.

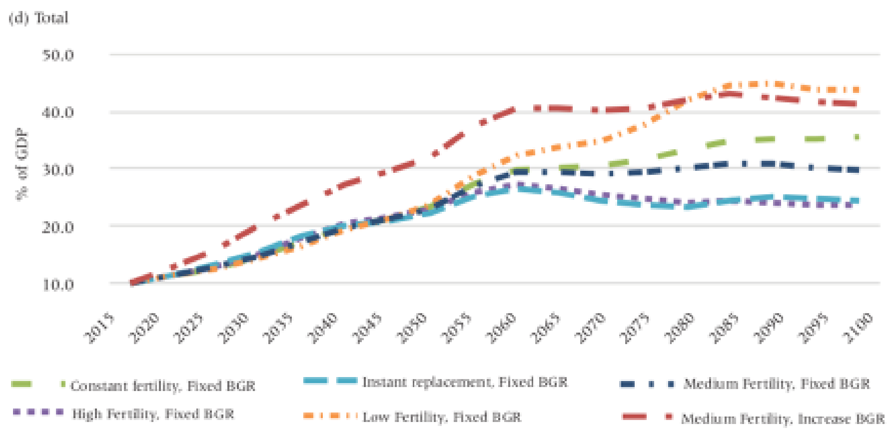
Under such conditions the demand for public resources to support health and pensions for the elderly could prove immense. In 2015, public social benefits accounted for roughly 10 percent of China's GDP. By 2060, NTA researchers hypothesize that social services for China could account for 25%, 35%, or even

40% of GDP—this despite a continuing modeled (by them) decrease in the share of GDP accruing to public support for health and education for China's youth.

Slide 395

The Coming Demand Chinese Social Welfare State?

FIGURE 6 Projected total public spending as a share of GDP under various scenarios—China, 2015–2100



NOTE: Y-axes are not on the same scale as Figure 5.

Source: Yong Cai, Wang Feng, Ke Shen, "Fiscal Implications of Population Aging and Social Sector Expenditure in China", *Population and Development Review* 44(4): 811-831 (December 2018) 92

4-v) Summary Comments

All other things being equal, the corpus of research known today as "the economics of the family" or "household economics" would suggest that China's extremely low fertility levels should be advantageous for human capital formation and "knowledge capital" augmentation for China's rising cohorts of children and youths. Over half a century of work in this field has documented the quantity-quality tradeoff in childbearing and child rearing. Given these findings, one would expect the past generation of sub-replacement fertility in China to have augured well for "knowledge capital" for the young in China—all the more so given the high regard in which education and studying is held in China, where the Confucian tradition's "*gaokao* culture" is still alive and well today.

But all things are not equal in the PRC. "Low fertility with Chinese characteristics" means *hukou* restriction and rural neglect; the withering of the extended family system that has served as the social safety net in China throughout history; and steep, pronounced aging of society, with an exploding population of senior citizens competing with children for limited family and public resources.

Although we suspect the adverse implications specifically of the “left behind children” phenomenon may be overstated as regards “knowledge capital” in some contemporary reportage[134], the overall tableau of tendencies and pressures we have described in this section seems to us qualitatively to lean in the direction of downward rather than upward adjustments to our expectations about the true all-China level of pupil achievement today.

[134] We are only talking here about the impact of parental migration of “knowledge capital” for the children in question: evidence of other adverse impacts (including emotional and psychological) appear to us to be more fully substantiated.

5. **Concluding Observations, Questions, and Next Steps**[135]

Our investigation of the terrain of knowledge and skills for China's school-age population has relied heavily on quantitative data, though we have also been alert to the sorts of qualitative hints and evidence that can always improve any assessment of human conditions, and that, we believe, certainly helps in this one.

While we concentrate on test-score data that purport to measure student achievement—specifically reading, math and science abilities—we hasten to underscore our conviction that there is more to "knowledge capital" than these quantities, important as they are. Un-measured and perhaps unmeasurable attitudes, qualities and talents—persistence and determination; sociability and disposition to cooperate with others; entrepreneurialism; optimism; the ability to come through under pressure—all of these matter more than incidentally to individual economic performance, and perhaps to national economic performance as well. None of those are examined in this long paper. It does not mean we believe they do not matter.

We confronted the puzzle in assessing the "knowledge capital" of China's school age population. The data that would ordinarily be our tools in such an exercise were highly problematic: limited and flawed—and deliberately so. The lack of transparency that bedeviled our investigation reflects the will of decision-makers in Beijing. In effect, the data at our disposal were akin to (perhaps even quantitative equivalents of) "hostile witnesses" in the courtroom.

But under proper questioning, meaningful information can be obtained from hostile witnesses on the dock—and as we have shown, it is possible to extract some meaningful information about the true state of knowledge and skills among China's school age boys and girls from the PRC's deliberately limited and opaque official data, too.

Our initial foray into international standardized tests of student achievement theorized that all-China test data, if accurate and faithfully recorded, would likely place China in the company of other (mainly) upper-middle-income economies, simply given international patterns of correspondence between developmental trends and student performance:

[135] This section of the paper refers to Slides 437-439 in the Online Appendix.

Slide 437

Selected Illustrative Comparator Countries: Actual HLO Scores vs. Modeled All-China Scores

Country	HLO Mean National Score	2020 Per Capita GDP (PPP 2017 Intl Dollars)
Vietnam	519	8,200
Mauritius	472	20,522
All-China Modeled 2015	468	12,612 (2015)
Malaysia	468	26,472
All-China Modeled 2012	461	10,370 (2012)
Turkey	459	28,393
Gabon	456	14,321
Kenya	455	4,340
All-China Modeled 2009	454	8,069 (2009)
Cambodia	452	4,192
Mexico	429	17,852

Sources: https://blogs.worldbank.org/opendata/harmonized-learning-outcomes-transforming-learning-assessment-data-national-education; https://data.worldbank.org/indicator/NY.GDP.PCAP.PP.CD ; Figure 15 in this report.

Turkey was the country we identified as the "baseline" comparator for the PRC in this regard: we theorized that additional information (such as reliable readings on the cognitive abilities of school-age children, for example) might ultimately prove that PRC performance was more favorable than Turkish performance, but that Turkey would be the "starting point" for placing China in international perspective.

In this much deeper and more extensive exploration, we seem to have returned to that Turkey analogy with China for "knowledge capital" for pre-college youth—again and again. And we note that Eric Hanushek and colleagues seem to offer at least the same possibility in their innovative new work on "global universal basic skills". *If* the real China were to fall at the midpoint of the rather larger range they demarcate in their "sensitivity analysis" for the PRC (they had to cope with the same data problems we did), then the proportion of school-age children in China lacking basic skills would be similar to the measured fraction authentically reported for the Republic of Turkey: actually, somewhat below Turkey.

Slide 438

Basic Skills Levels for China Youth: Comparators from Hanushek et al GUBS Estimates

China Estimate	Percentage Youth with Basic Skills	Countries with Similar Levels
High Basic Skills	81	Denmark (81) Lativia (81) Netherlands (81) Vietnam (81)
Low Basic Skills	31	Myanmar (30) Libya (32) Bangladesh (33) Brazil(35)
Midpoint of Estimates	56	Uruguay (51) Armenia (55) Malaysia (55) Turkey (66)

Source: https://www.sciencedirect.com/science/article/abs/pii/S030438782300161X?via%3Dihub

Of course, that is a rather big *if*. We would be more confident of the ultimate relevance of this baseline if there weren't counties markedly less developed than the contemporary PRC that look to be consistently matching, and even outscoring, OECD in standardized student achievement.[136]

Vietnam is the prime "contrary cautionary" against our "baseline" Turkey-normed assessment for China. Here is a predominantly rural society with half the PRCs' level of per capita output that appears, in the Hanushek et al. table immediately above, to "present" like a Northern European society as far as its measured basic skills profile for youth is concerned. More attention to the Vietnam case is surely

[136] Vietnam's 2018 PISA scores were among the very highest reported internationally—and these results came from nationally representative sampling for Vietnam, unlike China. But PISA did not "validate" the Vietnam findings because those tests were conducted via paper instruments (rather than electronically), so Vietnam was not included in PISA's official rankings. Vietnam's high achievement levels in the PISA tests, however, preceded the 2018 PISA "referee call" against it, and was also validated in the 2022 round of PISA. Cf. "VN gets high scores but not named in PISA 2018 ranking", *Viet Nam News*, December 6, 2019, https://vietnamnews.vn/society/569454/vn-gets-high-scores-but-not-named-in-pisa-2018-ranking.html#:~:text=H%C3%80%20N%E1%BB%98I%20%E2%80%94%20Vi%E1%BB%87t%20Nam%20has,(ranking%204th)%20in%20science

warranted—not only from what it may tell us about potentialities for improving "knowledge capital" on a low budget, but also for what it may reveal about that 100 million-plus population's economic potential in the decades immediately ahead.

Our assessment, and the Hanushek et al. assessment utilized in Slide 439 below, indicate that the PRC has a larger population of young people with at least basic levels of skills than any other country on earth.

Slide 439

Illustrative Estimates of Youth on Track for Basic Skills

(applying Hanushek et al GUBS basic skills estimates to 2022 youth population)

Country/Region	0-17 Population (million)	Percent with Basic Skills	Total in Millions
USA	74	75	56
Russian Federation	30	77	23
Europe minus Russia	110	70	77
Japan	18	87	16
India	434	11	49
Latin America&Caribbean	183	35	64
Sub-Saharan Africa	557	6	33
China: Low Skills Estimate	295	31	92
China: High Skills Estimate	295	81	239
China: Midpoint Estimate	295	56	165

Sources: https://population.un.org/wpp/ ; https://www.sciencedirect.com/science/article/abs/pii/S030438782300161X?via%3Dihub

Even with a decidedly more "pessimistic" reading than we would venture—i.e. the Hanushek et al. "China low skills" estimate above—the PRC would have larger absolute numbers of young men and women with basic skills in its rising pre-college cohort than any other country: half again as many as the USA; almost twice as many as India; more than all the EU-27; three times as many as the entire SSA sub-continent. At the "midpoint estimate"—numbers close to our "baseline" assessment—China would have more rising youth with basic skills than the USA, EU-27, and Japan *combined*.

It is worth asking about the sorts of advantages and opportunities that might accrue from the benefits of such scale, economically and strategically, even if that profile also implied that the PRC was not yet ready to advance into the ranks of the developed economies.

The comparisons above also throw a highly unflattering light on India—where barely over a tenth of the school-age population is said to have attained basic skills. This assessment of "knowledge capital" in India invites much further attention, examination, and discussion.

If very roughly accurate, it implies serious human resource constraints on India's great power prospects, despite being the world's most populous country today and for the foreseeable future. It also may speak to the limits of potentialities in any possible US alignment or alliance with India.

These observations are not in conflict with a recognition of sustained rapid economic in India over the past generation, or of India's increasing military might over this same period.

But we should also recognize that India's amazingly low proportion of pupils estimated to possess basic skills is consistent, in the real existing India, with the simultaneous existence of a significant and impressive cadre of highly educated intellectuals, technicians, and policymakers; arguably one of the foremost cadres in the world today.

The seeming contradiction of a large and vibrant cadre of highly trained scientists and professionals at the pinnacle of an enormous population of not-even-barely-trained adult men and women speaks directly to three generations of policy choices in India's democracy: of a nation-building effort that prioritized an educated elite over mass education. At home and abroad, India now must live with the path-dependent consequences of those priorities.

At least two additional questions arise from our study. First: how has the Covid pandemic affected "knowledge capital" for China's school-age children?

We know the impact of the coronavirus lockdown was severe for counterpart children in the USA. Hanushek (at the forefront in this research as well) estimated terrible economic costs for the "learning loss" in America due to the school shutdowns and other educational disruptions, especially for the youngest students: from California alone he estimates the economic losses at over $1 trillion.[137]

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[137] Eric A. Hanushek, "The Economic Cost of the Pandemic: State by State", Hoover Institution, 2023, https://hanushek.stanford.edu/sites/default/files/Hanushek%202022%20HESI%20EconomicCost.pdf

What about China? We know PRC could not report results for PISA 2022 because the schools that would have been tested in B-S-J-Z were not open.

What has been the scale of "learning loss" for China as a whole? Reportage speaks of serious difficulties for school-age students in China.[138] It is commonly thought, in the PRC at least, that schooling was especially hard hit in rural areas;[139] and of course the quality of education was poorest there to begin with. When, and how, will we come by information to help us understand the magnitude of "Covid learning losses with Chinese characteristics"?

This question begets another, more general question: how will Chinese authorities extricate themselves from the statistical corner they seem to have painted themselves into with their acceptance of unrealistic published assessments of scholastic performance of school age children in the PRC? How to make a transition to relatively accurate assessments of knowledge and skills for their rising youth cohorts for PRC authorities in the best positon to know just this?

In India, educational authorities unintentionally trapped themselves in their limited participation in PISA testing. The seemingly safe choice of "pilot" testing Tamil Nadu—India's famously most literate state—resulted in disastrous and embarrassingly poor metrics that placed India near the very bottom of tested populations around the world. To no surprise, Dehli has not consented to any further PISA testing, much less coverage of the entire country with PISA soundings.

Beijing's problem is somewhat different. Having gamed the PISA test so spectacularly, any accurate and genuinely random sample of the areas previously surveyed will almost inevitably mean a sharp drop in "performance" for China—and extending coverage to the entire population of school-age children would quite possibly be a public relations fiasco for the regime. Educational planners in China thus face the risk of their own version of a "Great Leap Forward" moment—when

[138] Jane Cai and He Huifeng, "Mass mental health crisis looms for young Chinese after 3 years of lockdowns, home school and zero-Covid", *South China Morning Post,* December 2022, https://www.scmp.com/news/china/politics/article/3202120/mass-mental-health-crisis-looms-young-chinese-after-3-years-lockdowns-home-school-and-zero-covid

[139] Chol-Kyun Shin, Youngeun An, & Soon-young Oh, "Reduced in-person learning in COVID-19 widens student achievement gaps in schools", *Asia Pacific Education Review*, June 2023, https://link.springer.com/article/10.1007/s12564-023-09862-0

completely unrealistic claims and promises about government-led national performance collide with unforgiving reality.

Officials are likely to do what they can to put off that moment of truth, of course—that is what cadres (and perhaps bureaucrats everywhere) do. But further questions also arise.

Do NAEQ cadres possess an accurate understanding of the true mean level, and extraordinarily wide distribution, of student achievement in their country?

A 2017 PRC NAEQ presentation to UNESCO, cited in this paper, indicates that NAEQ has its own unpublished version of an achievement test that tracks very closely with PISA 2015 results for 10 provinces. Have these internal tests continued? Do they cover all of China? Who is allowed to see such results as exist?

PRC NAEQ's National Large Scale Assessment (NLSA) of "learning poverty" allowed the World Bank's team to place China's school age learning deficits in international perspective. Have they done the same thing?

Do they know how far China needs to go before pupils from all-China (not the imaginary "China" in the PISA assessments) could perform at OECD levels? At Vietnam levels? At Taiwan levels? And if so, have they communicated that information (very possibly bad news) to the Minister of Education? The Premier of the State Council? The General Secretary of the CCP?

Top leaders in China have been woefully out of touch about general conditions in their country in the none too distant past in other—now fatefully obvious—respects. Consider only PRC population policies: enforced with merciless coercion under the now obviously highly mistaken presumption that 21st Century PRC peasants still wanted large numbers of children, and ostentatiously relaxed with the expectation that births would then surge, when in fact the conditional suspension of the One Child Policy was followed by a sharp and still continuing birth collapse.

Is Chinese leadership systematically misled about the true state of "knowledge capital" in the country in general, or for school-age children in particular? And if so, what sorts of miscalculations could those misimpressions contribute to?